NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

Contents

1. Conventions, definitions and the frozen statements 2
1.1. Conventions 2
1.2. Track-K definitions used 3
1.3. Frozen track-N definitions 3
1.4. The frozen theorems 5
2. Interfaces and conversion of hypotheses 6
2.1. External facts 7
2.2. Domains and norms 7
2.3. Holomorphy implies real smoothness 8
2.4. Quasi-convexity: the uniform Euclidean form 9
2.5. Normalisations 10
2.6. The dictionary of constants 11
3. N-QC and N-ZERO 13
3.1. N-QC 13
3.2. N-ZERO 14
4. Hamiltonian ODE infrastructure (shared by tracks I, N, P) 15
4.1. Conventions and external facts 15
4.2. The general solution predicate; Lipschitz data 17
4.3. Local existence 18
4.4. Uniqueness on every interval 19
4.5. Maximal solutions and continuation 19
4.6. First exit from an open set; barriers 21
4.7. Hamiltonian layer 22
4.8. The frozen solution predicates are instances 22
4.9. The track-N statements 24
4.10. Label index for tracks I and P 25
5. Weighted Fourier norms and Cauchy estimates (shared by tracks N, P, K) 26
5.1. External facts 26
5.2. Domains, function class, norm 26
5.3. Lemma B.1: sup norm versus weighted norm 27
5.4. Lemma B.2: algebra property 28
5.5. Lemma B.3: Cauchy estimates in the weighted norm 28

5.6. Lemma B.4: the Poisson bracket 29
5.7. Lemma C.1: the general Cauchy inequality 29
5.8. Projections and truncation 30
5.9. The stronger norm ♯ (sum of coefficient suprema) 30
6. Canonical transformations: time-one maps and the Lie transform (shared by tracks N, P) 31
6.1. Setting 32
6.2. The flow of a small holomorphic vector field 32
6.3. Taylor's formula along the flow 34
6.4. Lemma B.5: composition in the weighted norm 35
6.5. The canonical map $\Phi = X_\phi^1$ on weighted domains 37
6.6. Variants for the norm ♯ 38
7. The homological equation, the Iterative Lemma and the Normal Form Lemma (shared by tracks N, P) 38
7.1. External facts 38
7.2. Setting and the divisor bound 39
7.3. The homological equation 39
7.4. The Iterative Lemma 40
7.5. Composition of the steps 42
7.6. The Normal Form Lemma 42
7.7. The case $2\pi Ks < 6$ 44
7.8. Deviations, errata, uses 45
8. Stability estimates: nonresonant, convex, quasi-convex and resonant (shared by tracks N, I) 45
8.1. External facts 46
8.2. Real systems on a real neighbourhood of D 46
8.3. Transfer through a symplectic change of variables 47
8.4. Proposition 1: nonresonant stability 49
8.5. The energy argument 50
8.6. Lemma 1 and Lemma 2 51
8.7. Proposition 2: the Resonant Stability Lemma 52
8.8. Errata, deviations, dictionary, consumers 53

1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) are not yet frozen and are not part of this part. Results of track K's blueprint are cited as [K:`bp:p1:...`].

1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every n. A phase point is $z = (\theta, I)$ with angles θ and actions I, in this order.
- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise.

The norm $\|\cdot\|$ in every frozen statement is the max norm; on the phase space $\mathbb{R}^n \times \mathbb{R}^n$ it is $\|(\theta, I)\| = \max(|\theta|, |I|)$ (Lean product norm). The only Euclidean structure is the dot product $\sum_i \omega_i v_i$ used for orthogonality in quasi-convexity; Euclidean norms used later for comparison with the sources come with explicit inequalities ($|v| \le \|v\|_2 \le \sqrt{n}|v|$). For $n = 0$ all these spaces are $\{0\}$ and the norm is 0.

- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex h is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v|\le 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on f (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k\cdot\theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < s \ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

1.2. Track-K definitions used.

Definition 1.1 (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, J viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \le 0$, and for B open B_r is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 (D_s); track K, [K:`bp:p0:domains`].

Definition 1.2 (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at I. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at I (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

Definition 1.3 (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i,\ (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if H is not differentiable at z).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

Definition 1.4 (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with θ, I viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

Remark 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; for $n > 0$ it is empty when $s \le 0$, and for $n = 0$ it is the whole (one-point) space $\mathbb{C}^0$ for every s. No hypothesis $n > 0$ is added to N-ZERO or N-QC.
Frozen Lean declaration: `KAM.complexStrip`

1.3. Frozen track-N definitions.

Definition 1.6 (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.
Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

Definition 1.7 (Complex domain)**.** $D_s(R) = T_s \times B(0, R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\, R\, s$. It is the period-one lift of BM11's D_s, with a single width s for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 (D_s); ours for the lift.

Definition 1.8 (Real action domain)**.** $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0,R)_s\}$; `realActionDomain` $n\,R\,s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 (B_s).

Remark 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I-J|$ for real I, J is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I-J| < s\} = B(0, R+s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0,R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

Definition 1.10 (Real part of the integrable part)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` h.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

Definition 1.11 (Analytic integrable part, $h \in A_s(D)$)**.** `IsAnalyticIntegrable` $R\,s\,h$ holds if

(i) h is complex differentiable on $B(0,R)_s$;
(ii) h is continuous on $\overline{B(0,R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \geq 0$ with $|h(I)| \leq A$ for all $I \in B(0,R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class C^3 on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on h is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

Definition 1.12 (Analytic perturbation of size ε)**.** `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) f is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with 2ε. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

Definition 1.13 (Derivative bounds $B(M)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, U is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1,2,3\}$ and all $(i_1, \ldots, i_k) \in \{1, \ldots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \ldots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for C^3 functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

Definition 1.14 (Strict quasi-convexity $QC(m)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\,h\,m$ holds if $m > 0$, U is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v,v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.

Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

Definition 1.15 (Complex Hessian bound)**.** `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, h is C^2 in the complex sense on $B(0,R)_s$ and $\|D^2h(z)\| \le M_{\mathbb{C}}$ for all $z \in B(0,R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n,\mathbb{C}))$ for the max norm. Every entry $|\partial_i\partial_j h|$ is then $\le M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\le nM_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian h" (conversion by n recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

Definition 1.16 (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\,U\,J\,z$ holds if for every $t \in J$: $z(t)_2 \in U$, H is differentiable at $z(t)$, and z is differentiable at t (two-sided, also at endpoints of J, since z is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

Definition 1.17 (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0,I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\, I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

Remark 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) In N-HEAD-C and N-ZERO, ω_h and $D^2h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is C^3 (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones. N-QC uses them on its arbitrary open U, where its hypothesis gives C^2 (item (e)). The definitions `frequencyMap`, `integrableOrbit` themselves are total and keep Lean's default values outside the differentiability locus.
(b) D^2h for complex h is used only on $B(0,R)_s$, open, where h is C^2 (Definition 1.15).
(c) X_H is used only inside `IsSolutionOn`, which requires differentiability of H at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathbb{R}^n \times \mathcal{B}_s(R) \supseteq \mathbb{R}^n \times B(0,R)$ (angles first, actions restricted) because h, f are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda\omega_h(I'))$ is differentiable at (I, λ) because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

1.4. The frozen theorems.

Theorem 1.19 (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \ge 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\,\varepsilon_0^{1/(2n)} \le R/4$$

such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

(H1) `IsAnalyticIntegrable` $R\,s\,h$ *(Definition 1.11);*
(H2) `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ *(Definition 1.15);*
(H3) $\omega_{\min} \le |\omega_h(I)|$ *for every $I \in B(0,R)$ (max norm);*
(H4) `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\,h\,m$ *(Definition 1.14);*

(H5) `HasDerivativeBounds` $\mathcal{B}_s(R)\, h\, M$ *(Definition 1.13);*
(H6) `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12);*
(H7) $0 < \varepsilon \le \varepsilon_0$.

For every $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\tau := T\exp\bigl(c\,\varepsilon^{-1/(2n)}\bigr)$ *and* $H = H_{h,f}$, *there is* $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with*

(a) $z(0) = (\theta_0, I_0)$;
(b) `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$;
(c) $|z(t)_2 - I_0| \le C\varepsilon^{1/(2n)}$ *for all* t *with* $|t| \le \tau$;
(d) (uniqueness) for every $L \in [0, \tau]$ *and every* $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with* $y(0) = (\theta_0, I_0)$ *and* `IsSolutionOn` $H\, B(0,R)\, [-L, L]\, y$: $y(t) = z(t)$ *for all* $t \in [-L, L]$.

The order of choice is: n; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; *then* C, T, c, ε_0, *depending only on these seven data; then* h, f, ε, *then* θ_0, I_0; *the solution* z *may depend on all of them. There is no dependence on* A *in Definition 1.11(iii). In (H4)–(H5) the set is* $\mathcal{B}_s(R) = B(0, R+s)$ *(Remark 1.9), in (H3) the smaller ball* $B(0, R)$.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of `docs/tracks/N/sources.md`*); exponent* $1/(2n)$ *as BM11 p. 5 with* $\delta = (2n(n-1))^{-1}$; *ruling D18 Q2, Q4; deviation D20: the extra inputs* $M_{\mathbb{C}}$ *and* $\omega_{\min}$.

Source gap: The printed theorem is for a 2π*-periodic torus, Euclidean norms and exceptional initial frequencies* $\|\omega(I_0)\| \le mr_0/8$; *the conversion to the frozen statement (rescaling* $H \mapsto 2\pi H(\varphi/2\pi, I)$, *choice of* r_0, s_0 *from the seven data, exclusion of the exceptional branch by* $\omega_{\min}$, *existence and uniqueness of the solution) is given only in* `sources.md` *and is proved in Part 10 of* `PLAN.md`.

Theorem 1.20 (N-ZERO: exact integrable dynamics)**.** *Let* $n \in \mathbb{N}$, $R, s > 0$, h *with* `IsAnalyticIntegrable` $R\, s\, h$ *and* f *with* `IsAnalyticPerturbation` $R\, s\, 0\, f$. *For every* $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\gamma = \gamma_{\theta_0, I_0}$ *(Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`

Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes $\varepsilon = 0$*); from* $H = h(I)$, *BM11 p. 1.*

Theorem 1.21 (N-QC: nonsingular isoenergetic derivative)**.** *Let* $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ *with* `IsStrictlyQuasiConvex` $U\, h\, m$. *For* $I \in U$ *and* $\lambda > 0$, *the derivative at* (I, λ) *of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \bigl(h_{\mathbb{R}}(I'),\ \lambda'\omega_h(I')\bigr)$$

is a bijective linear map $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. *Explicitly it is* $D\Psi(I, \lambda)[v, u] = \bigl(\omega_h(I) \cdot v,\ u\,\omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I)v\bigr)$. *No uniform radius of invertibility is asserted.*

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).

2. Interfaces and conversion of hypotheses

This part connects the frozen hypotheses of track N (Definitions of Part 0) to the form in which the proofs use them. It proves: the real action domain and the inclusions of tubes (§2.2); that holomorphy gives a smooth real Hamiltonian with $X_H = X_{h+f}$ (§2.3); the translation of the derivative bounds $B(M)$, the complex Hessian bound and $QC(m)$ into the Euclidean, uniform form of Pöschel's Theorem 1 (§2.4); the change of normalisation 2π-periodic↔ 1-periodic and the sup-to-weighted Fourier norm (§2.5); and the dictionary of constants (§2.6). *Convention ruling.* All of Parts 1–10 work in the frozen conventions (period 1, max norms, characters $e^{2\pi i k\cdot\theta}$). Pöschel's normalisation (period 2π, Euclidean norms on actions, weights $e^{|k|_1 s}$) enters only through the labelled conversions Lemmas 2.8, 2.9, 2.7, 2.10; quantities in his normalisation carry a tilde or the

subscript P. This overrules the suggestion in `PLAN.md` to work in Pöschel's normalisation. The Euclidean norm is written $|\cdot|_2$, the max norm $|\cdot|$ (or $|\cdot|_\infty$), and $|k|_1 = \sum_i |k_i|$; the phase-space norm is $\max(|\theta|_\infty, |I|_\infty)$, Euclidean structure is used only for the dot product and where stated.

2.1. **External facts.**

(E1) *Finite-dimensional linear algebra.* An injective linear map $L : V \to W$ between finite-dimensional real vector spaces with $\dim V = \dim W$ is bijective (rank–nullity; Mathlib `LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, in `LinearAlgebra/FiniteDimensional/Lemmas.lean`; the endomorphism form `LinearMap.injective_iff_surjective` does not apply directly to $V \neq W$). The finranks used below are $\dim(\mathbb{R}^n \times \mathbb{R}) = \dim(\mathbb{R} \times \mathbb{R}^n) = n + 1$ (`Module.finrank_prod`, `Module.finrank_fin_fun`). Exists.

(E2) *Norm comparison and Cauchy–Schwarz.* For $x \in \mathbb{R}^n$ or $\mathbb{C}^n$: $|x|_\infty \le |x|_2 \le \sqrt{n}|x|_\infty$ and $|x|_1 \le \sqrt{n}|x|_2$, $|x|_2 \le |x|_1$; $|x \cdot y| \le |x|_2 |y|_2$ (for the bilinear, not Hermitian, product too). Exists.

(E3) *Restriction of scalars and chain rule.* A $\mathbb{C}$-differentiable (resp. C^k) map is $\mathbb{R}$-differentiable (resp. C^k) with the same derivative; the chain rule for the $\mathbb{R}$-linear inclusion $\iota : \mathbb{R}^N \to \mathbb{C}^N$ and for Re. (Mathlib `DifferentiableAt.restrictScalars`, `ContDiffOn.restrict_scalars`, `HasFDerivAt.comp`.) Exists.

(E4) *Product rule* for $(x, \lambda) \mapsto \lambda\, v(x)$ with v differentiable, and differentiability of a vector function coordinatewise. Exists.

Track-K results used: [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:ext-schwarz`], [K:`bp:p1:decay`], [K:`bp:p1:summation`], [K:`bp:p1:exp-sums`].

2.2. **Domains and norms.**

Lemma 2.1 (Norm and lattice comparisons)**.** *(a) For $x \in \mathbb{R}^n$ (or $\mathbb{C}^n$), every $n \ge 0$ (all norms are 0 when $n = 0$): $|x|_\infty \le |x|_2 \le \sqrt{n}\,|x|_\infty$, i.e. $|x|_2^2 \le n|x|_\infty^2$; for $n > 0$ this gives $|x|_\infty^2 \ge |x|_2^2/n$. (b) For $k \in \mathbb{Z}^n$: $|k|_2 \le |k|_1 \le \sqrt{n}|k|_2$. Consequently $\{|k|_1 \le K\} \subset \{|k|_2 \le K\}$ and $\{|k|_2 \le K\} \subset \{|k|_1 \le \sqrt{n}K\}$; so a geometric cutoff K_g in the Euclidean norm and an analytic cutoff K_a in $|\cdot|_1$ are compatible when $\sqrt{n}\,K_g \le K_a$. (c) The bilinear bound: if $|Q_{ij}| \le M_0$ for a matrix Q then $|x^\top Q y| \le n M_0 |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$.*
Source: Standard ((E2)); the lattice inclusions correct the direction stated in `PLAN.md` (review N06).

Proof. **Step S2.1: Vectors.** (a),(b) are (E2): $|x|_2^2 = \sum |x_i|^2 \le n \max |x_i|^2$, $|x|_1 \le \sqrt{n}|x|_2$ is Cauchy–Schwarz with $(1, \dots, 1)$.

Step S2.2: Matrices. $|x^\top Q y| \le M_0 |x|_1 |y|_1 \le M_0\, n |x|_2 |y|_2$ by (b). □

Lemma 2.2 (Domains)**.** *Let $R, s > 0$, $n \ge 0$.*

(a) $\mathcal{B}_s(R) = B(0, R + s)$; in particular it is open and $B(0, R) \subset \mathcal{B}_s(R)$ (Remark 1.9); $W_s := \mathbb{R}^n \times \mathcal{B}_s(R)$ is open in $\mathbb{R}^n \times \mathbb{R}^n$ and $D_s(R) \cap (\mathbb{R}^n \times \mathbb{R}^n) = W_s$ (real points of $D_s(R)$: $\theta \in \mathbb{R}^n \subset T_s$ since $s > 0$).
(b) $B(0, R)_s \cap \mathbb{R}^n = \mathcal{B}_s(R)$; $D_s(R)$ and $B(0, R)_s$ are open.
(c) (Euclidean tubes) Let $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$ and $0 < r_0 \le \min(s/2, R/8)$. If $I \in \mathbb{C}^n$ and $J \in P$ with $|I - J|_2 < r_0$, then $I \in B(0, R)_s$ and $|I - J|_\infty < s$. If moreover $I \in \mathbb{R}^n$ then $|I|_\infty < 5R/8 \le 3R/4$.
(d) $T_\rho \subset T_s$ for $\rho \le s$; $T_s = \mathbb{C}^0$ is the universal singleton for $n = 0$ (no $n > 0$ hypothesis is needed anywhere in N-ZERO, N-QC).

Source: Ours (Remark 1.9); BM11 p. 4 for the notation.

Proof. **Step S2.3: Real tube.** ($\subset$) $|I| \le |J| + |I-J| < R+s$. ($\supset$) If $|I| < R+s$ put $J = \frac{R}{R+s} I$; then $|J| < R$ and $|I-J| = \frac{s}{R+s}|I| < s$ (for $I = 0$ take $J = 0$).

Step S2.4: Openness and real points. $B(0,R)_s = \bigcup_{J \in B(0,R)} \{|I-J| < s\}$ is a union of open balls; T_s is open as a preimage of an open set; products of open sets are open; the rest is the definition of ι and $\mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ by Step S2.3.

Step S2.5: Euclidean tubes. By (E2) $|I-J|_\infty \le |I-J|_2 < r_0 \le s/2 < s$, and $|J| < R$ since $J \in P$; so $I \in B(0,R)_s$. For real I: $|I|_\infty < R/2 + R/8$. □

2.3. **Holomorphy implies real smoothness.** Throughout this subsection $R, s > 0$, h satisfies `IsAnalyticIntegrable` $R\, s\, h$, f satisfies `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ ($\varepsilon \ge 0$), $G(\theta, I) = h(I) + f(\theta, I)$ on $D_s(R)$, $H = H_{h,f}$ and $\iota : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}^n \times \mathbb{C}^n$ the componentwise inclusion. Complex partials are $\partial_{\theta_i} G(w) = DG(w)[(e_i, 0)]$, $\partial_{I_i} G(w) = DG(w)[(0, e_i)]$.

Lemma 2.3 (Real smoothness and real Hamiltonian field)**.** *(a) G is holomorphic and C^∞ on $D_s(R)$; h is holomorphic and C^∞ on $B(0,R)_s$.*

(b) H is C^∞ on the open set $W_s = \mathbb{R}^n \times B(0, R+s)$ (Lemma 2.2); on W_s it is real and $H = G \circ \iota$, and for all orders $D^k H(z)[\zeta_1, \dots, \zeta_k] = D^k G(\iota z)[\iota\zeta_1, \dots, \iota\zeta_k] \in \mathbb{R}$. Likewise $h_{\mathbb{R}} = h \circ \iota$ is C^∞ on $\mathcal{B}_s(R)$, with $D^k h_{\mathbb{R}}(I)[a_1, \dots, a_k] = D^k h(I)[\iota a_1, \dots]$.

(c) (Hamiltonian field) For $z = (\theta, I) \in W_s$: H is differentiable at z and $X_H(z) = \big((\partial_{I_i} G(\iota z))_i, (-\partial_{\theta_i} G(\iota z))_i\big) = X_{h+f}(\iota z)$, the complex Hamiltonian field of G at the real point ιz. The frequency map is $\omega_h(I)_i = \partial_i h(I) \in \mathbb{R}$ and $D^2 h_{\mathbb{R}}(I)[e_i, e_j] = \partial_i \partial_j h(I) = \partial_j \partial_i h(I) \in \mathbb{R}$ for $I \in \mathcal{B}_s(R)$.

(d) (Periodicity) $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$; consequently DH, $D^k H$ and X_H are 1-periodic in θ on W_s.

(e) If $\varepsilon = 0$ then $f = 0$ on $D_s(R)$ and $H(\theta, I) = h_{\mathbb{R}}(I)$ on W_s.

Source: BM11 p. 4 (regularity of h on B_s); [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`]*; ours for the packaging. Part 3 uses this as* [N:`bpN:p3:holo-input`].

Proof. **Step S2.6: Holomorphy.** h is complex differentiable on the open set $B(0,R)_s$ and f on $D_s(R)$ (Definitions 1.11(i), 1.12(i); openness Lemma 2.2(b)); the map $(\theta, I) \mapsto h(I)$ is holomorphic on $D_s(R)$ (composition with the linear projection; $B(0,R)_s$ is the second factor). So G is holomorphic on $D_s(R)$, hence C^∞ with holomorphic derivatives of all orders ([K:`bp:p1:holomorphic-regularity`]), which is (a); by (E3) G is C^∞ in the real sense.

Step S2.7: Reality. $\iota(W_s) \subset D_s(R)$ (Lemma 2.2(a)) and $W_s = \iota^{-1}(D_s(R))$ is open. For $z = (\theta, I) \in W_s$, $\operatorname{Im} h(I) = 0$ (Definition 1.11(iv), $I \in \mathcal{B}_s(R)$) and $\operatorname{Im} f(\theta, I) = 0$ (Definition 1.12(iii)); so $G(\iota z) \in \mathbb{R}$ and $H(z) = \operatorname{Re} G(\iota z) = G(\iota z)$. Likewise $h_{\mathbb{R}} = h \circ \iota$ on $\mathcal{B}_s(R)$.

Step S2.8: Derivatives of all orders. On the open set W_s, $H = G \circ \iota$ with ι $\mathbb{R}$-linear and G real-C^∞ (Step S2.6): chain rule (E3) gives $H \in C^\infty(W_s)$ and the displayed formula for $D^k H$, with values real because H is real-valued (the derivative of a real-valued function along real directions is a limit of real difference quotients). The same for $h_{\mathbb{R}}$.

Step S2.9: The field. By Definition 1.3, $X_H(z) = ((DH(z)[(0, e_i)])_i, (-DH(z)[(e_i, 0)])_i)$; by Step S2.8 these are $\partial_{I_i} G(\iota z)$ and $-\partial_{\theta_i} G(\iota z)$. This is [K:`bp:p1:hamiltonian-real`] (with $V = D_s(R)$, H agreeing with $\operatorname{Re} G$ on the real neighbourhood W_s). Taking $k = 1, 2$ for $h_{\mathbb{R}}$ gives $\omega_h(I)_i = Dh_{\mathbb{R}}(I)[e_i] = \partial_i h(I)$ and $D^2 h_{\mathbb{R}}(I)[e_i, e_j] = \partial_i \partial_j h(I)$, symmetric by [K:`bp:p1:holomorphic-regularity`] ([K:`bp:p1:ext-schwarz`]).

Step S2.10: Periodicity. For $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$ and real θ: $(\theta, I), (\theta + k, I) \in D_s(R)$ and $f(\theta + k, I) = f(\theta, I)$ (Definition 1.12(ii)), $h(I)$ does not depend on θ. So $H \circ \tau_k = H$ on W_s, $\tau_k(\theta, I) = (\theta + k, I)$, which maps W_s onto itself with $D\tau_k = \mathrm{id}$; differentiating (chain rule) gives $D^j H(\tau_k z) = D^j H(z)$ and hence $X_H(\tau_k z) = X_H(z)$.

Step S2.11: Zero perturbation. $|f(w)| \le 0$ on $D_s(R)$ gives $f = 0$ there, so $H = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ on W_s. □

Lemma 2.4 (Ordered partials and first-order bounds)**.** *Let $U \subset \mathbb{R}^n$ be open with* `HasDerivativeBounds` $U\, h\, M$ *(Definition 1.13). Then for $I \in U$, $a, b \in \mathbb{R}^n$: $|\omega_h(I)_i| \le M$, $|\omega_h(I)|_\infty \le M$, $|\omega_h(I)|_2 \le \sqrt{n}\, M$, and $|D^2 h_{\mathbb{R}}(I)[e_i, e_j]| \le M$. The ordered partial $D^k h_{\mathbb{R}}(I)[e_{i_1}, \dots, e_{i_k}]$ ($k \le 3$) is invariant under permutations of $(i_1, \dots, i_k)$, so it equals the multi-index partial of the source.*
Source: BM11 p. 4 ($B(M)$); [K:`bp:p1:ext-schwarz`].

Proof. **Step S2.12: First order.** $k = 1$ in the definition gives $|Dh_{\mathbb{R}}(I)[e_i]| = |\omega_h(I)_i| \le M$ (Definition 1.2); then $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ (Lemma 2.1).

Step S2.13: Symmetry. $h_{\mathbb{R}} \in C^3$ on the open set U: $D^2 h_{\mathbb{R}}(I)$ and $D^3 h_{\mathbb{R}}(I)$ are symmetric ([K:`bp:p1:ext-schwarz`] applied to $h_{\mathbb{R}}$, $Dh_{\mathbb{R}}$, $D^2 h_{\mathbb{R}}$), which is the permutation invariance; the entry bound is $k = 2$. □

Lemma 2.5 (Complex Hessian bound in Euclidean form)**.** *Let* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$. *For $I \in B(0,R)_s$ let $Q(I) = (\partial_i \partial_j h(I))_{ij}$ (complex symmetric). Then $|Q(I)_{ij}| \le M_{\mathbb{C}}$ and $|x^\top Q(I) y| \le n M_{\mathbb{C}} |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$; hence the operator norm of $Q(I) : (\mathbb{C}^n, |\cdot|_2) \to (\mathbb{C}^n, |\cdot|_2)$ is $\le n M_{\mathbb{C}}$. For real $I \in \mathcal{B}_s(R)$ this matrix is the real Hessian $D^2 h_{\mathbb{R}}(I)[e_i, e_j]$ (Lemma 2.3(c)), and for real x, y: $|D^2 h_{\mathbb{R}}(I)[x, y]| \le n M_{\mathbb{C}} |x|_2 |y|_2$.*
Source: Pos93 p. 190 (Hessian bound, anchor POS190-HESS); the conversion factor n is ours (`docs/tracks/N/sources.md`).
Source gap: Pos93 p. 190 says that for the symmetric Q the Euclidean operator norm "amounts to the spectral radius norm". For complex symmetric matrices this is false ($Q = \left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ has $Q^2 = 0$). The frozen definition uses the operator norm of $D^2 h$ for the max norm, and so does this blueprint; no spectral-radius claim is used.

Proof. **Step S2.14: Entries.** $D^2 h(I) = D(Dh)(I) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ has norm $\le M_{\mathbb{C}}$ by hypothesis and $|e_i| = 1$ for the max norm, so $|\partial_i \partial_j h(I)| = |D^2 h(I)[e_i, e_j]| \le M_{\mathbb{C}}$ (the order of arguments is immaterial by symmetry, [K:`bp:p1:holomorphic-regularity`]).

Step S2.15: Euclidean bound. Lemma 2.1(c). The operator-norm statement: $|Qx|_2 = \sup_{|y|_2=1} |\sum_i \bar{y}_i (Qx)_i| \le n M_{\mathbb{C}} |x|_2$ by the same estimate applied to $\bar{y}$. □

2.4. **Quasi-convexity: the uniform Euclidean form.**

Lemma 2.6 (Transverse coercivity gives conic coercivity (linear algebra core))**.** *Let $n \ge 1$, $B, q > 0$, $Q : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ bilinear (not necessarily symmetric) with $|Q[x, y]| \le B|x|_2|y|_2$, $\omega \in \mathbb{R}^n \setminus \{0\}$, and*

$$\omega \cdot v = 0 \implies Q[v, v] \ge q|v|_2^2 \qquad (v \in \mathbb{R}^n).$$

If $\ell' \ge 0$ and $\ell' \le |\omega|_2\, q/(8(B+q))$, then for every $\xi \in \mathbb{R}^n$:

$$|\omega \cdot \xi| \le \ell' |\xi|_2 \implies Q[\xi, \xi] \ge \tfrac{5}{8} q |\xi|_2^2 \quad (\ge \tfrac{q}{2}|\xi|_2^2).$$

Source: Ours; the computation is in `docs/tracks/N/sources.md` ("Uniform conversion"); all inequalities re-checked here.

Proof. **Step S2.16: Reduction to unit vectors.** Both hypothesis and conclusion are homogeneous (degrees 1 and 2) in ξ, and the case $\xi = 0$ is trivial; so let $|\xi|_2 = 1$.

Step S2.17: Decomposition. Put $e = \omega/|\omega|_2$, $t = \xi \cdot e$, $u = \xi - te$. Then $u \cdot \omega = 0$, $|u|_2^2 = 1 - t^2$, $|t| \le 1$, and $|t| = |\omega \cdot \xi|/|\omega|_2 \le \ell'/|\omega|_2 \le \kappa := q/(8(B+q)) < \frac{1}{8}$.

Step S2.18: Expansion. $Q[\xi,\xi] = Q[u,u] + tQ[u,e] + tQ[e,u] + t^2 Q[e,e]$. By hypothesis $Q[u,u] \geq q(1-t^2)$; by the bound on Q and $|u|_2, |e|_2 \leq 1$: $|Q[u,e]|, |Q[e,u]|, |Q[e,e]| \leq B$. So $Q[\xi,\xi] \geq q - (q+B)t^2 - 2B|t|$.

Step S2.19: Numerics. $(q+B)t^2 \leq (q+B)\kappa \cdot \kappa = \frac{q}{8}\kappa \leq \frac{q}{8}$ and $2B|t| \leq 2B\kappa = \frac{2Bq}{8(B+q)} \leq \frac{q}{4}$ (as $B \leq B+q$). Hence $Q[\xi,\xi] \geq q - \frac{q}{8} - \frac{q}{4} = \frac{5}{8}q$. □

Lemma 2.7 (Quasi-convexity in Pöschel's quantified form)**.** *Let $n \geq 1$, $R, s, m, M_{\mathbb{C}}, \omega_{\min} > 0$ and put*

$$q = \frac{m}{n}, \qquad B_c = nM_{\mathbb{C}} + q, \qquad \mu = \frac{q}{2}, \qquad \ell = \frac{\omega_{\min} q}{8(B_c + q)}.$$

Let h satisfy `IsAnalyticIntegrable` $R\, s\, h$ *and* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$, `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$, *and $\omega_{\min} \leq |\omega_h(I)|_\infty$ for all $I \in B(0,R)$. Then for every $I \in B(0,R)$, writing $\omega = \omega_h(I)$, $Q = D^2 h_{\mathbb{R}}(I)$:*

(a) $|Q[x,y]| \leq B_c |x|_2 |y|_2$; $\omega \cdot v = 0 \Rightarrow Q[v,v] \geq q|v|_2^2$;

(b) (Pöschel's alternative, conditional*) for every $\xi \in \mathbb{R}^n$: $|\omega \cdot \xi| \leq \ell|\xi|_2 \implies Q[\xi,\xi] \geq \mu|\xi|_2^2$; equivalently, either $|\omega \cdot \xi| > \ell|\xi|_2$ or $Q[\xi,\xi] \geq \mu|\xi|_2^2$ (Pos93 p. 190, "ℓ, m-quasi-convex" with $m := \mu$);*

(c) $|\omega|_2 \geq |\omega|_\infty \geq \omega_{\min}$ and $|\omega|_2 \leq \sqrt{n}|\omega|_\infty$.

The constants q, B_c, μ, ℓ depend only on $(n, m, M_{\mathbb{C}}, \omega_{\min})$. In Pöschel's normalisation ($\tilde{h} = 2\pi h$, Lemma 2.8) the same holds with $(\tilde{\ell}, \tilde{\mu}, \tilde{B}_c, \tilde{\omega}, \tilde{Q}) = 2\pi(\ell, \mu, B_c, \omega, Q)$. Warning. *Full positive definiteness $Q[\xi,\xi] \geq \mu|\xi|_2^2$ for all ξ is false in general and is not claimed: for $h(I) = I_1 + \frac{1}{2}(I_2^2 + I_3^2)$ one has $\omega(I) = (1, I_2, I_3) \neq 0$ (equal to e_1 on $I_2 = I_3 = 0$), $QC(m)$ holds on every bounded set for some $m > 0$ depending on the set (for $\omega \cdot v = 0$: $Q[v,v] = v_2^2 + v_3^2$ and $|v_1| \leq (|I_2| + |I_3|)\max(|v_2|, |v_3|)$, so $m = (1+2K)^{-2}$ works on $|I|_\infty \leq K$), but $Q[e_1, e_1] = 0$ at every point. The estimate (b) is only needed (and only true) in the cone $|\omega \cdot \xi| \leq \ell|\xi|_2$.*

Source: Pos93 p. 190 (anchors POS190-QC, POS190-HESS, POS191-LOW; I read the scan, p. 190, ℓ, m-quasi-convexity, hypothesis $r_0 \leq 4\ell/m$); conversion: `docs/tracks/N/sources.md`*, "Uniform conversion"; BM11 p. 4 for $QC(m)$. Review N05.*

Proof. **Step S2.20: Regularity and Hessian.** Since $B(0,R) \subset \mathcal{B}_s(R)$, $I \in \mathcal{B}_s(R)$; Q is the real Hessian of Lemma 2.5, so $|Q[x,y]| \leq nM_{\mathbb{C}}|x|_2|y|_2 \leq B_c|x|_2|y|_2$, the first half of (a).

Step S2.21: Transverse coercivity in the Euclidean norm. Definition 1.14 at $I \in \mathcal{B}_s(R)$ gives $\omega \cdot v = 0 \Rightarrow Q[v,v] \geq m|v|_\infty^2 \geq \frac{m}{n}|v|_2^2 = q|v|_2^2$ (Lemma 2.1(a)). Also $\omega \neq 0$.

Step S2.22: Frequency size. (c): $|\omega|_\infty \geq \omega_{\min}$ on $B(0,R)$ by hypothesis, $|\omega|_2 \geq |\omega|_\infty$; $|\omega|_2 \leq \sqrt{n}|\omega|_\infty$ by Lemma 2.1.

Step S2.23: Conic coercivity. Apply Lemma 2.6 with $B = B_c$, q and $\ell' = \ell$: $\ell = \omega_{\min} q/(8(B_c + q)) \leq |\omega|_2 q/(8(B_c+q))$ by (c). The conclusion is $Q[\xi,\xi] \geq \frac{5}{8}q|\xi|_2^2 \geq \mu|\xi|_2^2$.

Step S2.24: Scaling. Multiplying h by 2π multiplies ω, Q, B_c and the three of μ, q, ℓ by 2π, and the implications are homogeneous, so they persist. □

2.5. **Normalisations.**

Lemma 2.8 (Period conversion)**.** *Let $\Lambda(\theta, I) = (2\pi\theta, I)$ and, for $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and arbitrary $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, define*

$$\tilde{H}(\varphi, I) = 2\pi H(\varphi/2\pi, I), \qquad \tilde{h} = 2\pi h, \qquad \tilde{f}(\varphi, I) = 2\pi f(\varphi/2\pi, I),$$

so $\tilde{H} = H_{\tilde{h}, \tilde{f}}$ when $H = H_{h,f}$. Clauses (a), (b) are unconditional (no analyticity or periodicity is assumed). Clauses (c), (d) and the periodicity statement (e) are about f satisfying `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12), $R, s > 0$. Then:*

(a) *(Dynamics) For $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$: z satisfies* `IsSolutionOn` $H\,U\,J\,z$ *iff* $\Lambda \circ z$ *satisfies* `IsSolutionOn` $\tilde{H}\,U\,J$ *(same time, same actions). In particular* $|(\Lambda z)_2 - I_0| = |z_2 - I_0|$.

(b) *(Quantities)* $\omega_{\tilde{h}} = 2\pi\omega_h$, $D^2\tilde{h}_{\mathbb{R}} = 2\pi D^2 h_{\mathbb{R}}$, *and so:* `IsStrictlyQuasiConvex` $U\,h\,m \Rightarrow$ `IsStrictlyQuasiConvex` $U\,\tilde{h}\,2\pi m$; $B(M) \Rightarrow$ *coordinate partials of* $\tilde{h}_{\mathbb{R}}$ *of order* $1,2,3$ *bounded by* $2\pi M$; *the complex Hessian bound* $M_{\mathbb{C}} \to 2\pi M_{\mathbb{C}}$; $\omega_{\min} \to 2\pi\omega_{\min}$; *Euclidean sizes of* ω: $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. *Time is unchanged.*

(c) *(Domains; needs holomorphy of f on $D_s(R)$ and the bound $|f| \le \varepsilon$)* $\tilde{f}$ *is holomorphic on* $\{|\mathrm{Im}\,\varphi| < 2\pi s\} \times B(0,R)_s$ *and* $|\tilde{f}| \le 2\pi\varepsilon$ *there; the strip* T_ρ *(ρ in θ) corresponds to the strip of width* $2\pi\rho$ *in* φ.

(d) *(Periodicity; needs $f(\theta + k, I) = f(\theta, I)$ on $D_s(R)$)* $\tilde{f}(\varphi + 2\pi k, I) = \tilde{f}(\varphi, I)$ *for* $k \in \mathbb{Z}^n$ *and* φ *in that strip. (Without this premise the claim fails: $f = \theta_1$ gives $\tilde{f} = \varphi_1$.)*

(e) *(Fourier; needs (c), (e))* $\tilde{f}(\varphi, I) = \sum_k \hat{\tilde{f}}_k(I) e^{ik\cdot\varphi}$ *with* $\hat{\tilde{f}}_k(I) = 2\pi\hat{f}_k(I)$, $\hat{f}_k(I) = \int_{[0,1]^n} f(\theta, I) e^{-2\pi i k\cdot\theta}\, d\theta$, *whenever the Fourier series of $f(\cdot, I)$ converges to it (*[K:`bp:p1:summation`]*); the coefficient radius s' of Pöschel's weight $e^{|k|_1 s'}$ equals $2\pi\rho$.*

Source: `docs/tracks/N/sources.md` *("For the period conversion"); Pos93 p. 189 (set-up: $\theta \in \mathbb{R}^n/2\pi\mathbb{Z}^n$, strip W_{s_0} of width s_0 in the max norm of $\mathrm{Im}\,\theta$).*

Proof. **Step S2.25: Chain rule.** $\tilde{H} = 2\pi\, H \circ \Lambda^{-1}$; Λ is a linear isomorphism, so $\tilde{H}$ is differentiable at Λz iff H is at z, with $\partial_{I_i}\tilde{H}(\Lambda z) = 2\pi\partial_{I_i}H(z)$ and $\partial_{\varphi_i}\tilde{H}(\Lambda z) = \partial_{\theta_i}H(z)$.

Step S2.26: Vector field. Hence $X_{\tilde{H}}(\Lambda z) = (2\pi\partial_I H(z), -\partial_\theta H(z)) = (2\pi\dot{\theta}, \dot{I}) = \frac{d}{dt}\Lambda z$ when $\dot{z} = X_H(z)$, and conversely; the condition $z(t)_2 \in U$ is unchanged. This is (a). (Rescaling the angle without the factor 2π in H would change time.)

Step S2.27: Scalars. (b) is linearity of $h \mapsto \omega_h, D^2 h_{\mathbb{R}}, D^k h_{\mathbb{R}}$ and of the operator norm. (c)–(f) are, under the stated premises, the substitution $\varphi = 2\pi\theta$, $|\mathrm{Im}\,\varphi| = 2\pi|\mathrm{Im}\,\theta|$, and $e^{ik\cdot\varphi} = e^{2\pi i k\cdot\theta}$. □

Lemma 2.9 (From sup norm on a strip to the weighted Fourier norm)**.** *Let $u : T_\rho \to \mathbb{C}$ be holomorphic, bounded and 1-periodic, $\|u\|_\rho = \sup_{T_\rho}|u|$, and $0 < \delta < \rho$. Then*

$$\sum_{k\in\mathbb{Z}^n} |\hat{u}_k|\, e^{2\pi|k|_1(\rho-\delta)} \;\le\; \coth(\pi\delta)^n\, \|u\|_\rho,$$

the left side is Pöschel's weighted norm with weight $e^{|k|_1 s'}$, $s' = 2\pi(\rho - \delta)$, and the factor is $\coth^n\sigma$ with $\sigma = \pi\delta$ (Pos93 p. 190: $|u|_{D,r,s} \le \coth^n\sigma\, |u|_{D,r,s+2\sigma}$, where the right side is the sup norm on the strip of width $s + 2\sigma$ in φ, i.e. $\rho = (s' + 2\sigma)/2\pi$). Conversely $\|u\|_\rho \le \sum_k |\hat{u}_k| e^{2\pi|k|_1\rho}$. For I-dependent $u(\theta, I)$ the statement holds for each fixed I with the same constant, so the sup over I in the weighted norm is $\le \coth(\pi\delta)^n \sup_{\theta,I}|u|$.

Source: Pos93 p. 190 (the displayed relation between the weighted and sup norms); frozen-convention statement: [K:`bp:p1:summation`], [K:`bp:p1:decay`], [K:`bp:p1:exp-sums`]*(b).*

Proof. **Step S2.28: Decay.** For $\rho' \in (\rho - \delta, \rho)$, [K:`bp:p1:decay`] gives $|\hat{u}_k| \le e^{-2\pi|k|_1\rho'}\|u\|_\rho$; hence $|\hat{u}_k| e^{2\pi|k|_1(\rho-\delta)} \le e^{-2\pi|k|_1(\rho'-\rho+\delta)}\|u\|_\rho$, and letting $\rho' \uparrow \rho$ (termwise, then summing, or directly [K:`bp:p1:summation`], converse part) the bound is $e^{-2\pi\delta|k|_1}\|u\|_\rho$ per k.

Step S2.29: Sum. $\sum_{k\in\mathbb{Z}^n} e^{-2\pi\delta|k|_1} = \coth(\pi\delta)^n$ ([K:`bp:p1:exp-sums`](b) with $x = 2\pi\delta$). The case with parameters is pointwise in I. □

2.6. **The dictionary of constants.**

Lemma 2.10 (Constants dictionary for N-HEAD-C)**.** *Let $n \geq 3$ (only $n \geq 2$ is used by Pöschel) and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$, $a = \frac{1}{2n}$. Define successively*

$$q = \tfrac{m}{n},\ B_c = nM_{\mathbb{C}} + q,\ \mu = \tfrac{q}{2},\ \ell = \tfrac{\omega_{\min} q}{8(B_c+q)}, \qquad r_0 = \min\Big(\tfrac{s}{2}, \tfrac{R}{8}, \tfrac{2\ell}{\mu}, \tfrac{4\omega_{\min}}{\mu}\Big), \qquad A = \tfrac{11B_c}{\mu},$$

$$s_0 = \pi s, \quad \sigma = \tfrac{\pi s}{4}, \quad F = 2\pi \coth^n\big(\tfrac{\pi s}{4}\big), \quad \varepsilon_* = \frac{2\pi\mu\, r_0^2}{2^{10} A^{2n}},$$

$$C = \tfrac{r_0}{A}\Big(\tfrac{F}{\varepsilon_*}\Big)^a, \quad c = \tfrac{s_0}{6}\Big(\tfrac{\varepsilon_*}{F}\Big)^a, \quad T = \frac{A^2 s_0}{4\pi n M},$$

$$\varepsilon_0 = \min\Big(\tfrac{\varepsilon_*}{F},\ \big(\tfrac{R}{4C}\big)^{2n}\Big).$$

Pöschel's data (his normalisation, for $\tilde{h} = 2\pi h$): $m_P = 2\pi\mu$, $\ell_P = 2\pi\ell$, $M_P = 2\pi B_c$, $A_P = 11M_P/m_P = A$, $\varepsilon_{0,P} = m_P r_0^2/(2^{10}A_P^{2n}) = \varepsilon_*$, $R_0 = r_0/A$, and the set $P = \{J \in \mathbb{R}^n : |J| \leq R/2\}$. Then:

- *(a) All constants are positive and finite and depend only on $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ — not on $h, f, \varepsilon, \theta_0, I_0$, nor on the bound A of Definition 1.11(iii). Order of choice: $q, B_c, \mu, \ell, r_0, A$; then s_0, σ, F; ε_*; C, c; T; ε_0. And $C\varepsilon_0^a \leq R/4$, $0 < \varepsilon_0 \leq \varepsilon_*/F$.*
- *(b) $r_0 \leq 4\ell_P/m_P$ $(= 4\ell/\mu)$; $A \geq 22$; $R_0 = r_0/A \leq r_0/22 \leq R/176$.*
- *(c) (Tubes) With $V_{r_0}P = \{I \in \mathbb{C}^n : \exists J \in P,\ |I - J|_2 < r_0\}$ and $U_{r_0}P = V_{r_0}P \cap \mathbb{R}^n$: $V_{r_0}P \subset B(0,R)_s$ and $U_{r_0}P \subset B(0, 5R/8) \subset B(0,R)$.*
- *(d) (Hypotheses of Theorem 1) If h satisfies (H1)–(H5) of Theorem 1.19, then for $I \in U_{r_0}P$: $\tilde{h}$ is ℓ_P, μ_P-quasi-convex (Lemma 2.7) with $\mu_P = 2\pi\mu$; the complex Hessian operator norm (Euclidean) of $\tilde{h}$ on $V_{r_0}P$ is $\leq M_P$ (Lemma 2.5); Pöschel's m, M are m_P, M_P and his hypothesis $r_0 \leq 4\ell_P/m_P$ holds.*
- *(e) (Perturbation) If f satisfies (H6) with ε, then Pöschel's norm of $\tilde{f}$ on $V_{r_0}P \times$(strip of width s_0) is $\leq F\varepsilon$; and $F\varepsilon \leq \varepsilon_{0,P}$ if $\varepsilon \leq \varepsilon_0$.*
- *(f) (Exceptional branch is empty) For $I_0 \in B(0, R/2)$: $|\tilde{\omega}(I_0)|_2 \geq 2\pi\omega_{\min} > \pi\omega_{\min} \geq m_P r_0/8$.*
- *(g) (Time prefactor) For $I_0 \in B(0, R/2)$, $0 < 2\pi\omega_{\min} \leq \Omega_0 = \sup_{|I-I_0|_2 \leq R_0} |\tilde{\omega}(I)|_2 \leq 2\pi\sqrt{n}M \leq 2\pi n M$, so Pöschel's $T_0 = A_P^2 s_0/\Omega_0 \geq 2T$.*
- *(h) (Conclusion shape, for $\varepsilon > 0$; $\varepsilon = 0$ is N-ZERO) Pöschel's $R_0(\varepsilon_P/\varepsilon_{0,P})^a$ and $T_0 \exp\big(\frac{s_0}{6}(\varepsilon_{0,P}/\varepsilon_P)^a\big)$ with $\varepsilon_P = F\varepsilon$ are $C\varepsilon^a$ and $T_0 \exp(c\varepsilon^{-a})$, and $T\exp(c\varepsilon^{-a}) \leq \frac{1}{2}T_0 \exp(c\varepsilon^{-a})$.*

Source: `docs/tracks/N/sources.md` "Uniform conversion" (all constants), checked against Pos93 p. 190 (hypotheses, $A = 11M/m$, $\varepsilon_0 = mr_0^2/(2^{10}A^{2n})$, $r_0 \leq 4\ell/m$), p. 191 (parameters a, R_0, T_0, exceptional branch) and p. 204 (assembly: $K = (\varepsilon_0/\varepsilon)^a$). Pos93 requires $\|f_\varepsilon\| \leq \varepsilon$ (p. 190) with $\varepsilon \leq \varepsilon_{0,P}$.

Source gap: Pos93 states no theorem about the $\omega_{\min}$-conversion and the frozen-to-source dictionary: it exists only in `sources.md` and here (Part 0 `bpN:p0:head`). The choice of T with the factor $1/2$ is slack, not needed by the source.

Proof. **Step S2.30: Positivity and dependence.** Each constant is a positive finite expression in earlier ones (denominators: $n, \mu, 8(B_c+q), A, F, \varepsilon_*, 4C, 4\pi nM$, all > 0); $\varepsilon_0 > 0$ as a minimum of positive numbers, and $C\varepsilon_0^a \leq C \cdot \frac{R}{4C} = R/4$ since $((R/4C)^{2n})^{1/(2n)} = R/(4C)$. The list of inputs in each formula contains only $n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$. This is (a).

Step S2.31: Size relations. $B_c = nM_{\mathbb{C}} + q \geq q = 2\mu$ so $A = 11B_c/\mu \geq 22$. $4\ell_P/m_P = 4\ell/\mu \geq 2\ell/\mu \geq r_0$, and $R_0 = r_0/A \leq r_0/22 \leq R/176$. This is (b).

Step S2.32: Tubes. $r_0 \leq \min(s/2, R/8)$: Lemma 2.2(c) gives (c).

Step S2.33: Quasi-convexity and Hessian. For (d): $U_{r_0}P \subset B(0,R)$ by (c), so Lemma 2.7(b) applies at every point of $U_{r_0}P$ and, after Lemma 2.8(b), gives the (ℓ_P, μ_P)-alternative for $\tilde{h}$. For

$I \in V_{r_0}P \subset B(0,R)_s$ the Euclidean operator norm of $\tilde{Q}(I)$ is $\le 2\pi n M_{\mathbb{C}} \le 2\pi B_c = M_P$ (Lemma 2.5). Pöschel's $A = 11M/m$ with $M = M_P$ and $m = \mu_P = m_P$ is $11B_c/\mu = A$.

Step S2.34: Perturbation. By (H6) $|f| \le \varepsilon$ on $D_s(R) \supset T_{3s/4} \times V_{r_0}P$ ($V_{r_0}P \subset B(0,R)_s$, (c)). For each fixed $I \in V_{r_0}P$ apply Lemma 2.9 to $u = f(\cdot, I)$ with $\rho = 3s/4$, $\delta = s/4$: $\sum_k |\hat{f}_k(I)| e^{2\pi|k|_1 s/2} \le \coth(\pi s/4)^n \varepsilon$. Multiplying by 2π (Lemma 2.8(f); f satisfies (H6), so (c), (e) apply) and noting $2\pi \cdot s/2 = \pi s = s_0$ gives $\le F\varepsilon$. (In Pöschel's terms $s_0 + 2\sigma = 3\pi s/2 = 2\pi \cdot \frac{3s}{4}$.) $\varepsilon \le \varepsilon_*/F$ gives $F\varepsilon \le \varepsilon_* = \varepsilon_{0,P}$. This is (e).

Step S2.35: Exceptional branch. $I_0 \in B(0, R/2) \subset B(0,R)$ so $|\omega(I_0)|_2 \ge \omega_{\min}$ (Lemma 2.7(c)) and $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. And $m_P r_0/8 = 2\pi\mu r_0/8 \le 2\pi\mu \cdot \frac{4\omega_{\min}}{8\mu} = \pi\omega_{\min}$ by $r_0 \le 4\omega_{\min}/\mu$. This is (f).

Step S2.36: Time. The Euclidean ball $\{|I - I_0|_2 \le R_0\}$ with $R_0 \le r_0$ lies in $\{|I|_\infty < R/2 + R_0\} \subset B(0, 5R/8)$ ($|I - I_0|_\infty \le |I - I_0|_2$), inside $\mathcal{B}_s(R)$ where Lemma 2.4 gives $|\omega(I)|_2 \le \sqrt{n}M$. So $\Omega_0 \le 2\pi\sqrt{n}M \le 2\pi n M$ (finite); also I_0 belongs to the closed ball, $I_0 \in B(0,R)$, so $\Omega_0 \ge |\tilde{\omega}(I_0)|_2 \ge 2\pi\omega_{\min} > 0$ (Lemma 2.7(c)). Inverting the positive quantities, $T_0 = A^2 s_0/\Omega_0 \ge A^2 s_0/(2\pi n M) = 2T$. (g).

Step S2.37: Exponent and constants. $R_0(F\varepsilon/\varepsilon_*)^a = \frac{r_0}{A}(F/\varepsilon_*)^a \varepsilon^a = C\varepsilon^a$ and $\frac{s_0}{6}(\varepsilon_*/(F\varepsilon))^a = c\varepsilon^{-a}$. (h) follows from $T \le T_0/2$. □

Remark 2.11 (Forward use and an unused constant). Part 10 chooses C, T, c, ε_0 as in Lemma 2.10, verifies Pos93's hypotheses by (d),(e) and excludes the exceptional branch by (f). The time prefactor T enters only through (g); the remaining slack $\varepsilon_0 \le (R/4C)^{2n}$ gives $C\varepsilon^a \le R/4$ so that the a-priori confinement keeps $I(t) \in B(0, R/2 + R/4) = B(0, 3R/4)$ (continuation, [N:`bpN:p3:continuation`]). The value $M_{\mathbb{C}}$ enters only through B_c; M only through T.

3. N-QC and N-ZERO

This part proves two frozen theorems completely: N-QC (Theorem 1.21, the nonsingular isoenergetic derivative) and N-ZERO (Theorem 1.20, the exact integrable dynamics). Neither uses the normal-form machinery. The linear-algebra core of N-QC is stated in a reusable form (Lemma 3.1, cited by track I as the isoenergetic gate, [N:`bpN:p2:qc-kernel`]). Frozen conventions throughout (max norms, period 1); Euclidean structure appears only as the dot product $\omega \cdot v = \sum \omega_i v_i$.

3.1. N-QC.

Lemma 3.1 (Kernel core: bordered Hessian is nonsingular). *Let $n \ge 1$, $\omega \in \mathbb{R}^n$, A a real $n \times n$ matrix (not necessarily symmetric, no norm bound), $\lambda \ne 0$, and suppose*

$$\omega \cdot v = 0,\ v \ne 0 \implies v^\top A v \ne 0 \qquad \text{and} \qquad \omega \ne 0.$$

Let $L : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$, $L(v,u) = (\omega \cdot v,\ u\,\omega + \lambda A v)$. Then $\ker L = 0$, hence L is bijective. In particular this holds if $\lambda > 0$, $\omega \ne 0$ and $v^\top A v \ge m|v|^2$ for some norm and $m > 0$ whenever $\omega \cdot v = 0$. (For $n = 0$ no $\omega \ne 0$ exists; N-QC is then vacuous, Remark 3.2.)
Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (kernel computation); linear algebra (E1).

Proof. **Step S3.1: Kernel equations.** Let $L(v,u) = 0$: $\omega \cdot v = 0$ and $u\omega + \lambda A v = 0$.

Step S3.2: Pairing with v. Dot the second equation with v: $u(\omega \cdot v) + \lambda v^\top A v = 0$, so $\lambda v^\top A v = 0$, i.e. $v^\top A v = 0$ as $\lambda \ne 0$. Since also $\omega \cdot v = 0$, the hypothesis forces $v = 0$.

Step S3.3: The scalar. With $v = 0$: $u\omega = 0$; as $\omega \ne 0$, $u = 0$.

Step S3.4: Bijectivity. L is a linear map between spaces of dimension $n+1$ with trivial kernel, hence bijective by (E1) (`LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, with $\dim(\mathbb{R}^n \times \mathbb{R}) = \dim(\mathbb{R} \times \mathbb{R}^n) = n + 1$). □

Remark 3.2 ($n = 0$). For $n = 0$ the space $\mathbb{R}^0$ is a point, so $\omega = 0$ and Definition 1.14 requires $\omega_h(I) \neq 0$: no I satisfies it. Hence for $n = 0$ the hypotheses of N-QC are never met with $I \in U$ and the theorem is vacuous; no restriction $n \geq 1$ is added to it (review N03).

Lemma 3.3 (The derivative of Ψ)**.** *Let $U \subset \mathbb{R}^n$ be open, h with $h_{\mathbb{R}} \in C^2(U)$, $I \in U$, $\lambda \in \mathbb{R}$, $\Psi(I', \lambda') = (h_{\mathbb{R}}(I'), \lambda' \omega_h(I'))$. Then Ψ is differentiable at (I, λ) (so* `fderiv` *is the genuine derivative; Remark 1.18(e)) and*

$$D\Psi(I, \lambda)[v, u] = \big(\omega \cdot v,\ u\,\omega + \lambda Q v\big), \qquad \omega = \omega_h(I),\ Q_{ij} = D^2 h_{\mathbb{R}}(I)[e_i, e_j] = Q_{ji}.$$

Source: BM11 p. 10; (E4), [K:`bp:p1:ext-schwarz`].

Proof. **Step S3.5: First component.** $(I', \lambda') \mapsto h_{\mathbb{R}}(I')$ is differentiable at (I, λ) ($h_{\mathbb{R}}$ is C^2 on the open set $U \ni I$) with derivative $[v, u] \mapsto Dh_{\mathbb{R}}(I)[v] = \sum_i v_i Dh_{\mathbb{R}}(I)[e_i] = \omega \cdot v$.

Step S3.6: The frequency map is C^1. $\omega_h(I')_i = Dh_{\mathbb{R}}(I')[e_i]$ is the composition of $I' \mapsto Dh_{\mathbb{R}}(I') \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ (which is C^1 on U since $h_{\mathbb{R}} \in C^2(U)$ and U is open) with the continuous linear evaluation $\ell \mapsto \ell(e_i)$. So ω_h is differentiable on U with $D\omega_h(I)[v]_i = D(Dh_{\mathbb{R}})(I)[v][e_i] = D^2 h_{\mathbb{R}}(I)[v, e_i]$.

Step S3.7: Symmetry. $D^2 h_{\mathbb{R}}(I)$ is symmetric ([K:`bp:p1:ext-schwarz`]), so $D\omega_h(I)[v] = Qv$ with Q the symmetric Hessian matrix.

Step S3.8: Product. By the product rule (E4), $(I', \lambda') \mapsto \lambda' \omega_h(I')$ is differentiable at (I, λ) with derivative $[v, u] \mapsto u\,\omega + \lambda Q v$. Both components are differentiable, so Ψ is, with the stated derivative. □

Theorem 3.4 (N-QC)**.** *Theorem 1.21 holds: for* `IsStrictlyQuasiConvex` $U\,h\,m$*, $I \in U$, $\lambda > 0$, the map $D\Psi(I, \lambda)$ is bijective.*
Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`
Source: BM11 Lemma 3.3 and proof, pp. 9–10.

Proof. **Step S3.9: Unpack the hypothesis.** Definition 1.14: $m > 0$, U open, $h_{\mathbb{R}} \in C^2(U)$, $\omega := \omega_h(I) \neq 0$, and $\omega \cdot v = 0 \Rightarrow m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v, v]$ (max norm on v). Note $n \geq 1$ because $\omega \neq 0$ (Remark 3.2).

Step S3.10: The derivative. By Lemma 3.3, $D\Psi(I, \lambda) = L$ of Lemma 3.1 with $A = Q$ the Hessian matrix, $v^\top Q v = D^2 h_{\mathbb{R}}(I)[v, v]$ (bilinearity).

Step S3.11: Positivity on $\omega^\perp$. For $\omega \cdot v = 0$, $v \neq 0$: $v^\top Q v \geq m|v|_\infty^2 > 0$, since $m > 0$ and $|v|_\infty > 0$. So the hypothesis of Lemma 3.1 holds (with $\lambda > 0$).

Step S3.12: Conclusion. Lemma 3.1 gives bijectivity of $L = D\Psi(I, \lambda) =$ `fderiv` $\mathbb{R}\,\Psi\,(I, \lambda)$. □

Remark 3.5 (Reusable form for track I). Lemma 3.1 needs no norm, no symmetry of A and no uniform constant: any quasi-convexity-type hypothesis ("$v^\top A v > 0$ on $\omega^\perp \setminus 0$") together with $\omega \neq 0$ and $\lambda \neq 0$ gives nonsingularity of the bordered matrix $\begin{pmatrix} \omega^\top & 0 \\ \lambda A & \omega \end{pmatrix}$. A quantitative inverse bound is not asserted by N-QC and is not provided here; it would follow, if needed, from the compactness of $\{v : |v|_2 = 1,\ \omega \cdot v = 0\}$ for a fixed (I, λ) but is not uniform in (I, λ).

3.2. **N-ZERO.**

Lemma 3.6 (The field at the integrable orbit)**.** *Let $R, s > 0$,* `IsAnalyticIntegrable` $R\,s\,h$*,* `IsAnalyticPerturbation` $R\,s\,0\,f$*, $H = H_{h,f}$, and $W_s = \mathbb{R}^n \times B(0, R + s)$. Then $H(\theta, I) = h_{\mathbb{R}}(I)$ on W_s; H is differentiable at every $z = (\theta, I) \in W_s$ with $DH(z)[(a, b)] = Dh_{\mathbb{R}}(I)[b]$ and*

$$X_H(\theta, I) = (\omega_h(I), 0).$$

Source: BM11 p. 1 ($H = h(I)$ when $f = 0$); ours.

Proof. **Step S3.13: f vanishes.** Definition 1.12(iv) with $\varepsilon = 0$: $|f| \le 0$ on $D_s(R)$, so $f = 0$ there. For $z = (\theta, I) \in W_s$, $(\theta, I) \in D_s(R)$ (Lemma 2.2(a)), so $H(z) = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ (Lemma 2.3(e)).

Step S3.14: Differentiability. W_s is open and on it $H = h_{\mathbb{R}} \circ \mathrm{pr}_2$ with $h_{\mathbb{R}} \in C^3(\mathcal{B}_s(R))$ (Definition 1.11(v)), $\mathcal{B}_s(R) = B(0, R + s)$ open. So H is differentiable at each point of W_s with $DH(z)[(a, b)] = Dh_{\mathbb{R}}(I)[b]$ (chain rule with the continuous linear projection). The point-wise hypotheses of the totality remark 1.18 are met.

Step S3.15: Field. By Definition 1.3, $\partial_{I_i} H = Dh_{\mathbb{R}}(I)[e_i] = \omega_h(I)_i$ (Definition 1.2) and $\partial_{\theta_i} H = DH(z)[(e_i, 0)] = 0$. □

Theorem 3.7 (N-ZERO)**.** *Theorem 1.20 holds.*
Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`
Source: Ours (supplement to BM11 Theorem 2.1, p. 5).

Proof. **Step S3.16: Initial value and action.** $\gamma(t) = (\theta_0 + t\omega_h(I_0), I_0)$ (Definition 1.17) gives $\gamma(0) = (\theta_0, I_0)$ and $\gamma(t)_2 = I_0$ for all t: the first and third conclusions.

Step S3.17: The action stays in $B(0, R)$. $I_0 \in B(0, R/2) \subset B(0, R)$ since $R > 0$ (max norm: $|I_0| < R/2 < R$).

Step S3.18: Differentiability of H along the orbit. $\gamma(t) \in \mathbb{R}^n \times B(0, R) \subset W_s$ (Lemma 2.2(a): $B(0, R) \subset B(0, R + s)$). By Lemma 3.6 H is differentiable at $\gamma(t)$ for each t.

Step S3.19: Derivative of the orbit. γ is affine in t, so differentiable at every $t \in \mathbb{R}$ (two-sided) with $\gamma'(t) = (\omega_h(I_0), 0)$.

Step S3.20: The equation. Lemma 3.6 gives $X_H(\gamma(t)) = (\omega_h(I_0), 0) = \gamma'(t)$ for every $t \in \mathbb{R}$. With Steps S3.17–S3.19, every clause of Definition 1.16 holds with $U = B(0, R)$ and $J = \mathbb{R}$. □

Remark 3.8 (Scope). N-ZERO uses neither the holomorphy of Part 1 (beyond the stated clauses of the definitions) nor the ODE layer of Part 3; Part 3 supplies uniqueness if ever required for $\varepsilon = 0$ (γ is the unique solution by [N:`bpN:p3:gen-uniqueness`]), which is not asserted by the frozen statement. The case $\varepsilon = 0$ is excluded from N-HEAD-C by (H7).

4. Hamiltonian ODE infrastructure (shared by tracks I, N, P)

This part is the single ODE layer of the three blueprints I, N, P. It has two levels. The *general layer* (§4.2–§4.6) is about an autonomous vector field F of class C^1 on an open subset W of a finite-dimensional real normed space E: local existence, uniqueness on *every* interval, maximal solutions, continuation (compact sets, and closed sets with uniform bounds), first exit from an open set. The *Hamiltonian layer* (§4.7) adds $F = X_H$: smoothness of X_H, energy conservation, translation in the angles. §4.8 shows that the frozen solution predicates of the three tracks (`IsSolutionOn`, `IsHamiltonianSolutionOn`, `IsRestrictedSolutionOn`, `IsNewtonSolution`, `IsFixedSolution`) are instances of the general predicate, and §4.9 gives the six track-N results of `PLAN.md`. Tracks I and P cite the general results as [N:`bpN:p3:gen-...`].

4.1. Conventions and external facts.

- E is a finite-dimensional real normed space (hence complete and proper: closed bounded sets are compact), $F : E \to E$ a *total* function, $W \subset E$ open. In the applications $E = \mathbb{R}^n \times \mathbb{R}^n$ with the max norm $|(\theta, I)| = \max(|\theta|, |I|)$ (the product norm of the Lean text, equal to the max norm on the $2n$ coordinates), or $E = (\mathbb{R}^2)^N \times (\mathbb{R}^2)^N$ for the Newtonian problems.
- F is C^1 *on* W means `ContDiffOn` $\mathbb{R}$ 1 F W; as W is open this gives $F \in C^1$ at each point of W, and no information off W. Off W the values of F never enter (Lean totality, below).
- An *interval* is an order-connected subset of $\mathbb{R}$ (Lean `OrdConnected`); open, closed, half-open, bounded or not, possibly a single point. $\overline{B}(x, \delta)$ is the closed ball.

- All derivatives $\dot{z}(t)$ in solution predicates are two-sided (Lean `HasDerivAt`) at every $t \in J$, including endpoints of J, exactly as in the frozen predicates ("z is defined on $\mathbb{R}$").

External facts. Mathlib-level statements, with the declarations that exist (checked in `Mathlib/Analysis/ODE/`); what is *not* in Mathlib is listed after them.

(E1) *Picard–Lindelöf, quantitative form.* Let $F : E \to E$, $x \in E$, $a, M, L \geq 0$ with $\|F(y)\| \leq M$ and F L-Lipschitz on $\overline{B}(x, a)$, $T \geq 0$ with $MT \leq a$. Then there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and α has derivative $F(\alpha(t))$ within $[t_0 - T, t_0 + T]$ at each t of that interval. (Mathlib: the root-namespace structure `IsPicardLindelof` $f\, t_0\, x_0\, a\, r\, L\, K$ of `Analysis/ODE/PicardLindelof.lean`, with $r = 0$ and a time-independent field, and `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt`$_0$ (Unicode subscript zero) of `Analysis/ODE/ExistUnique.lean`; Mathlib's norm bound is the parameter L (our M) and its Lipschitz parameter is K (our L); the time constraint is the field `mul_max_le`: $M \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$. Parameters are `NNReal`. The contraction on iterates means no smallness of LT is needed.) Exists.

(E2) *Uniqueness with a Lipschitz constant on a set.* If $S \subset E$ and F is L-Lipschitz on S, $f, g : [a, b] \to E$ are continuous, have right derivative $F(f(t))$, $F(g(t))$ on $[a, b)$, take values in S and $f(a) = g(a)$, then $f = g$ on $[a, b]$; and the mirror statement with $f(b) = g(b)$ and left derivatives. (Mathlib: `ODE_solution_unique_of_mem_Icc_right`, `ODE_solution_unique_of_mem_Icc_left`.) Exists.

(E3) C^1 *implies locally Lipschitz.* If F is C^1 at p there are L and a neighbourhood on which F is L-Lipschitz (Mathlib: `ContDiffAt.exists_lipschitzOnWith`); and a function locally Lipschitz on a compact set is Lipschitz there (Mathlib: `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact`, `Topology/Algebra/MetricSpace/Lipschitz.lean`), which packages Lemma 4.4 once local Lipschitzness on K is extracted from C^1 on W. Exists.

(E4) *Mean value inequality*: [K:`bp:p1:ext-mvt`]; for a differentiable map on a convex set with $\|DF\| \leq L$, F is L-Lipschitz there; for a function of one variable with derivative bounded by M on an interval, $\|\alpha(u) - \alpha(s)\| \leq M|u - s|$. Exists.

(E5) *Topology.* Heine–Borel in E (`ProperSpace`); continuous images of compact sets are compact; sequential compactness; for a *nonempty* closed set A, $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive exactly on $E \setminus A$ (`Metric.infDist`, `IsClosed.notMem_iff_infDist_pos`, which requires `A.Nonempty`; for $A = \emptyset$ Mathlib has $\operatorname{infDist}(x, \emptyset) = 0$, `Metric.infDist_empty`, so no statement below uses infDist of an empty set: thickenings are defined by $K_\delta = \{y : \exists x \in K,\ \|y - x\| \leq \delta\}$, which is empty for $K = \emptyset$). Exists.

(E6) *Calculus.* Chain rule `HasFDerivAt.comp_hasDerivAt`; a function continuous on $[s, t]$ with right derivative 0 on $[s, t)$ is constant (`constant_of_has_deriv_right_zero`); two one-sided derivatives agree $\Rightarrow$ two-sided one (`HasDerivWithinAt.union` with $(-\infty, b] \cup [b, \infty) = \mathbb{R}$ and `hasDerivWithinAt_univ`); `ContDiffOn.fderiv_of_isOpen`: if H is C^{k+1} on an open set then DH is C^k there. Exists.

(E7) *Uniform existence time near a point.* If F is C^1 at x_0 and $t_0 \in \mathbb{R}$ there are $r, \varepsilon > 0$ such that for every $x \in \overline{B}(x_0, r)$ there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ (Mathlib: `ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt`, `ExistUnique.lean:145`; note the solution is for the total F and is not asserted to stay in the neighbourhood of x_0). Exists. It covers Theorem 4.10 for a single point after the staying-in-W step; the explicit $\delta/\max(M, 1)$ time of Proposition 4.9, uniform over arbitrary sets Φ with uniform data, is derived from (E1).

Not in Mathlib (checked in `Analysis/ODE` and `Geometry/Manifold/IntegralCurve`: no maximal-solution, continuation or first-exit statement, and no uniqueness on arbitrary intervals

with only local Lipschitz constants): the following must be proved from the blueprint below: uniform data on arbitrary sets and its compact and periodic instances (Lemmas 4.6, 4.7), Lemma 4.8, Proposition 4.9 (derived from (E1); the existing statement (E7) may be reused for the single-point case), Theorems 4.11, 4.13, 4.14, Lemma 4.19 and the Hamiltonian layer.

Remark 4.1 (Lean totality). `KAM.hamiltonianVectorField` H uses `fderiv`, which is 0 at a point of non-differentiability, and the Newtonian fields use $1/\ell^3$ with $\ell =$ `planeLength` or `radius`. In every use below these are guarded: X_H occurs only on an open set where H is C^2 (or inside the frozen predicates, which assert differentiability at every visited point), and $\ell > 0$ at every visited point because the frozen predicates require it. The general layer never uses F off W.

4.2. **The general solution predicate; Lipschitz data.**

Definition 4.2 (Solutions in W)**.** For $F : E \to E$, $W \subset E$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to E$: `IsSolIn` $F\,W\,J\,z$ holds if for every $t \in J$, $z(t) \in W$ and z has derivative $F(z(t))$ at t (two-sided). Suggested Lean name `KAM.ODE.IsSolutionIn`; it is *not* a frozen declaration.
Source: Standard; shape dictated by the frozen predicates of Definition 1.16 and §4.8.

Elementary consequences (each a one-line Lean lemma, (E6)): (i) restriction: `IsSolIn` $F\,W\,J\,z$ and $J' \subset J$ give `IsSolIn` $F\,W\,J'\,z$; (ii) z is continuous on J (differentiable at each point of J); (iii) *local nature*: if J' is a neighbourhood of each of its points and $z = y$ on it, then `IsSolIn` $F\,W\,J'\,y \Leftrightarrow$ `IsSolIn` $F\,W\,J'\,z$ for open J'.

Lemma 4.3 (Symmetries)**.** *Let* `IsSolIn` $F\,W\,J\,z$.

(a) (time shift) For $c \in \mathbb{R}$, $t \mapsto z(t + c)$ is a solution on $J - c$.
(b) (time reversal) $t \mapsto z(-t)$ is a solution of $-F$ on $-J$, and $-F$ is C^1 on W if F is.
(c) (space translation) If $v \in E$ satisfies $W + v = W$ and $F(x + v) = F(x)$ for $x \in W$, then $t \mapsto z(t) + v$ is a solution of F in W on J.

Source: Standard.

Proof. **Step S4.1: Chain rule.** (a) and (b) are the chain rule for $t \mapsto t + c$, $t \mapsto -t$ ((E6)); (c): $z + v$ has the same derivative $F(z(t)) = F(z(t) + v)$, and $z(t) + v \in W + v = W$. □

Lemma 4.4 (Bounds and Lipschitz constants on compact sets)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then F is bounded on K and there is $L \geq 0$ with $\|F(x) - F(y)\| \leq L\|x - y\|$ for $x, y \in K$.*
Source: Standard.

Proof. **Step S4.2: Via Mathlib.** Alternatively: F is locally Lipschitz at every point of W ((E3)), hence locally Lipschitz on K, and `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact` ((E3)) gives L. The remaining steps are the explanatory proof of that fact.

Step S4.3: Bounded. F is continuous on $W \supset K$, so $\|F\| \leq M$ on the compact K ((E5)).

Step S4.4: Contradiction hypothesis. If no L works, for every $k \in \mathbb{N}$, $k \geq 1$, there are $x_k \neq y_k$ in K with $\|F(x_k) - F(y_k)\| > k\|x_k - y_k\|$; then $\|x_k - y_k\| < 2M/k \to 0$.

Step S4.5: Compactness. A subsequence $x_{k_j} \to p \in K$, hence also $y_{k_j} \to p$. By (E3) there are L_p and a neighbourhood N_p of p on which F is L_p-Lipschitz. For large j both points lie in N_p and $k_j > L_p$, which contradicts Step S4.4. □

Definition 4.5 (Uniform data)**.** Let F be C^1 on the open set W. A set $\Phi \subset E$ has *uniform data* (δ, M, L), with $\delta > 0$, $M, L \geq 0$, if for every $x \in \Phi$: $\overline{B}(x, \delta) \subset W$, $\|F\| \leq M$ on $\overline{B}(x, \delta)$ and F is L-Lipschitz on $\overline{B}(x, \delta)$. (Balls are convex, so by (E4) a bound $\|DF\| \leq L$ on $\overline{B}(x, \delta) \subset W$ gives the Lipschitz clause.)

Lemma 4.6 (Compact sets have uniform data)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then K has uniform data (δ, M, L) for some $\delta > 0$, and $K_\delta = \{y : \exists x \in K,\ \|y - x\| \le \delta\} \subset W$ is compact.*

Proof. **Step S4.6: Choice of δ.** If $K = \emptyset$ put $\delta = 1$: then $K_\delta = \emptyset$ and all assertions are vacuous (no infDist of the empty set is used). If $W = E$ put $\delta = 1$. Otherwise $K \neq \emptyset$ and $A = E \setminus W$ is closed and nonempty, so $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive on K ((E5)); it has a minimum $2\delta > 0$ on the compact nonempty K, and then $K_\delta \subset W$. K_δ is the image of the compact $K \times \overline{B}(0, \delta)$ under $(x, w) \mapsto x + w$, hence compact.

Step S4.7: Data. Apply Lemma 4.4 to K_δ: $\|F\| \le M$ and F is L-Lipschitz on $K_\delta \supset \overline{B}(x, \delta)$ for $x \in K$. □

Lemma 4.7 (Uniform data for angle-periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$ (max norm), $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. If $K \subset U$ is compact, then $\Phi = \mathbb{R}^n \times K$ has uniform data (δ, M, L) for some $\delta \in (0, 1]$.*

Proof. **Step S4.8: A compact cell.** If $K = \emptyset$ then $\Phi = \emptyset$ and any δ, M, L work. Otherwise choose $\delta \in (0, 1]$ with $K_\delta = \{I : \exists J \in K,\ |I - J| \le \delta\} \subset U$ (as in Step S4.6, using the nonempty closed set $\mathbb{R}^n \setminus U$, or $\delta = 1$ if $U = \mathbb{R}^n$). $C = [-1, 2]^n \times K_\delta$ is compact and contained in W; let M, L_C be the constants of Lemma 4.4 for C.

Step S4.9: Translating a ball into the cell. Let $x = (\theta, I) \in \Phi$, $k = (\lfloor \theta_i \rfloor)_i \in \mathbb{Z}^n$ and $x' = x - (k, 0) \in [0, 1)^n \times K$. Since the norm is the max norm and $\delta \le 1$, $\overline{B}(x', \delta) \subset C$, and $\overline{B}(x, \delta) = \overline{B}(x', \delta) + (k, 0) \subset W$. By periodicity $F(y + (k, 0)) = F(y)$ on $\overline{B}(x', \delta)$, so $\|F\| \le M$ and F is L_C-Lipschitz on $\overline{B}(x, \delta)$. □

4.3. **Local existence.**

Lemma 4.8 (A priori short-time confinement)**.** *Let J be an interval, $t_0 \in J$, $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha'(t) = F(\alpha(t))$ for all $t \in J$ (derivative two-sided; F is only assumed to be a function), and suppose $\|F\| \le M$ on $\overline{B}(x, \delta)$ with $M > 0$. Then $\|\alpha(t) - x\| \le M|t - t_0|$ for all $t \in J$ with $|t - t_0| < \delta/M$; in particular $\alpha(t) \in \overline{B}(x, \delta)$.*
Source: Standard.

Proof. **Step S4.10: Forward bootstrap.** Let $t \in J$, $t_0 < t < t_0 + \delta/M$ (the case $t < t_0$ follows by Lemma 4.3(b) and the case $t = t_0$ is trivial). Let $A = \{s \in [t_0, t] : \|\alpha(u) - x\| \le M(u - t_0)\ \forall u \in [t_0, s]\}$ and $s^* = \sup A$. $t_0 \in A$ and A is closed by continuity of α on J, so $s^* \in A$.

Step S4.11: Extension. If $s^* < t$, then $\|\alpha(s^*) - x\| \le M(s^* - t_0) < \delta$, so by continuity there is $\eta > 0$, $s^* + \eta \le t$, with $\alpha([s^*, s^* + \eta]) \subset \overline{B}(x, \delta)$. On this interval $\|\alpha'\| = \|F(\alpha)\| \le M$, so by (E4) $\|\alpha(u) - \alpha(s^*)\| \le M(u - s^*)$ and hence $\|\alpha(u) - x\| \le M(u - t_0)$ there, contradicting the definition of s^*. So $s^* = t$. □

Proposition 4.9 (Local existence with uniform time)**.** *Let F be C^1 on the open set W and Φ have uniform data (δ, M, L); put $M' = \max(M, 1)$ and $\varepsilon = \delta/M'$. Then for every $x \in \Phi$ and every $t_0 \in \mathbb{R}$ there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$, $\|\alpha(t) - x\| < \delta$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ and* `IsSolIn` $F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$.
Source: Standard (Picard–Lindelöf).

Proof. **Step S4.12: Picard–Lindelöf.** Apply (E1) with $a = \delta$, bound M', Lipschitz constant L and $T = \varepsilon$ ($M'T = \delta \le a$): there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ within $[t_0 - \varepsilon, t_0 + \varepsilon]$ at each point of it. At points t of the open interval this is a two-sided derivative, since the closed interval is a neighbourhood of t.

Step S4.13: Staying in W. By Lemma 4.8 on $J = (t_0 - \varepsilon, t_0 + \varepsilon)$, $\|\alpha(t) - x\| \le M'|t - t_0| < \delta$, so $\alpha(t) \in \overline{B}(x, \delta) \subset W$. □

Theorem 4.10 (Local existence, both time directions)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $t_0 \in \mathbb{R}$. There are $\varepsilon > 0$ and $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x_0$ and $\texttt{IsSolIn}\, F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$. Moreover ε may be taken uniform on any compact $K \subset W$ (for all $x_0 \in K$).*
Source: Standard.

Proof. **Step S4.14: Reduction.** $\{x_0\}$ (resp. K) is compact; Lemma 4.6 gives uniform data and Proposition 4.9 the claim. □

4.4. Uniqueness on every interval.

Theorem 4.11 (Uniqueness on arbitrary intervals)**.** *Let F be C^1 on the open set W, $J \subset \mathbb{R}$ an interval (of any kind: open, closed, half-open, unbounded, or a point), $t_0 \in J$, and $y_1, y_2 : \mathbb{R} \to E$ with $\texttt{IsSolIn}\, F\, W\, J\, y_i$ and $y_1(t_0) = y_2(t_0)$. Then $y_1 = y_2$ on J. No confinement, boundedness or smallness of either solution is assumed. In particular, if y_1, y_2 are solutions on intervals J_1, J_2 and agree at one point of $J_1 \cap J_2$, they agree on $J_1 \cap J_2$ (an interval), and two solutions which agree at one time of J agree at all times of J.*
Source: Standard; used for BM11 Theorem 2.1 (uniqueness) and in the maximal-solution argument of BM11 p. 13.

Proof. **Step S4.15: A compact piece.** Let $t_1 \in J$, $t_1 > t_0$ ($t_1 < t_0$ is the mirror image, using the left-derivative form of (E2); $t_1 = t_0$ is the hypothesis). Then $[t_0, t_1] \subset J$, each y_i is continuous on it, so $K = y_1([t_0, t_1]) \cup y_2([t_0, t_1])$ is compact ((E5)) and $K \subset W$.

Step S4.16: Lipschitz on K. Lemma 4.4 gives L with F L-Lipschitz on K.

Step S4.17: Apply (E2). Both y_i are continuous on $[t_0, t_1]$, have right derivative $F(y_i(t))$ on $[t_0, t_1)$ (a two-sided derivative gives a right derivative) and take values in K; so $y_1(t_1) = y_2(t_1)$. □

4.5. Maximal solutions and continuation.

Lemma 4.12 (Extension to an open interval)**.** *Let F be C^1 on the open set W, $J \neq \emptyset$ an interval and $\texttt{IsSolIn}\, F\, W\, J\, y$. Then there are an open interval $J' \supset J$ and y' with $y' = y$ on J and $\texttt{IsSolIn}\, F\, W\, J'\, y'$.*

Proof. **Step S4.18: Right end.** If J has a largest element b, apply Theorem 4.10 at $x_0 = y(b) \in W$, $t_0 = b$: α on $(b - \varepsilon, b + \varepsilon)$. Put $y_1(s) = y(s)$ for $s \le b$ and $y_1(s) = \alpha(s)$ for $s > b$. Then y_1 is a solution on $J_1 = J \cup (b, b + \varepsilon)$: at $t \in J$, $t < b$, and at $t \in (b, b + \varepsilon)$ it agrees with y, resp. α, near t; at $t = b$ it has derivative $F(y(b))$ within $(-\infty, b]$ (it equals y there) and within $[b, \infty)$ (it equals α there, as $\alpha(b) = y(b)$), so by (E6) it has the two-sided derivative. If J has no largest element, $J_1 = J$, $y_1 = y$.

Step S4.19: Left end. Apply Step S4.18 to the reversed solution $y_1(-\cdot)$ (Lemma 4.3(b)). The result has no endpoint, so is an open interval. □

Theorem 4.13 (Maximal solution)**.** *Let F be C^1 on the open set W and $x_0 \in W$. There are $T_- \in [-\infty, 0)$, $T_+ \in (0, +\infty]$ and $z : \mathbb{R} \to E$ with $z(0) = x_0$ and $\texttt{IsSolIn}\, F\, W\, (T_-, T_+)\, z$, such that every solution y with $y(0) = x_0$ on an interval $J \ni 0$ satisfies $J \subset (T_-, T_+)$ and $y = z$ on J. Then (T_-, T_+) is the* maximal interval *and $z|_{(T_-, T_+)}$ the* maximal solution*; both are unique. For initial time t_0 instead of 0 use Lemma 4.3(a).*
Source: BM11 p. 13 ("maximal time of existence"); standard.

Proof. **Step S4.20: The union.** Let $\mathcal{J}$ be the set of open intervals $J \ni 0$ carrying a solution y_J with $y_J(0) = x_0$; $\mathcal{J} \neq \emptyset$ by Theorem 4.10. The union I of $\mathcal{J}$ is an open interval $(T_-, T_+) \ni 0$ (a union of intervals containing 0).

Step S4.21: Gluing. For $t \in I$ choose $J \in \mathcal{J}$ with $t \in J$ and set $z(t) = y_J(t)$ (and $z(t) = x_0$ outside I). This is independent of J: for $J, J' \in \mathcal{J}$, $y_J = y_{J'}$ on $J \cap J' \ni 0$ by Theorem 4.11. Each $t \in I$ has an open neighbourhood $J \in \mathcal{J}$ on which $z = y_J$, so z is a solution on I (local nature, (iii) above).

Step S4.22: Maximality. If y is a solution on an interval $J \ni 0$ with $y(0) = x_0$, Lemma 4.12 gives an open $J' \supset J$ and a solution y' extending y, so $J' \in \mathcal{J}$, $J \subset J' \subset I$, and $y = y' = z$ on J (Theorem 4.11). Uniqueness of the interval (T_-, T_+), and of z as a function on it (equality on this interval of any two representatives; values outside it are arbitrary), is immediate from the maximality property applied twice. □

Theorem 4.14 (Continuation criterion, uniform form)**.** *Let F be C^1 on the open set W, $x_0 \in W$, (T_-, T_+) and z the maximal interval and solution of Theorem 4.13. Let $\Phi \subset E$ have uniform data (δ, M, L) (Definition 4.5). If $T_+ < \infty$ there is no $t_* \in [0, T_+)$ with $z(t) \in \Phi$ for all $t \in [t_*, T_+)$. Symmetrically, if $T_- > -\infty$ there is no $t_* \in (T_-, 0]$ with $z(t) \in \Phi$ for all $t \in (T_-, t_*]$. Equivalently: if the solution stays in Φ near the right (left) end of its maximal interval, that end is $+\infty$ ($-\infty$).*
Source: Standard; BM11 p. 13 ("this solution is defined for all time"), Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.23: Set-up.** Suppose $T_+ < \infty$ and $z(t) \in \Phi$ on $[t_*, T_+)$. Let $\varepsilon = \delta / \max(M, 1)$ from Proposition 4.9. Choose $t_1 \in [t_*, T_+)$ with $t_1 > T_+ - \varepsilon$.

Step S4.24: A solution through $z(t_1)$. Proposition 4.9 with $x = z(t_1) \in \Phi$ and $t_0 = t_1$ gives α on $(t_1 - \varepsilon, t_1 + \varepsilon)$ with $\alpha(t_1) = z(t_1)$. By Theorem 4.11 on $(t_1 - \varepsilon, t_1 + \varepsilon) \cap (T_-, T_+)$, $\alpha = z$ there.

Step S4.25: Gluing past T_+. Put $y = z$ on (T_-, T_+) and $y = \alpha$ on $[T_+, t_1 + \varepsilon)$; then $y = \alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ and $y = z$ on (T_-, T_+), two open sets covering $(T_-, t_1 + \varepsilon)$ on each of which y is a solution. So y is a solution on $(T_-, t_1 + \varepsilon) \ni 0$ with $y(0) = x_0$, and $t_1 + \varepsilon > T_+$ contradicts Theorem 4.13. The left end follows by Lemma 4.3(b). □

Corollary 4.15 (Compact continuation)**.** *In the setting of Theorem 4.14, if $K \subset W$ is compact and $z(t) \in K$ for all $t \in [t_*, T_+)$ (some $t_* \in [0, T_+)$), then $T_+ = +\infty$; likewise on the left.*

Proof. Lemma 4.6 gives uniform data for K; apply Theorem 4.14. □

Corollary 4.16 (Closed sets, periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$, $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $k \in \mathbb{Z}^n$. If the maximal solution through $x_0 \in W$ satisfies $z(t)_2 \in K$ for $t \in [t_*, T_+)$ with $K \subset U$ compact, then $T_+ = +\infty$ (likewise on the left). The angle components $z(t)_1$ may be unbounded.*

Proof. Lemma 4.7 gives uniform data for $\Phi = \mathbb{R}^n \times K$; apply Theorem 4.14. □

Remark 4.17 (What the closed-set variant really needs). Staying in a closed non-compact set $\Phi \subset W$ does *not* suffice in general (the solution of $\dot{x} = x^2$ on $E = \mathbb{R}$, $W = \mathbb{R}$, $x_0 = 1$, $x(t) = 1/(1 - t)$, stays in the closed set $[1, \infty)$ for $0 \le t < 1$ and blows up at $T_+ = 1$). What suffices is the uniform data (δ, M, L) of Definition 4.5: a δ-neighbourhood of Φ inside W on which F is bounded and Lipschitz, uniformly. This holds for compact sets, for angle-periodic fields over a compact action set (Lemma 4.7), and for the Newtonian regions bounded away from collisions with bounded velocities (Corollary 4.27).

Theorem 4.18 (Confinement glue)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $\tau > 0$, and $\Phi \subset E$ a set with uniform data (e.g. compact $\Phi \subset W$, or as in Lemma 4.7). Assume the* a priori confinement*: every solution y on an open interval $J \ni 0$ with $y(0) = x_0$ satisfies $y(t) \in \Phi$ for all $t \in J$ with $|t| \le \tau$.*

Then the maximal interval contains $[-\tau,\tau]$ in its interior, so z is a solution on $[-\tau,\tau]$ with $z(t)\in\Phi$ for $|t|\le\tau$, and every solution on $[-L,L]$ $(0\le L\le\tau)$ with initial value x_0 equals z there.
Source: BM11 p. 13 (the two-case argument "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.26: No exit before τ.** Suppose $T_+\le\tau$. The maximal solution z on (T_-,T_+) satisfies, for $t\in[0,T_+)$ ($|t|<\tau$), $z(t)\in\Phi$ by confinement. Theorem 4.14 gives $T_+=\infty$, a contradiction. So $T_+>\tau$, and likewise $T_-<-\tau$.

Step S4.27: Conclusion. $[-\tau,\tau]\subset(T_-,T_+)$ so z is a solution on $[-\tau,\tau]$ and $z(t)\in\Phi$ for $|t|\le\tau$ by confinement applied to z itself. Uniqueness is Theorem 4.11. □

4.6. **First exit from an open set; barriers.**

Lemma 4.19 (First exit)**.** *Let `IsSolIn` $F\,W\,J\,z$ with J an interval, $0\in J$, $V\subset E$ open and $z(0)\in V$. Put $A=\{t\in J: t\ge 0,\ z(t)\notin V\}$. Then exactly one of the following holds.*

(i) $A=\emptyset$: $z(t)\in V$ for all $t\in J$, $t\ge 0$.

(ii) $A\neq\emptyset$. Then $t^*=\min A$ exists, $t^*>0$, $t^*\in J$, $z(t)\in V$ for $t\in[0,t^*)$ and $z(t^*)\in\partial V=\overline{V}\setminus V$. The number t^* is the first exit time *and equals* $\sup\{t\in J: t\ge 0,\ z([0,t])\subset V\}$*; in particular t^* is finite and lies in J (t^* is "inside the solution's interval" and the solution is defined, continuous, at t^*).*

The same holds backwards in time (replace $z(t)$ by $z(-t)$, Lemma 4.3(b)). Only continuity of z on J is used; F and W play no role.
Source: Standard; used for BM11 p. 13 ("smallest time t_* such that $\omega(t_*)\in R_K$") and Pöschel 1993 p. 198–199 (stopping-time arguments).

Proof. **Step S4.28: Closedness.** z is continuous on J (consequence (ii)), $E\setminus V$ is closed, so A is closed in $J\cap[0,\infty)$.

Step S4.29: The minimum. If $A\neq\emptyset$, $t^*=\inf A\in[0,\infty)$; $t^*\in J$ since J is an interval containing 0 and a point of A that is $\ge t^*$. By closedness $t^*\in A$, i.e. $z(t^*)\notin V$. As $z(0)\in V$, $t^*>0$.

Step S4.30: Boundary point. For $t\in[0,t^*)$, $t\notin A$, so $z(t)\in V$. By continuity $z(t^*)=\lim_{t\uparrow t^*}z(t)\in\overline{V}$, and $z(t^*)\notin V$; as V is open, $\overline{V}\setminus V=\partial V$.

Step S4.31: The supremum formula. $z([0,t])\subset V$ iff $t<t^*$ (for $t\in J$, $t\ge 0$), so the supremum is t^*. □

Definition 4.20 (Invariant sets)**.** $S\subset E$ is *invariant* for (F,W) if for every interval J' and every y with `IsSolIn` $F\,W\,J'\,y$: if $y(t_1)\in S$ for one $t_1\in J'$ then $y(t)\in S$ for all $t\in J'$. (Equilibria $\{x:F(x)=0\}$ are invariant by Theorem 4.11; so is any union of such sets and any set $\{G=c\}$ for a first integral G.)

Corollary 4.21 (Barrier)**.** *Let F be C^1 on the open set W, V open,* `IsSolIn` $F\,W\,J\,z$ *with J an interval, $0\in J$, $z(0)\in V$. Let $G\subset E$ with $z(J)\subset G$, and suppose $\partial V\cap G\subset S$ where S is invariant and $z(0)\notin S$ (e.g. G an energy level, Theorem 4.23, and $\partial V\cap G$ a union of invariant sets not meeting the initial point). Then $z(t)\in V$ for all $t\in J$ (both time directions).*

Proof. **Step S4.32: Forward.** If not, Lemma 4.19(ii) gives $t^*\in J$ with $z(t^*)\in\partial V\cap G\subset S$; invariance gives $z(0)\in S$, a contradiction.

Step S4.33: Backward. Apply Step S4.32 to $t\mapsto z(-t)$, a solution of $-F$ on $-J$; G, ∂V are unchanged and S is invariant for $-F$ as well (reversal maps solutions of F to solutions of $-F$ and back). □

4.7. **Hamiltonian layer.** Phase space $P = \mathbb{R}^n \times \mathbb{R}^n$, points $z = (\theta, I)$, max norm. X_H is the frozen `KAM.hamiltonianVectorField`, Definition 1.3 (convention of [K:`bp:p0:hamiltonian`]): $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, with $\partial_{I_i} H(z) = DH(z)(0, e_i)$ and $\partial_{\theta_i} H(z) = DH(z)(e_i, 0)$.

Lemma 4.22 (Regularity of X_H)**.** *Let $W \subset P$ be open and $H : P \to \mathbb{R}$ be C^{k+1} on W, $k \geq 0$. Then X_H is C^k on W; for $k \geq 1$ it is C^1 (hence locally Lipschitz) on W, and H is differentiable at every point of W.*
Source: Standard.

Proof. **Step S4.34: Derivative is C^k.** $DH : W \to \mathcal{L}(P, \mathbb{R})$ is C^k on the open W ((E6), `ContDiffOn.fderiv_of_isOpen`).

Step S4.35: Components. Each coordinate of X_H is $\pm$ the composition of DH with the continuous linear evaluation $\ell \mapsto \ell(0, e_i)$ or $\ell \mapsto \ell(e_i, 0)$; hence C^k, and a map into $\mathbb{R}^n \times \mathbb{R}^n$ with C^k coordinates is C^k. □

Theorem 4.23 (Energy conservation)**.** *Let $H : P \to \mathbb{R}$, $J \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to P$ such that for every $t \in J$, H is differentiable at $z(t)$ and z has derivative $X_H(z(t))$ at t. Then $t \mapsto H(z(t))$ is constant on J. (No regularity of H beyond the stated pointwise differentiability, and no confinement.)*
Source: BM11 p. 13 and Pöschel 1993 p. 198 ("energy conservation", "$\Delta H = H(I(t), \theta(t))|_0^T$ vanishes"); standard.

Proof. **Step S4.36: Chain rule.** For $t \in J$, $\frac{d}{dt} H(z(t)) = DH(z(t))\big[X_H(z(t))\big]$ ((E6)).

Step S4.37: Cancellation. Write $z = z(t)$, $\ell = DH(z)$ and $X_H(z) = (a, b)$ with $a_i = \ell(0, e_i)$, $b_i = -\ell(e_i, 0)$. By linearity, $\ell(a, b) = \sum_i a_i\, \ell(e_i, 0) + \sum_i b_i\, \ell(0, e_i) = \sum_i \ell(0, e_i)\ell(e_i, 0) - \sum_i \ell(e_i, 0)\ell(0, e_i) = 0$.

Step S4.38: Constancy. Let $s < t$ in J. $H \circ z$ is continuous on $[s, t] \subset J$ and has derivative 0 at every point, so $H(z(t)) = H(z(s))$ ((E6)). □

Lemma 4.24 (Periodicity in the angles passes to solutions)**.** *Let $W \subset P$ be open with $W + (k, 0) = W$ for all $k \in \mathbb{Z}^n$, and $H : P \to \mathbb{R}$ with $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. Then $X_H(\theta + k, I) = X_H(\theta, I)$ on W. Consequently, if `IsSolIn` $X_H\, W\, J\, z$ then `IsSolIn` $X_H\, W\, J\, (z + (k, 0))$ for every $k \in \mathbb{Z}^n$, and $H \circ (z + (k, 0)) = H \circ z$.*
Source: Standard.

Proof. **Step S4.39: Field.** Let $\tau_k(x) = x + (k, 0)$. On the open set W, $H \circ \tau_k = H$, so $DH = D(H \circ \tau_k) = DH \circ \tau_k$ at each point of W (the derivative of a translate; both sides are 0 if H is not differentiable, by the definition of `fderiv`). Evaluating on $(0, e_i)$ and $(e_i, 0)$ gives $X_H \circ \tau_k = X_H$ on W.

Step S4.40: Solutions. Lemma 4.3(c) with $v = (k, 0)$ applies to $F = X_H$. □

4.8. **The frozen solution predicates are instances.**

Proposition 4.25 (Track N and track I Hamiltonian predicates)**.** *Let $H : P \to \mathbb{R}$, $U \subset \mathbb{R}^n$ open, $W \subset P$ open with $\mathbb{R}^n \times U \subset W$, and H be C^2 on W. Put $W_U = \mathbb{R}^n \times U$. For every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$,*

$$\texttt{IsSolutionOn}\, H\, U\, J\, z \iff \texttt{IsSolIn}\, X_H\, W_U\, J\, z \iff \texttt{IsHamiltonianSolutionOn}\, H\, U\, J\, z,$$

and X_H is C^1 on W_U. (The two frozen Hamiltonian predicates have the same formula up to the letter naming the action set, and under the regularity hypotheses both are equivalent to `IsSolIn` as shown below, `KAM.Nekhoroshev.IsSolutionOn` and `KAM.Iso.IsHamiltonianSolutionOn`.)
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`, `KAM.Iso.IsHamiltonianSolutionOn`
Source: Ours (bookkeeping).

Proof. **Step S4.41: Membership.** $z(t)_2 \in U \iff z(t) \in W_U$ by definition of W_U.

Step S4.42: Differentiability is free. For $z(t) \in W_U \subset W$, H is C^2 on the open W, hence differentiable at $z(t)$ (Lemma 4.22); so the clause `DifferentiableAt` $\mathbb{R}\, H\,(z\,t)$ is implied by $z(t) \in W_U$, and the two predicates coincide. Smoothness of X_H: Lemma 4.22, $k = 1$, restricted to $W_U \subset W$. □

Proposition 4.26 (Newtonian predicates)**.** *Let $N \geq 2$, $G \in \mathbb{R}$, `mass` : $Fin\,N \to \mathbb{R}$. Let $E_N = (Fin\,N \to \mathbb{R}^2) \times (Fin\,N \to \mathbb{R}^2)$ (max norm), $\ell =$ `planeLength`,*

$$W_N = \{(q,v) : \ell(q_j - q_i) > 0 \ \forall i \neq j\}, \qquad F_N(q,v) = \Big(v,\ \big(\textstyle\sum_{j\neq i} \frac{G\,\mathrm{mass}_j}{\ell(q_j-q_i)^3}(q_j - q_i)\big)_i\Big).$$

Then W_N is open, F_N is C^∞ on W_N, and for $Q, V : \mathbb{R} \to (Fin\,N \to \mathbb{R}^2)$ with $x(t) = (Q(t), V(t))$: `IsNewtonSolution` $G\,\mathrm{mass}\,Q\,V \iff$ `IsSolIn` $F_N\,W_N\,\mathbb{R}\,x$. Likewise for `IsFixedSolution` $G\,m_0\,m\,\varepsilon$ with $W_N' = W_N \cap \{\ell(q_i) > 0 \ \forall i\}$ and the field $F'(q,v) = (v, (-\frac{Gm_0}{\ell(q_i)^3}q_i + \varepsilon\sum_{j\neq i}\frac{Gm_j}{\ell(q_j-q_i)^3}(q_j-q_i))_i)$, and for `IsRestrictedSolutionOn` $\mu\,J$ with $W_\mu = \{(q,p) :$ `CollisionFree` $\mu\,(q,p)\}$ and the total extension F_μ of `restrictedVectorField` μ by the same formula (with the values off W_μ irrelevant): `IsRestrictedSolutionOn` $\mu\,J\,z \iff$ `IsSolIn` $F_\mu\,W_\mu\,J\,z$. In all three cases F is C^∞ on the open set W and the general layer applies with $J = \mathbb{R}$ (resp. the given J).

Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsFixedSolution`, `KAM.Iso.IsRestrictedSolutionOn`

Source: Ours (bookkeeping); Fejoz Section 6.1, Newton equations as in the frozen text.

Proof. **Step S4.43: Open domain and smoothness.** $\ell(w)^2 =$ `planeSq`$(w) = w_0^2 + w_1^2$ is a polynomial, so $\{\ell > 0\} =$ `{planeSq > 0}` is open, and $\ell = \sqrt{\cdot}$ is C^∞ where `planeSq` > 0. Hence $\ell(q_j - q_i)^{-3}$ is C^∞ on W_N and F_N is a finite sum of products of C^∞ maps. Same for F' and F_μ (`radius` $= \sqrt{\texttt{radiusSq}}$ with the shifts by $\texttt{primary}_{1,2}\mu$).

Step S4.44: Derivatives of pairs. A map into a finite product has derivative (a,b) at t iff each component has the corresponding derivative (`hasDerivAt_pi`, `HasDerivAt.prodMk` and converse). The first block of the frozen definition (distance positivity) is membership in W_N; the second block says Q has derivative $V(t)$; the third says V has derivative the force. Together they say exactly that x has derivative $F_N(x(t))$ at t. □

Corollary 4.27 (Newtonian continuation)**.** *In the setting of Proposition 4.26, let x be the maximal solution through $x_0 \in W_N$ (Theorem 4.13), $\rho > 0$, $\bar V \geq 0$ and*

$$\Phi_{\rho,\bar V} = \{(q,v) :\ \ell(q_j - q_i) \geq \rho\ (i \neq j),\ |v| \leq \bar V\}.$$

If $T_+ < \infty$ then for every $\rho, \bar V$ and every $t_* \in [0, T_+)$ there is $t \in [t_*, T_+)$ with $x(t) \notin \Phi_{\rho,\bar V}$ (a finite-time singularity needs near-collisions or unbounded velocities; positions may be unbounded). The same for F' with the additional condition $\ell(q_i) \geq \rho$, and for F_μ with $\ell \geq \rho$ from both primaries and $|q|, |p| \leq \bar V$.

Proof. **Step S4.45: Uniform data.** Let $\delta = \rho/8$. If $|y - x| \leq \delta$ (max norm) with $x \in \Phi_{\rho,\bar V}$ then each $\ell(q_j' - q_i') \geq \rho - 2\sqrt{2}\,\delta \geq \rho/2$, so $y \in W_N$ and $\overline{B}(x,\delta) \subset W_N$. On this ball $\ell \geq \rho/2$ and $|v'| \leq \bar V + \delta$; the entries of F_N and DF_N are polynomials in v', the differences $q_j' - q_i'$ (Euclidean length $\leq \ell(q_j' - q_i')$) and the inverse powers ℓ^{-3}, ℓ^{-5} ($D(w/\ell^3) = \ell^{-3}I - 3\ell^{-5}ww^\top$), hence bounded by a constant $C(N, G, \mathrm{mass}, \rho, \bar V)$ *independent of the position of x*. By (E4) on the convex ball, F_N is C-Lipschitz there. So $\Phi_{\rho,\bar V}$ has uniform data.

Step S4.46: Apply. Theorem 4.14 gives the claim; the other two fields are identical in structure (for F_μ the extra terms are linear in (q,p) and bounded on $|q|, |p| \leq \bar V$). □

4.9. **The track-N statements.** Throughout, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$ with $\texttt{IsAnalyticIntegrable}\, R\, s\, h$, $\varepsilon \geq 0$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with $\texttt{IsAnalyticPerturbation}\, R\, s\, \varepsilon\, f$ (Definitions 1.11, 1.12), $H = H_{h,f}$ (Definition 1.4), $W_s = \mathbb{R}^n \times B(0, R+s) \subset P$ (open; $B(0,R+s) = \mathcal{B}_s(R)$ by Remark 1.9), and $U \subset B(0,R+s)$ open (in N-HEAD-C, $U = B(0,R)$).

Lemma 4.28 (Input from Part 1: H is smooth on W_s)**.** *$H_{h,f}$ is C^∞ (in particular C^2) on the open set W_s, $X_H = X_{h+f}$ on $W_s \cap (\mathbb{R}^n \times \mathbb{R}^n)$, and $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$.*
Source: Part 1, Lemma 2.3(b)–(d) (real smoothness on W_s, field $X_H = X_{h+f} \circ \iota$, periodicity); ingredients [K:bp:p1:holomorphic-regularity], [K:bp:p1:hamiltonian-real].

Proof. **Step S4.47: Citation.** (b), (c), (d) of Lemma 2.3 are exactly the three claims. For orientation, the argument is as follows.

Step S4.48: Regularity. The function $G(\theta, I) = h(I) + f(\theta, I)$ is holomorphic on the open set $D_s(R) \supset W_s$ (real points (θ, I) with $I \in \mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ lie in $T_s \times B(0,R)_s$), hence C^∞ in the real sense ([K:bp:p1:holomorphic-regularity]); it is real on W_s by Definition 1.11(iv) and 1.12(iii). So $H = \operatorname{Re} G = G$ on W_s is the restriction of C^∞ map to a real subspace: C^∞ on W_s; $X_H = X_G$ at real points is [K:bp:p1:hamiltonian-real]. (Packaged as Lemma 2.3.)

Step S4.49: Periodicity. $f(\theta + k, I) = f(\theta, I)$ for real θ, $I \in \mathcal{B}_s(R)$, and h depends on I only. □

Theorem 4.29 (X_H is C^1, hence locally Lipschitz)**.** *$X_{H_{h,f}}$ is C^1 on $W_s \supset \mathbb{R}^n \times B(0,R)$; consequently it is Lipschitz on every compact subset of W_s, and on every set $\mathbb{R}^n \times K$ with $K \subset B(0,R+s)$ compact there are δ, M, L as in Lemma 4.7.*
Source: BM11 p. 13 (implicit in "the solution"); standard.

Proof. **Step S4.50: Smoothness.** Lemma 4.28 and Lemma 4.22 ($k = 1$).

Step S4.51: Lipschitz. Lemma 4.4 on compact sets, Lemma 4.7 (with $U = B(0,R+s)$, using the periodicity of X_H from Lemma 4.24 and Lemma 4.28) for the cylinders. □

Theorem 4.30 (Local existence and uniqueness for $\texttt{IsSolutionOn}$)**.** *For every $t_0 \in \mathbb{R}$ and $z_0 \in \mathbb{R}^n \times U$ there are $\epsilon > 0$ and $y : \mathbb{R} \to P$ with $y(t_0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, (t_0 - \epsilon, t_0 + \epsilon)\, y$ (both time directions). If J is any interval, $t_0 \in J$ and y_1, y_2 satisfy $\texttt{IsSolutionOn}\, H\, U\, J\, y_i$ with $y_1(t_0) = y_2(t_0)$, then $y_1 = y_2$ on J.*
Source: BM11 p. 13 and Theorem 2.1 p. 5; Picard–Lindelöf (external fact (E1)).

Proof. **Step S4.52: Reduction.** Proposition 4.25 (with $W = W_s$, H C^2 there by Lemma 4.28) turns $\texttt{IsSolutionOn}\, H\, U$ into $\texttt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)$ with X_H C^1 on $\mathbb{R}^n \times U$.

Step S4.53: Existence, uniqueness. Theorem 4.10 and Theorem 4.11. □

Theorem 4.31 (Energy conservation for $\texttt{IsSolutionOn}$)**.** *If $\texttt{IsSolutionOn}\, H\, U\, J\, z$ for an interval J (here H may be any function $P \to \mathbb{R}$ and U any set), then $H(z(t)) = H(z(t'))$ for $t, t' \in J$. In particular, for $H = H_{h,f}$ and $J = [0,T]$ the quantity $\Delta H = H(z(t))|_0^T$ of Pöschel's Lemma 1 vanishes.*
Source: Pöschel 1993 p. 198 ("by energy conservation ... ΔH vanishes"; page read); BM11 p. 13.

Proof. **Step S4.54: Direct.** The definition gives differentiability of H at every $z(t)$ and $z'(t) = X_H(z(t))$; Theorem 4.23. □

Theorem 4.32 (Maximal solutions and continuation)**.** *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in U$. There is a unique maximal interval $(T_-, T_+) \ni 0$ and a solution z with $z(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, (T_-, T_+)\, z$, such that every y with $y(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, J\, y$ on an interval $J \ni 0$ has $J \subset (T_-, T_+)$ and $y = z$ on J (Theorem 4.13 through Proposition 4.25). If $K \subset U$ is compact and $z(t)_2 \in K$ for all $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$; likewise*

on the left. The angle components $z(t)_1$ may be unbounded: no obstruction arises because X_H is 1-periodic in θ.
Source: BM11 p. 13 ("From this estimate we can deduce that the solution $I(t)$ belongs to some compact ball ... this solution is defined for all time"); Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.55: General statements.** By Proposition 4.25, `IsSolutionOn` $H\,U\,J\,y \iff$ `IsSolIn` $X_H\,(\mathbb{R}^n \times U)\,J\,y$ and X_H is C^1 on $W = \mathbb{R}^n \times U$; maximality is Theorem 4.13.

Step S4.56: Periodic cylinder. X_H is 1-periodic in θ on W (Lemmas 4.24, 4.28). Corollary 4.16 yields $T_+ = \infty$. □

Theorem 4.33 (Uniqueness on closed intervals)**.** *For $L \geq 0$ and y_1, y_2 with $y_1(0) = y_2(0)$ and `IsSolutionOn` $H\,U\,[-L,L]\,y_i$ one has $y_1 = y_2$ on $[-L,L]$. In particular a solution on $[-L,L]$ agrees with every solution on a larger interval containing $[-L,L]$ and the same initial value (clause (d) of Theorem 1.19, where $U = B(0,R)$, $L \in [0,\tau]$, y arbitrary and z the exhibited solution). No confinement of y is assumed: Gronwall is replaced by Theorem 4.11, whose Lipschitz constant comes from the compact set $y_1([-L,L]) \cup y_2([-L,L])$.*
Source: Standard (Gronwall); clause (d) of the frozen N-HEAD-C.

Proof. **Step S4.57: Apply.** $J = [-L,L]$, $t_0 = 0$ in Theorem 4.30 (uniqueness part). □

Theorem 4.34 (Confinement glue)**.** *Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in U$, $\tau > 0$ and $\rho > 0$ with $\overline{B}(I_0,\rho) \subset U$ (e.g. $|I_0| < R/2$, $\rho \leq R/4$, $U = B(0,R)$). Assume that for every open interval $J \ni 0$ and every y with $y(0) = (\theta_0, I_0)$ and `IsSolutionOn` $H\,U\,J\,y$ one has $|y(t)_2 - I_0| \leq \rho$ for all $t \in J$ with $|t| \leq \tau$. Then there is z with $z(0) = (\theta_0, I_0)$, `IsSolutionOn` $H\,U\,[-\tau,\tau]\,z$ and $|z(t)_2 - I_0| \leq \rho$ for $|t| \leq \tau$; and for every $L \in [0,\tau]$ and every y with $y(0) = z(0)$ and `IsSolutionOn` $H\,U\,[-L,L]\,y$, $y = z$ on $[-L,L]$. This is the assembly of clauses (a)–(d) of Theorem 1.19 from an a priori estimate (Part 10); $\rho = C\varepsilon^{1/(2n)}$ there.*
Source: BM11 p. 13 (two cases, "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.58: Compact confinement set.** $K = \overline{B}(I_0,\rho)$ is compact and $\subset U$; $\Phi = \mathbb{R}^n \times K$ has uniform data by Lemma 4.7 (with $W = \mathbb{R}^n \times U$, $F = X_H$ periodic by Lemma 4.24).

Step S4.59: General glue. By Proposition 4.25 the hypothesis reads: every general solution on an open $J \ni 0$ with this initial value stays in Φ for $|t| \leq \tau$. Theorem 4.18 gives z on $[-\tau,\tau]$ in Φ, i.e. $|z(t)_2 - I_0| \leq \rho$, and uniqueness on $[-L,L]$ (it is also Theorem 4.33). □

4.10. Label index for tracks I and P.

`bpN:p3:gen-sol`, `gen-symmetry`	predicate; time shift, reversal, space translation
`gen-lip`, `gen-uniform`, `gen-uniform-compact`, `gen-uniform-periodic`	Lipschitz and uniform data
`gen-short`, `gen-uniform-exist`, `gen-local`	local existence, uniform time
`gen-uniqueness`	uniqueness on all intervals, no confinement
`gen-extend`, `gen-maximal`	maximal solutions
`gen-continuation`, `-compact`, `-periodic`, `gen-confinement`	continuation, closed sets with uniform bounds, confinement glue
`gen-exit`, `gen-invariant`, `gen-barrier`	first exit, barrier
`ham-smooth`, `ham-energy`, `ham-periodic`	Hamiltonian layer
`ham-reduction`, `newton-reduction`, `newton-continuation`	frozen predicates of I, N, P
`holo-input`, `lipschitz`, `local-exist`, `energy`, `continuation`, `uniqueness-closed`, `confinement-glue`	track N specializations

5. Weighted Fourier norms and Cauchy estimates (shared by tracks N, P, K)

This part is Appendix B and Appendix C of Pöschel 1993 (pp. 213–215) in the frozen conventions. It is shared infrastructure: track P (averaging, two-scale KAM) cites its results as [N:`bpN:p4:...`]. The statements are therefore made for functions $u(\theta, I)$ with $\theta \in \mathbb{T}^n$ and I in a neighbourhood of an *arbitrary* set $D \subset \mathbb{C}^m$, with m independent of n; only the Poisson bracket (Lemma 5.7) needs $m = n$. All norms on $\mathbb{R}^N$, $\mathbb{C}^N$ are max norms; no Euclidean norm occurs in this part.
Normalisation. Angles have period 1, characters are $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, and the strip width s is the *geometric* width, $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta|_\infty < s\}$ ([K:`bp:p0:domains`]); the weight is $e^{2\pi|k|_1 s}$, that is, the coefficient radius is $2\pi s$ as in `docs/fourier-conventions.md`. With this choice every formula of Pöschel's Appendix B holds *verbatim*, with no factor 2π (Remark 5.11); for the θ-derivative the factor 2π of $\partial_{\theta_j}\chi_k = 2\pi i k_j \chi_k$ is absorbed by the weight. Widths are r (actions) and s (angles); the *losses* are ρ (actions) and σ (angles), as in Pöschel's (ρ, σ). (The brief's $|u|_{D,r,\sigma}$ is $|u|_{D,r,s}$ here.)

5.1. External facts.

Besides [K:`bp:p1:ext-cauchy`], [K:`bp:p1:ext-integral`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-mvt`], the proofs use:

(E1) *Holomorphic dependence on a parameter.* Let $Q \subset \mathbb{R}^p$ be a compact box, $U \subset \mathbb{C}^q$ open, $G : Q \times U \to \mathbb{C}^d$ continuous, holomorphic in $I \in U$ for each $x \in Q$, with $D_I G$ continuous on $Q \times U$. Then $g(I) = \int_Q G(x, I)\,dx$ is holomorphic on U and $Dg(I) = \int_Q D_I G(x, I)\,dx$. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le` with a bound on a compact polydisc.) Exists.

(E2) *Uniform limits.* If K_j are holomorphic on an open $U \subset \mathbb{C}^q$ and $K_j \to K$ uniformly on U, then K is holomorphic on U, and $DK_j \to DK$ uniformly on $\{z : \bar{P}(z, \eta) \subset U\}$ for each $\eta > 0$ (the derivatives by Lemma 5.8 applied to $K_j - K_l$; holomorphy by [K:`bp:p1:uniform-limits`]). Exists in this form modulo that lemma.

(E3) *Series and integrals.* Tonelli for nonnegative double series and for integrals of nonnegative functions; the Cauchy product of two absolutely convergent series; termwise integration of a uniformly convergent series over a box ([K:`bp:p1:ext-integral`]). Exists.

5.2. Domains, function class, norm.

Definition 5.1 (Domains)**.** Let $m, n \geq 0$, $D \subset \mathbb{C}^m$ any subset, $r, s > 0$. Put $V_r D = \{I \in \mathbb{C}^m : \exists I' \in D,\ |I - I'|_\infty < r\}$ and $\mathcal{D}_{r,s} = T_s \times V_r D \subset \mathbb{C}^n \times \mathbb{C}^m$ (Pöschel's $V_{r,s}$). For $z = (\theta, I)$ let $\bar{P}(z; a, b) = \{(\theta', I') : |\theta' - \theta|_\infty \leq a, |I' - I|_\infty \leq b\}$. Then: (a) $V_r D$ and $\mathcal{D}_{r,s}$ are open, and $\mathcal{D}_{r',s'} \subset \mathcal{D}_{r,s}$ if $r' \leq r$, $s' \leq s$. (b) If $0 < r' < r$, $0 < s' < s$ and $z \in \mathcal{D}_{r',s'}$, then $\bar{P}(z; s - s', r - r') \subset \mathcal{D}_{r,s}$.
Source: Pos93 p. 213 ("analytic on complex neighbourhoods of a fixed domain $D \times \mathbb{T}^n$"); ours (max norms).

Proof. **Step S5.1: Openness and monotonicity.** $V_r D = \bigcup_{I' \in D} P(I', r)$ is a union of open polydiscs, and T_s is open; monotonicity is clear.

Step S5.2: Polydisc. Let $|\theta' - \theta|_\infty \leq s - s'$ and $|I' - I|_\infty \leq r - r'$. Then $|\operatorname{Im}\theta'_j| \leq |\operatorname{Im}\theta_j| + s - s' < s$; and with $I_0 \in D$, $|I - I_0| < r'$: $|I' - I_0| < r' + (r - r') = r$. □

Definition 5.2 (The class $\mathcal{O}(r, s)$ and the weighted norm)**.** $\mathcal{O}(r, s)$ is the set of $u : \mathcal{D}_{r,s} \to \mathbb{C}$ that are holomorphic (complex differentiable on the open set $\mathcal{D}_{r,s} \subset \mathbb{C}^{n+m}$) and 1-periodic in each θ_j. For $u \in \mathcal{O}(r, s)$ and $k \in \mathbb{Z}^n$, $I \in V_r D$, put

$$\hat{u}_k(I) = \int_{[0,1]^n} u(x, I)\, e^{-2\pi i k\cdot x}\,dx, \qquad |u|_{I,s} = \sum_{k \in \mathbb{Z}^n} |\hat{u}_k(I)|\, e^{2\pi|k|_1 s}, \qquad |u|_{r,s} = \sup_{I \in V_r D} |u|_{I,s} \in [0, \infty].$$

(This is $|u|_{D,r,s}$; the integrand is continuous because $\mathbb{R}^n \subset T_s$.) For a vector- or matrix-valued $u = (u_a)_a$ (components in $\mathcal{O}(r, s)$) the coefficients are taken componentwise and $|\hat{u}_k|_\infty = \max_a |\hat{u}_{a,k}|$ replaces $|\hat{u}_k|$ in $|u|_{I,s}$. If $|u|_{r,s} < \infty$ we say u has *finite norm*.

Source: Pos93 p. 213 (definition of $|u|_{I,s}$, $|u|_{r,s}$), in the frozen convention of `docs/fourier-conventions.md` ("Bridge to strip sup norms": weight $e^{2\pi|k|_1 s}$).

Monotonicity: $|u|_{r',s'} \le |u|_{r,s}$ for $r' \le r$, $s' \le s$ (restriction to $\mathcal{D}_{r',s'}$; the coefficients at $I \in V_{r'}D$ are unchanged). For a vector, $\max_a |u_a|_{r,s} \le |u|_{r,s} \le \sum_a |u_a|_{r,s}$.

Lemma 5.3 (Coefficients, derivatives, series)**.** *Let $u \in \mathcal{O}(r,s)$.*

(a) Each $\hat{u}_k$ is holomorphic on V_rD.

(b) $\partial_{\theta_j} u$, $\partial_{I_i} u \in \mathcal{O}(r,s)$, $\widehat{(\partial_{\theta_j} u)}_k = 2\pi i k_j \hat{u}_k$, and $\widehat{(\partial_{I_i} u)}_k = \partial_{I_i} \hat{u}_k$.

(c) (Series) Let $a_k : V_rD \to \mathbb{C}$ ($k \in \mathbb{Z}^n$) be holomorphic with $S := \sup_{I \in V_rD} \sum_k |a_k(I)| e^{2\pi|k|_1 s} < \infty$. Then $v = \sum_k a_k \chi_k$ converges absolutely and uniformly on $\mathcal{D}_{r,s'}$ for each $s' < s$, $v \in \mathcal{O}(r,s)$, $\hat{v}_k = a_k$ and $|v|_{r,s} = S$.

(d) (Representation) If $|u|_{r,s} < \infty$ then $u(\theta, I) = \sum_k \hat{u}_k(I)\chi_k(\theta)$ on $\mathcal{D}_{r,s}$, and $|\hat{u}_k(I)| e^{2\pi|k|_1 s} \le |u|_{r,s}$.

Source: Standard; the one-variable ingredients are [K:`bp:p1:fourier-basic`], [K:`bp:p1:summation`].

Proof. **Step S5.3: (a).** Fix $I_0 \in V_rD$ and $\eta > 0$ with $\bar{P}(I_0, \eta) \subset V_rD$. The integrand $G(x, I) = u(x, I) e^{-2\pi i k \cdot x}$ is continuous on $[0,1]^n \times P(I_0, \eta)$, holomorphic in I, and $D_I G$ is continuous there, because Du is holomorphic ([K:`bp:p1:holomorphic-regularity`]) hence continuous. (E1) gives holomorphy of $\hat{u}_k$ near I_0.

Step S5.4: (b). The derivatives are holomorphic ([K:`bp:p1:holomorphic-regularity`]) and periodic (differentiate the identity $u(\theta + e_j, I) = u(\theta, I)$). The formula for θ_j is [K:`bp:p1:fourier-basic`](a) at fixed I (the restriction of $u(\cdot, I)$ to $\mathbb{R}^n$ is continuous, periodic and C^1). The formula for I_i is the derivative formula of (E1) as in Step S5.3.

Step S5.5: (c). Each hypothesis $|a_k(I)| e^{2\pi|k|_1 s} \le S$ gives, on $\mathcal{D}_{r,s'}$, $|a_k(I)\chi_k(\theta)| \le |a_k(I)| e^{2\pi|k|_1 s'} \le S e^{-2\pi|k|_1 (s - s')}$, which is summable ([K:`bp:p1:exp-sums`](b)). By the Weierstrass test the series converges uniformly on $\mathcal{D}_{r,s'}$, and each term is holomorphic, so v is holomorphic there ((E2)); as $s' < s$ is arbitrary, v is holomorphic on $\mathcal{D}_{r,s}$. It is periodic since each χ_k is. Integrating termwise ((E3)) against χ_{-k} over $[0,1]^n$ gives $\hat{v}_k = a_k$ (as in the coefficient step of [K:`bp:p1:summation`]); the norm identity is then the definition.

Step S5.6: (d). The bound is a single term of the sum $|u|_{I,s} \le |u|_{r,s}$. Fix I. The function $v = \sum_k \hat{u}_k(I)\chi_k$ is holomorphic and 1-periodic on T_s with coefficients $\hat{u}_k(I)$ ([K:`bp:p1:summation`], first part), and $u(\cdot, I) - v$ is holomorphic, 1-periodic, with all coefficients zero, so $u(\cdot, I) = v$ by [K:`bp:p1:fourier-basic`](c). □

5.3. **Lemma B.1: sup norm versus weighted norm.**

Lemma 5.4 (B.1)**.** *Let $u \in \mathcal{O}(r,s)$ (scalar, vector or matrix valued;* $\sup|u|$ *is the sup of the max norm of the values).*

(a) $\sup_{\mathcal{D}_{r,s}} |u| \le |u|_{r,s}$.

(b) If $\sigma > 0$ and $|u| \le M$ on $\mathcal{D}_{r,s+\sigma}$ (so $u \in \mathcal{O}(r, s+\sigma)$), then $|u|_{r,s} \le \coth(\pi\sigma)^n M$.

In Pöschel's form: $|u|^{\sup}_{r,s} \le |u|_{r,s} \le \coth^n \sigma_P \, |u|^{\sup}_{r,s+2\sigma_P}$ with $\sigma_P = \pi\sigma$ (the coefficient radius grows by $2\sigma_P = 2\pi\sigma$); see Remark 5.11.

Source: Pos93 p. 213, Lemma B.1; frozen-convention form: [K:`bp:p1:summation`], [K:`bp:p1:decay`], [K:`bp:p1:exp-sums`]*(b); this is Lemma 2.9 with the I-dependence made explicit.*

Proof. **Step S5.7: (a).** If $|u|_{r,s} = \infty$ there is nothing to prove. Otherwise Lemma 5.3(d) gives, for $(\theta, I) \in \mathcal{D}_{r,s}$, $|u(\theta, I)| \le \sum_k |\hat{u}_k(I)||\chi_k(\theta)| \le \sum_k |\hat{u}_k(I)| e^{2\pi|k|_1 s}$, using $|\chi_k(\theta)| \le e^{2\pi|k|_1 |\mathrm{Im}\,\theta|_\infty}$ ([K:`bp:p0:domains`]). For vector values apply this to each component and note $|\hat{u}_{a,k}| \le |\hat{u}_k|_\infty$.

Step S5.8: (b). Fix $I \in V_r D$. The function $u(\cdot, I)$ is holomorphic, periodic and bounded by M on $T_{s+\sigma}$, so [K:bp:p1:decay] gives $|\hat{u}_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ (limit $\rho' \uparrow s+\sigma$), and

$$|u|_{I,s} \le M \sum_k e^{-2\pi|k|_1\sigma} = \coth(\pi\sigma)^n M \qquad ([\text{K:bp:p1:exp-sums}](b),\ x = 2\pi\sigma).$$

Take the supremum over I. □

5.4. **Lemma B.2: algebra property.**

Lemma 5.5 (B.2)**.** *Let $u, v \in \mathcal{O}(r,s)$ be scalar with finite norms. Then $uv \in \mathcal{O}(r,s)$, $\widehat{(uv)}_k(I) = \sum_l \hat{u}_{k-l}(I)\hat{v}_l(I)$ (absolutely convergent), and $|uv|_{I,s} \le |u|_{I,s}\,|v|_{I,s}$ for every $I \in V_r D$; hence $|uv|_{r,s} \le |u|_{r,s}|v|_{r,s}$, and $|u^j|_{r,s} \le |u|_{r,s}^j$.*
Source: Pos93 p. 213, Lemma B.2. (The coefficient-level inequality is proved in Lean: `docs/fourier-conventions.md`, "Verified coefficient multiplication", `KAM/Fourier/Convolution.lean`; its identification with the product of functions is the remaining bridge.)

Proof. **Step S5.9: Product is in the class.** uv is holomorphic and periodic.

Step S5.10: Coefficients. Fix I. By Lemma 5.3(d), at real x, $u(x,I) = \sum_k \hat{u}_k \chi_k(x)$ and $v(x,I) = \sum_l \hat{v}_l \chi_l(x)$, both absolutely convergent ((E3)). The Cauchy product gives $u(x,I)v(x,I) = \sum_k c_k \chi_k(x)$ with $c_k = \sum_l \hat{u}_{k-l}\hat{v}_l$, and $\sum_k |c_k| \le |u|_{I,0}|v|_{I,0} < \infty$, so the convergence is uniform in x and termwise integration gives $\widehat{(uv)}_k(I) = c_k$.

Step S5.11: Norm. Since $|k|_1 \le |k-l|_1 + |l|_1$, $|uv|_{I,s} = \sum_k |c_k| e^{2\pi|k|_1 s} \le \sum_k \sum_l |\hat{u}_{k-l}| e^{2\pi|k-l|_1 s} |\hat{v}_l| e^{2\pi|l|_1 s} = |u|_{I,s}|v|_{I,s}$ (Tonelli, with the bijection $(k,l) \mapsto (k-l, l)$). Take the supremum over I; induct for u^j. □

5.5. **Lemma B.3: Cauchy estimates in the weighted norm.**

Lemma 5.6 (B.3)**.** *Let $u \in \mathcal{O}(r_0, s_0)$ have finite norm, $0 < r < r_0$, $0 < s < s_0$. Then*

(a) (pointwise in I) for every $I \in V_{r_0} D$: $\displaystyle\sum_{j=1}^n |\partial_{\theta_j} u|_{I,s} \le \frac{1}{e(s_0 - s)} |u|_{I,s_0}$; consequently $\displaystyle\sup_{I \in V_{r_0} D} \sum_j |\partial_{\theta_j} u|_{I,s} \le \frac{|u|_{r_0,s_0}}{e(s_0-s)}$ and, for each single j (global norm), $|\partial_{\theta_j} u|_{r_0,s} \le \frac{1}{e(s_0-s)} |u|_{r_0,s_0}$. The sum over j of the global norms $\sum_j \sup_I |\partial_{\theta_j} u|_{I,s}$ is not bounded by this constant (for D disconnected it can be larger by a factor up to n; with the norm $\sharp$ of Lemma 5.10 it is bounded);

(b) $\displaystyle\max_{1 \le i \le m} |\partial_{I_i} u|_{r,s_0} \le \frac{1}{r_0 - r} |u|_{r_0,s_0}$.

In particular (taking $r_0 = r+\rho$, $s_0 = s+\sigma$): $\sup_{I \in V_r D} \sum_j |\partial_{\theta_j} u|_{I,s-\sigma} \le (e\sigma)^{-1} |u|_{r,s}$, $|\partial_{\theta_j} u|_{r,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$ for each j, and $\max_i |\partial_{I_i} u|_{r-\rho,s} \le \rho^{-1}|u|_{r,s}$ for $0 < \sigma < s$, $0 < \rho < r$. The same holds for vector- and matrix-valued u, componentwise (and with the pointwise sum over the θ-index outside, never the sum of global norms).
Source: Pos93 pp. 213–214, Lemma B.3 (with s geometric; Pöschel's $1/(e\sigma)$ is the same number, Remark 5.11).

Proof. **Step S5.12: (a): coefficients.** Fix $I \in V_{r_0} D$. By Lemma 5.3(b), $\sum_j |\partial_{\theta_j} u|_{I,s} = \sum_k \sum_j 2\pi|k_j| |\hat{u}_k| e^{2\pi|k|_1 s} = \sum_k 2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)} |\hat{u}_k| e^{2\pi|k|_1 s_0}$. By [K:bp:p1:exp-sums](a) with $\tau = 1$, $a = 2\pi(s_0 - s)$: $2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)} \le 2\pi/(e \cdot 2\pi(s_0-s))$. Hence the sum is $\le (e(s_0-s))^{-1}|u|_{I,s_0}$ for this I. The two global consequences follow by taking the supremum over I of the left side (resp. of a single summand, which is at most the sum) and using $|u|_{I,s_0} \le |u|_{r_0,s_0}$.

Step S5.13: (b): Cauchy integral. Fix $I \in V_r D$, i, and put $\rho' = r_0 - r$. For $|\zeta| \le \rho'$ the point $I + \zeta e_i$ lies in $V_{r_0} D$: if $I_0 \in D$ with $|I - I_0| < r$ then $|I + \zeta e_i - I_0| < r + \rho' = r_0$. By Lemma 5.3(a),(b) and [K:bp:p1:ext-cauchy], $\partial_{I_i} \hat{u}_k(I) = \frac{1}{2\pi i} \oint_{|\zeta|=\rho'} \hat{u}_k(I + \zeta e_i) \zeta^{-2}\, d\zeta$, so $|\partial_{I_i} \hat{u}_k(I)| \le \frac{1}{2\pi} \oint |\hat{u}_k(I + \zeta e_i)| \rho'^{-2} |d\zeta|$. Multiply by $e^{2\pi |k|_1 s_0}$, sum over k and use Tonelli ((E3)): $|\partial_{I_i} u|_{I,s_0} \le \frac{1}{2\pi} \oint \rho'^{-2} |u|_{I+\zeta e_i, s_0} |d\zeta| \le \rho'^{-1} |u|_{r_0,s_0}$. The vector case is componentwise, the matrix case likewise. □

5.6. **Lemma B.4: the Poisson bracket.** For $m = n$ and u, v holomorphic on an open subset of $\mathbb{C}^n \times \mathbb{C}^n$, $\{u, v\} = \sum_{i=1}^n \big(\partial_{\theta_i} u\, \partial_{I_i} v - \partial_{I_i} u\, \partial_{\theta_i} v\big)$ (Definition 6.2; $\{u, v\} = Du \cdot X_v$ for the frozen field $X_v = (\partial_I v, -\partial_\theta v)$).

Lemma 5.7 (B.4)**.** *Let $m = n$, $u \in \mathcal{O}(r_0, s_0)$, $v \in \mathcal{O}(r, s)$ with finite norms, $0 < \rho < r$, $0 < \sigma < s$, $r - \rho < r_0$, $s - \sigma < s_0$. Then $\{u, v\} \in \mathcal{O}(r - \rho, s - \sigma)$ and*

$$|\{u, v\}|_{r-\rho, s-\sigma} \le \frac{1}{e} \Big(\frac{1}{(r_0 - r + \rho)\, \sigma} + \frac{1}{(s_0 - s + \sigma)\, \rho} \Big) |u|_{r_0,s_0} |v|_{r,s}.$$

Source: Pos93 p. 214, Lemma B.4 (same formula, with Pöschel's $(r_0, s_0, r, s, \rho, \sigma)$ read as our geometric quantities).

Proof. Write $\{u, v\} = \sum_i \partial_{\theta_i} u\, \partial_{I_i} v - \sum_i \partial_{I_i} u\, \partial_{\theta_i} v$, all terms restricted to $\mathcal{D}_{r-\rho,s-\sigma} \subset \mathcal{D}_{r_0,s_0} \cap \mathcal{D}_{r,s}$ (monotonicity).

Step S5.14: Pointwise product estimate. Fix $I \in V_{r-\rho} D$ and write $|\cdot|_{I,\cdot} = |\cdot|_{I,s-\sigma}$, $t = s - \sigma$. By Lemma 5.5 (pointwise in I) and the triangle inequality for $|\cdot|_{I,t}$ (coefficients of a sum are sums), $|\{u, v\}|_{I,t} \le \sum_i |\partial_{\theta_i} u|_{I,t} |\partial_{I_i} v|_{I,t} + \sum_i |\partial_{I_i} u|_{I,t} |\partial_{\theta_i} v|_{I,t} \le \Big(\sum_i |\partial_{\theta_i} u|_{I,t}\Big) \max_i |\partial_{I_i} v|_{I,t} + \max_i |\partial_{I_i} u|_{I,t} \Big(\sum_i |\partial_{\theta_i} v|_{I,t}\Big)$. The angular sums are kept *pointwise in I*.

Step S5.15: Four derivative bounds. By Lemma 5.6, with (a) pointwise and (b) global, and monotonicity of the norm in both widths: $\sum_i |\partial_{\theta_i} u|_{I,t} \le \frac{|u|_{I,s_0}}{e(s_0 - s + \sigma)} \le \frac{|u|_{r_0,s_0}}{e(s_0 - s + \sigma)}$ ((a) with $s \to s - \sigma$; $I \in V_{r_0} D$ since $r - \rho < r_0$); $\max_i |\partial_{I_i} u|_{I,t} \le \frac{|u|_{r_0,s_0}}{r_0 - r + \rho}$ ((b) with $r \to r - \rho$); $\sum_i |\partial_{\theta_i} v|_{I,t} \le \frac{|v|_{I,s}}{e\sigma} \le \frac{|v|_{r,s}}{e\sigma}$ ((a), pointwise) and $\max_i |\partial_{I_i} v|_{I,t} \le \frac{|v|_{r,s}}{\rho}$ ((b)). Inserting the four bounds in the previous display and taking the supremum over $I \in V_{r-\rho} D$ *last* gives the claim; the constant is unchanged and no factor n appears. □

5.7. **Lemma C.1: the general Cauchy inequality.**

Lemma 5.8 (C.1)**.** *Let A, B be finite-dimensional complex normed spaces, $v \in A$, $R > 0$, and $G : P_A(v, R) = \{a : |a - v|_A < R\} \to B$ holomorphic with $|G|_B \le M$ on this ball. Then the operator norm satisfies $|DG(v)|_{B,A} \le M/R$.*
Source: Pos93 p. 215, Lemma C.1 ("analytic maps of Banach spaces"; only the finite-dimensional case is needed).

Proof. **Step S5.16: Reduction to one variable.** Let $u \in A$, $u \ne 0$. $f(\zeta) = G(v + \zeta u)$ is holomorphic on $|\zeta| < R/|u|_A$ with $|f| \le M$ and $f'(0) = DG(v)u$.

Step S5.17: Cauchy. For every $\varrho < R/|u|_A$, [K:bp:p1:ext-cauchy] gives $|f'(0)| \le M/\varrho$; let $\varrho \uparrow R/|u|_A$: $|DG(v)u|_B \le M|u|_A/R$. □

This also gives (E2)'s derivative statement: apply it to $K_j - K_l$ on the ball of radius η.

5.8. **Projections and truncation.**

Lemma 5.9 (P_S, T_K, averages)**.** *Let $u \in \mathcal{O}(r,s)$ have finite norm and $S \subset \mathbb{Z}^n$. Put $P_S u = \sum_{k\in S} \hat{u}_k \chi_k$. Then $P_S u \in \mathcal{O}(r,s)$, $\widehat{(P_S u)}_k = \hat{u}_k$ for $k \in S$ and 0 otherwise, and $|P_S u|_{r,s} \le |u|_{r,s}$. In particular, for $K \ge 0$ the truncation $T_K u = P_{\{|k|_1 \le K\}} u$ (a trigonometric polynomial in θ) and the average $\langle u\rangle(I) = \hat{u}_0(I) = P_{\{0\}} u$ (independent of θ) satisfy $|T_K u|_{r,s} \le |u|_{r,s}$, $|\langle u\rangle|_{r,s} \le |u|_{r,s}$, and*

$$u - T_K u = P_{\{|k|_1 > K\}} u, \qquad |u - T_K u|_{r,s-\sigma} \le e^{-2\pi K\sigma} |u|_{r,s} \qquad (0 < \sigma < s).$$

If $\Lambda \subset \mathbb{Z}^n$ is any set (e.g. a lattice, Pos93 p. 192) and u is said to be in Λ-normal form when $u = P_\Lambda u$, then $P_\Lambda u$, $u - P_\Lambda u$ are in the class with norms $\le |u|_{r,s}$. For vector-valued u everything is componentwise.
Source: Pos93 p. 192 (definition of T_K, P_Λ) and p. 212 ("$|f - T_K f|_{r,s-\sigma} \le e^{-K\sigma}|f|_{r,s}$"), geometric form.

Proof. **Step S5.18: Class and coefficients.** The coefficient family $a_k = \hat{u}_k \mathbf{1}_S(k)$ consists of holomorphic functions (Lemma 5.3(a)) with $\sup_I \sum_k |a_k| e^{2\pi|k|_1 s} \le |u|_{r,s} < \infty$; Lemma 5.3(c) gives all claims including $|P_S u|_{r,s} \le |u|_{r,s}$.

Step S5.19: Tail. For $|k|_1 > K$: $e^{2\pi|k|_1(s-\sigma)} = e^{-2\pi|k|_1\sigma} e^{2\pi|k|_1 s} \le e^{-2\pi K\sigma} e^{2\pi|k|_1 s}$. Sum over k with $|k|_1 > K$ and take the supremum over I. Also $u = \sum_k \hat{u}_k \chi_k$ (Lemma 5.3(d)) gives $u - T_K u = P_{\{|k|_1 > K\}} u$. □

5.9. **The stronger norm ♯ (sum of coefficient suprema).** Track P's averaging uses $\|f\|_{r,\Sigma} = \sum_k \sup_I |f_k(I)| e^{\Sigma|k|_1}$. With $s = \Sigma/(2\pi)$ this is, in our notation,

$$|u|^\sharp_{r,s} = \sum_{k\in\mathbb{Z}^n} \sup_{I \in V_r D} |\hat{u}_k(I)|_\infty \, e^{2\pi|k|_1 s} \ \in [0,\infty].$$

Always $|u|_{r,s} \le |u|^\sharp_{r,s}$; the reverse fails by an unbounded factor on disconnected D (on M disjoint action components carry one distinct mode each: $|\cdot|_\sharp = 1$, $|\cdot|^\sharp_\sharp = M$), so the two norms are *not* interchangeable and a bound for one does not give a bound for the other. *Width conversion for track P:* every width and loss of P (including a non-loss width $\bar{s}$) is converted by $s = \Sigma/(2\pi)$, $\sigma = \Delta\Sigma/(2\pi)$; the action losses are unchanged. Then the angular Cauchy constant $1/(e\sigma)$ reads $2\pi/(e\,\Delta\Sigma)$, the tail factor $e^{-2\pi K\sigma}$ reads $e^{-K\Delta\Sigma}$, and the composition parameter $\eta = 2|\phi|/(\rho\sigma)$ reads $4\pi|\phi|^\sharp/(\rho\,\Delta\Sigma)$.

Lemma 5.10 (Variants for ♯)**.** *All of Lemma 5.4(a),(b), Lemma 5.5, Lemma 5.6, Lemma 5.7 and Lemma 5.9 hold with $|\cdot|_\sharp$ replaced by $|\cdot|^\sharp_\sharp$, with the same constants. In Lemma 5.6(a) the stronger* global *form holds: $\sum_j |\partial_{\theta_j} u|^\sharp_{r_0,s} \le |u|^\sharp_{r_0,s_0}/(e(s_0 - s))$. For finite-norm u the series $\sum_k \hat{u}_k \chi_k$ of Lemma 5.3(c) may be taken with $S^\sharp := \sum_k \sup_I |a_k| e^{2\pi|k|_1 s} < \infty$ (the hypothesis is then stronger, and implies the one used there).*
Source: Ours; the blueprint's reading of Pos93 App. B for the norm of track P's two-scale iteration.

Proof. In every proof above the supremum over I is taken of a sum over k; for ♯ it is taken inside, coefficient by coefficient, and each step survives.

Step S5.20: B.1. (a) $|u(\theta,I)| \le \sum_k |\hat{u}_k(I)| e^{2\pi|k|_1 s} \le |u|^\sharp_{r,s}$. (b) the decay $\sup_I |\hat{u}_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ is uniform in I, so $|u|^\sharp_{r,s} \le \coth(\pi\sigma)^n M$.

Step S5.21: B.2. $\sup_I |\sum_l \hat{u}_{k-l}(I)\hat{v}_l(I)| \le \sum_l \sup_I |\hat{u}_{k-l}| \sup_I |\hat{v}_l|$, and then the weights as in Step S5.11.

Step S5.22: B.3. (a): $\sum_j \sup_I |2\pi k_j \hat{u}_k(I)| = 2\pi|k|_1 \sup_I |\hat{u}_k|$ (the sum over j is of the coefficient factors $|k_j|$ only), then Step S5.12 coefficientwise. (b): $\sup_{I\in V_r D} |\partial_{I_i} \hat{u}_k(I)| \le (r_0 - r)^{-1} \sup_{I' \in V_{r_0} D} |\hat{u}_k(I')|$ by the Cauchy integral of Step S5.13; then sum over k.

Step S5.23: B.4. The convolution bound of the previous steps gives, coefficientwise in the suprema, $|\{u,v\}|^\sharp_{;} \le \sum_i |\partial_{\theta_i} u|^\sharp_{;} |\partial_{I_i} v|^\sharp_{;} + \dots$, and the global forms of (a),(b) bound the four factors as in Step S5.15.

Step S5.24: Projections. $|P_S u|^\sharp_{;} = \sum_{k\in S} \sup_I |\hat{u}_k| e^{2\pi|k|_1 s} \le |u|^\sharp_{;}$ and the tail estimate is the same. □

Remark 5.11 (Dictionary with Pöschel's normalisation and with Part 1). Pöschel works with 2π-periodic angles φ, characters $e^{ik\cdot\varphi}$, weight $e^{|k|_1 s_P}$ and losses (ρ_P, σ_P). Under the symplectic scaling $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (so $d\varphi \wedge dI_P = d\theta \wedge dI$; time and the values of Hamiltonians are unchanged) his objects become ours with $s_P = 2\pi s$, $\sigma_P = 2\pi\sigma$, $r_P = r/2\pi$, $\rho_P = \rho/2\pi$. Then $e^{|k|_1 s_P} = e^{2\pi|k|_1 s}$ and $\rho_P \sigma_P = \rho\sigma$; the quantities in which Lemmas B.3–B.5 and the Iterative Lemma are written (for instance $|\phi|/(\rho\sigma)$ and the norm $|W\cdot|$) are therefore the same numbers. The conversion of Part 1, $\tilde{H} = 2\pi H(\varphi/2\pi, I)$ with I unchanged (Lemma 2.8), gives the same smallness ratio: $\tilde{\phi}/(\rho\sigma_P) = 2\pi\phi/(\rho \cdot 2\pi\sigma) = \phi/(\rho\sigma)$. So a condition "$|\phi| < \rho\sigma/2$" means the same in all three conventions, and the 2π-conversion does not affect the constants of the Iterative Lemma beyond the scalings of Lemma 2.8(b) (that bookkeeping is Part 6's). Where `docs/fourier-conventions.md` writes the derivative loss as $2\pi/(e\delta)$ for coefficient radius s, Lemma 5.6(a) is the same statement for the geometric width: a coefficient-radius loss $\delta = 2\pi\sigma$ gives $2\pi/(e \cdot 2\pi\sigma) = 1/(e\sigma)$.

6. Canonical transformations: time-one maps and the Lie transform (shared by tracks N, P)

This part is the canonical-transformation toolkit of Pöschel 1993, Appendix A (pp. 211–213: Taylor's formula, the flow X^t_ϕ of a small Hamiltonian field, its displacement and derivative bounds) and Lemma B.5 (pp. 214–215), in the frozen conventions of Part 5 (period 1, geometric widths, max norms). Track P cites it as [N:`bpN:p5:...`]. Generating functions ϕ live on general domains $\mathcal{D}_{r,s} = T_s \times V_r D$ with $D \subset \mathbb{C}^n$ arbitrary (Definition 5.1); $m = n$ because a Hamiltonian structure is needed. The ODE existence, uniqueness and continuation used are those of Part 4 (general layer); the weighted-norm estimates are those of Part 5.

Remark 6.1 (Decision for Lemma B.5: the Lie series, not the sup-norm fallback). **Route.** `PLAN.md` left open whether $|u \circ X^1_\phi| \le (1 - 2|\phi|/(\rho\sigma))^{-1}|u|$ is proved by Pöschel's convergent series $u \circ X^1_\phi = \sum_h \frac{1}{h!} \operatorname{ad}^h_\phi u$ or by a sup-norm estimate of the displaced function and a conversion back with Lemma 5.4(b). *We take the series*, for three reasons. (1) The Normal Form Lemma iterates $N \approx Ks/(12\ln 2)$ times and every step feeds the *weighted* norm of f_+ into the next step. The fallback re-enters the weighted norm through $\coth(\pi\sigma)^n$ in every step; this factor exceeds 1 by a fixed amount, so the contraction $\varepsilon_{i+1} \le \frac{1}{4}\varepsilon_i$ of the Iterative Lemma would be destroyed and not merely perturbed in constants. (2) The fallback needs $u \circ X^1_\phi$ to be holomorphic and periodic *before* one can speak of its Fourier coefficients, that is, holomorphic dependence of the flow on the initial point (Lemma 6.4(b)), a statement not in Mathlib. The series route gets holomorphy of $u \circ X^1_\phi$ for free (it is a norm-convergent series of holomorphic functions); the dependence on initial data is then needed only for the claims on the map $\Phi = X^1_\phi$ itself (Proposition 6.13). (3) The ingredients of the series route are already labelled results: the bracket estimate (Lemma 5.7), a one-line induction (Lemma 6.9), a Taylor remainder along a solution of $\dot{z} = X_\phi(z)$ (Part 4 and Lemma 6.8) and a geometric series. **What is lost.** The equality $u \circ X^1_\phi = \sum_h \frac{1}{h!} \operatorname{ad}^h_\phi u$ is proved here for $\eta := 2|\phi|/(\rho\sigma) \le \frac{1}{2}$ instead of Pöschel's $\eta < 1$ (Theorem 6.10); the bound itself is $(1-\eta)^{-1}$ as in the source. For $\eta < 1$ the equality also holds, by real-analyticity of $t \mapsto u(X^t_\phi z)$ and the identity theorem, but real-analytic dependence on time is another external fact; since the frozen headline only asserts the existence of constants, Part 6 simply uses $|\phi| \le \rho\sigma/4$ in place of $|\phi| < \rho\sigma/2$.

6.1. **Setting.**

Definition 6.2 (Hamiltonian field, bracket, scaled norm)**.** Points of $\mathbb{C}^n \times \mathbb{C}^n$ are $z = (\theta, I)$ (angles first, as in the frozen field); $J = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}$, so $J^2 = -\mathbf{1}$, $J^\top = -J$. For u holomorphic on an open set let $\nabla u = (\partial_\theta u, \partial_I u)^\top$,

$$X_u = J\nabla u = (\partial_I u, -\partial_\theta u), \qquad \{u, v\} = \sum_{i=1}^{n} \big(\partial_{\theta_i} u\, \partial_{I_i} v - \partial_{I_i} u\, \partial_{\theta_i} v\big) = Du \cdot X_v = \nabla u^\top J \nabla v = -\{v, u\}.$$

X_u is the frozen Hamiltonian field (`KAM.hamiltonianVectorField`, Definition 1.3; for a real-valued function on real points see [K:`bp:p1:hamiltonian-real`]); the bracket is normalised so that $\frac{d}{dt} u(X_\phi^t(z)) = \{u, \phi\}(X_\phi^t(z))$ (Lemma 6.7). For $\rho, \sigma > 0$ the *scaled norm* on $\mathbb{C}^n \times \mathbb{C}^n$ is $\nu_{\rho,\sigma}(a, b) = \max(|a|_\infty/\sigma, |b|_\infty/\rho)$ (Pöschel's $|W\,\cdot|$, $W = \mathrm{diag}(\sigma^{-1}, \rho^{-1})$ in angle-first order); it is a complex norm, and its closed ball of radius c about $z = (\theta, I)$ is $\bar{P}(z; c\sigma, c\rho)$. Iterated brackets: $\mathrm{ad}_\phi^0 u = u$, $\mathrm{ad}_\phi^j u = \{\mathrm{ad}_\phi^{j-1} u, \phi\}$.

Source: Pos93 pp. 211–212 (flow X_ϕ^t, W), p. 214 (ad).

Source gap: We did not locate the printed definition of the bracket's sign in pp. 211–215; only the two-term structure (Lemma B.4) and $\{\phi, h\} = \omega \cdot \partial_\theta \phi$ for $h = h(I)$ (p. 211, homological equation, divisors $ik \cdot \omega$) are used. With the convention above $\{\phi, h\} = \partial_\theta \phi \cdot \partial_I h$, which is that equation.

The bracket of holomorphic periodic functions is holomorphic and periodic ([K:`bp:p1:holomorphic-regularity`]); $\mathrm{ad}_\phi^j u$ is holomorphic on the common domain of u and ϕ; $\{u, v\} = 0$ if u, v depend only on I.

Lemma 6.3 (Size of the field)**.** *Let $\phi \in \mathcal{O}(r, s)$ have finite norm, $0 < \rho < r$, $0 < \sigma < s$. On $\mathcal{D}_{r-\rho, s-\sigma}$: $|\partial_{I_i}\phi| \le |\phi|_{r,s}/\rho$, $|\partial_{\theta_i}\phi| \le |\phi|_{r,s}/(e\sigma)$, and*

$$\nu_{\rho,\sigma}(X_\phi) = \max\Big(\frac{\max_i |\partial_{I_i}\phi|}{\sigma}, \frac{\max_i |\partial_{\theta_i}\phi|}{\rho}\Big) \le \frac{|\phi|_{r,s}}{\rho\sigma}.$$

Source: Pos93 p. 212 ("$|\phi_I|_\infty \le q\delta/(\alpha\rho)$, $|WX_\phi| \le q\delta/(\alpha\rho\sigma)$").

Proof. **Step S6.1: Derivatives.** Lemma 5.6 (with $r_0 = r$, $s_0 = s$ and the target widths $r - \rho$, $s - \sigma$) gives, for each single index, $|\partial_{I_i}\phi|_{r-\rho,s} \le \rho^{-1}|\phi|_{r,s}$ and $|\partial_{\theta_j}\phi|_{r,s-\sigma} \le (e\sigma)^{-1}|\phi|_{r,s}$ (the individual global angular estimate, not a sum of global norms). Restrict to $\mathcal{D}_{r-\rho,s-\sigma}$ (monotonicity), and apply Lemma 5.4(a) to each derivative.

Step S6.2: The field. Insert into the definition of $\nu_{\rho,\sigma}$: $\frac{1}{\sigma} \cdot \frac{|\phi|}{\rho} = \frac{|\phi|}{\rho\sigma}$ and $\frac{1}{\rho} \cdot \frac{|\phi|}{e\sigma} \le \frac{|\phi|}{\rho\sigma}$. □

6.2. **The flow of a small holomorphic vector field.**

Lemma 6.4 (Flow of a small holomorphic field)**.** *Let ν be a complex norm on $\mathbb{C}^N$, $\Omega \subset \mathbb{C}^N$ open, $F : \Omega \to \mathbb{C}^N$ holomorphic with $\nu(F) \le \varepsilon$ on Ω and $0 \le \varepsilon \le \frac{1}{2}$. For $c > 0$ let $\Omega^c = \{z : \bar{B}_\nu(z, c) \subset \Omega\}$ (an open set).*

- *(a) (Existence, uniqueness, displacement) For $z \in \Omega^{1/2}$ there is $\gamma_z : \mathbb{R} \to \mathbb{C}^N$ with `IsSolIn` $F\,\Omega\,[-1,1]\,\gamma_z$ and $\gamma_z(0) = z$; any solution in Ω through z on an interval $J \ni 0$ equals γ_z on $J \cap [-1, 1]$. Put $X^t(z) = \gamma_z(t)$ ($|t| \le 1$). Then $\nu(X^t(z) - z) \le |t|\,\varepsilon$.*
- *(b) (Regularity) $(t, z) \mapsto X^t(z)$ is continuous on $[-1, 1] \times \Omega^{1/2}$; for each t the map $X^t : \Omega^{1/2} \to \Omega$ is holomorphic; $(t, z) \mapsto D_z X^t(z)$ is continuous, and $Y(t) = D_z X^t(z)$ satisfies $Y(t) = \mathbf{1} + \int_0^t DF(X^\tau z) Y(\tau)\, d\tau$.*
- *(c) (Inverse, derivative) For $z \in \Omega^1$: $X^1(z) \in \Omega^{1/2}$ and $X^{-1}(X^1 z) = z$; and the ν-operator norm satisfies $|DX^t(z) - \mathbf{1}|_\nu \le 2\varepsilon$ for $|t| \le 1$. In the same way $X^1(X^{-1} z) = z$.*

(d) *(Translations) If* $\tau \in \mathbb{C}^N$, $\Omega + \tau = \Omega$ *and* $F(w+\tau) = F(w)$ *on* Ω, *then* $\Omega^{1/2} + \tau = \Omega^{1/2}$ *and* $X^t(z+\tau) = X^t(z) + \tau$.

(e) *(Reality) Let* $N = 2n$ *and suppose* $F(x) \in \mathbb{R}^N$ *for* $x \in \Omega \cap \mathbb{R}^N$. *Then* $X^t(x) \in \mathbb{R}^N$ *for* $x \in \Omega^{1/2} \cap \mathbb{R}^N$.

Source: Pos93 p. 212 ("X^t_ϕ *maps* $V_{r-\frac{3}{2}\rho, s-\frac{3}{2}\sigma}$ *into* $V_{r-\rho,s-\sigma}$ *for* $0 \le t \le 1$ *and satisfies* $|W(X^t_\phi - \mathrm{id})| \le \ldots$*"; the derivative estimate by Lemma C.1). The general-*ν *formulation and the holomorphy in (b) are ours (Pöschel asserts "real analytic").*

Proof. Regard $\mathbb{C}^N$ as a real normed space of dimension $2N$ with norm ν, extend F by 0 outside Ω (Remark 4.1); F is C^1 on Ω ([K:`bp:p1:holomorphic-regularity`], restriction of scalars).

Step S6.3: (a) The maximal solution stays in a compact set. Let $z \in \Omega^{1/2}$, (T_-, T_+) and γ the maximal interval and solution (Theorem 4.13). For $t \in [0, T_+)$, $\nu(\gamma'(t)) \le \varepsilon$, so $\nu(\gamma(t) - z) \le \varepsilon t$ ([K:`bp:p1:ext-mvt`] in one variable). If $T_+ \le 1$, then $\gamma(t) \in K = \bar{B}_\nu(z, \varepsilon) \subset \bar{B}_\nu(z, \frac{1}{2}) \subset \Omega$ for $t \in [0, T_+)$, with K compact, so $T_+ = \infty$ by Corollary 4.15, a contradiction. Hence $T_+ > 1$; symmetrically (time reversal, Lemma 4.3(b)) $T_- > -1$. Uniqueness on an interval $J \ni 0$ is Theorem 4.11; the displacement bound was shown.

Step S6.4: (b) Picard iterates. Fix $z_0 \in \Omega^{1/2}$. As $\bar{B}_\nu(z_0, \frac{1}{2})$ is compact in the open Ω, there is $\eta > 0$ with $\bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$. Let $\Lambda = \sup\{|DF(w)|_\nu : w \in \bar{B}_\nu(z_0, \frac{1}{2} + \eta)\} < \infty$; F is Λ-Lipschitz on this convex ball ([K:`bp:p1:ext-mvt`]). For $(t, z) \in [-1, 1] \times B_\nu(z_0, \eta)$ put $y_0(t, z) = z$, $y_{k+1}(t, z) = z + \int_0^t F(y_k(\tau, z))\, d\tau$. By induction on k: $\nu(y_k(t, z) - z) \le |t|\varepsilon$ (so $y_k(t, z) \in \bar{B}_\nu(z, \varepsilon) \subset \bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$, and the integral is defined); y_k is continuous; $y_k(t, \cdot)$ is holomorphic; $D_z y_k$ is continuous in (t, z). The last two follow from (E1) applied to $t \int_0^1 F(y_k(t\tau', z))\, d\tau'$, whose z-derivative is $t \int_0^1 DF(y_k) D_z y_k\, d\tau'$.

Step S6.5: (b) Convergence and identification. $\nu(y_{k+1} - y_k)(t, z) \le \varepsilon \Lambda^k |t|^{k+1}/(k+1)!$ by induction, so $y_k \to y_\infty$ uniformly on $[-1, 1] \times B_\nu(z_0, \eta)$. The limit is continuous and satisfies $y_\infty(t, z) = z + \int_0^t F(y_\infty)\, d\tau$ (F is Λ-Lipschitz), so $y_\infty(\cdot, z)$ is C^1 with derivative $F(y_\infty)$, i.e. a solution in Ω on $[-1, 1]$ through z, and $y_\infty(t, z) = X^t(z)$ by (a). Hence X^t is continuous in (t, z) and, by (E2), holomorphic on $B_\nu(z_0, \eta)$ with $D_z y_k \to D_z X^t$ uniformly on $[-1, 1] \times B_\nu(z_0, \eta/2)$, so $D_z X^t(z)$ is jointly continuous. As z_0 was arbitrary, (b) holds except for the integral equation. Differentiating $X^t(w) = w + \int_0^t F(X^\tau w)\, d\tau$ in w under the integral ((E1); the integrand's w-derivative $DF(X^\tau w) D_w X^\tau(w)$ is jointly continuous) gives the equation for Y.

Step S6.6: (c). Let $z \in \Omega^1$, $w = X^1(z)$. Then $\nu(w - z) \le \varepsilon \le \frac{1}{2}$, so $\bar{B}_\nu(w, \frac{1}{2}) \subset \bar{B}_\nu(z, 1) \subset \Omega$, i.e. $w \in \Omega^{1/2}$. The map $t \mapsto \gamma_z(1 + t)$ is a solution on $[-1, 0]$ through w, so it equals γ_w there (Theorem 4.11) and $X^{-1}(w) = \gamma_z(0) = z$. The statement for $X^1 \circ X^{-1}$ is this one for $-F$ (Lemma 4.3(b)). For the derivative: $B_\nu(z, \frac{1}{2}) \subset \Omega^{1/2}$ (as above), $G = X^t - \mathrm{id}$ is holomorphic there by (b) with $\nu(G) \le \varepsilon$ by (a); Lemma 5.8 with $R = \frac{1}{2}$ gives $|DG(z)|_\nu \le 2\varepsilon$.

Step S6.7: (d),(e). (d): $\Omega^{1/2} + \tau = \Omega^{1/2}$ since the ν-balls translate; $t \mapsto \gamma_z(t) + \tau$ is a solution through $z + \tau$ (Lemma 4.3(c)), so equals $\gamma_{z+\tau}$ by uniqueness. (e): apply (a) to $E = \mathbb{R}^N$, $W = \Omega \cap \mathbb{R}^N$, the real field $F|_W$ (it is C^1 and real by hypothesis, [K:`bp:p1:real-restriction`]) and the compact set $\bar{B}_\nu(x, \varepsilon) \cap \mathbb{R}^N \subset W$: this gives a real solution on $[-1, 1]$ through x, which is also a solution in Ω, hence equals γ_x. □

Lemma 6.5 (Symplectic flow)**.** *In Lemma 6.4 let* $N = 2n$, $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, *and* $F = X_\phi$ *with* ϕ *holomorphic on* Ω. *Then for* $z \in \Omega^{1/2}$, $|t| \le 1$, $Y = DX^t(z)$ *satisfies*

$$Y^\top J Y = J, \qquad Y J Y^\top = J, \qquad \det Y = \pm 1$$

($Y^\top$ the plain transpose of the complex matrix).

Source: Standard (Hamiltonian flows are symplectic; used implicitly by Pos93 pp. 211–212, "canonical transformation Φ").

Proof. **Step S6.8: Variational equation.** By Lemma 6.4(b), $Y \in C^1$ with $Y' = DF(X^t z)Y$, $Y(0) = \mathbf{1}$.

Step S6.9: The field. $F = J\nabla\phi$, so $DF = JS$ with $S = D\nabla\phi$ the Hessian, symmetric ([K:`bp:p1:ext-schwarz`]). Then $DF^\top J = S^\top J^\top J = S$ and $J\,DF = J^2 S = -S$, so $DF^\top J + J\,DF = 0$.

Step S6.10: Conservation. $\frac{d}{dt}(Y^\top JY) = Y^\top(DF^\top J + J\,DF)Y = 0$, so $Y^\top JY = J$ for all t. Taking determinants, $\det(Y)^2 = 1$; Y is invertible and inverting $Y^\top JY = J$ gives $Y^{-1}J^{-1}Y^{-\top} = J^{-1}$, that is, $J = YJY^\top$ ($J^{-1} = -J$). □

Corollary 6.6 (Canonical maps: bracket invariance and conjugation)**.** *In the setting of Lemma 6.5, let $z \in \Omega^{1/2}$, $\Phi = X^t$ ($|t| \le 1$), and u, v, H holomorphic on a neighbourhood of $\Phi(z)$. Then $u \circ \Phi$ is holomorphic near z and*

(a) $\{u \circ \Phi, v \circ \Phi\}(z) = \{u, v\}(\Phi(z))$;
(b) $D\Phi(z)\, X_{H\circ\Phi}(z) = X_H(\Phi(z))$;
(c) if $y : J \to \Omega^{1/2}$ satisfies $\dot y = X_{H\circ\Phi}(y)$ with H holomorphic on a neighbourhood of $\Phi(y(J))$, then $x = \Phi \circ y$ satisfies $\dot x = X_H(x)$.

Source: Standard.

Proof. **Step S6.11: Chain rule.** Φ is holomorphic (Lemma 6.4(b)) and $\nabla(u\circ\Phi) = D\Phi^\top(\nabla u)\circ\Phi$.

Step S6.12: (a). $\{u\circ\Phi, v\circ\Phi\} = \nabla u^\top\, YJY^\top\, \nabla v = \nabla u^\top J \nabla v$ at $\Phi(z)$ by Lemma 6.5.

Step S6.13: (b),(c). $D\Phi\, X_{H\circ\Phi} = Y\,J\,Y^\top(\nabla H)\circ\Phi = J(\nabla H)\circ\Phi = X_H \circ \Phi$; (c) is the chain rule $\dot x = D\Phi(y)\dot y = D\Phi(y) X_{H\circ\Phi}(y)$. □

For real data: if ϕ is real on real points then so is $F = X_\phi$ (Step S6.7), Φ maps real points to real points, its derivative at a real point is a real matrix ([K:`bp:p1:real-restriction`]), and (a)–(c) hold for real u, v, H with the real restriction of X ([K:`bp:p1:hamiltonian-real`]).

6.3. Taylor's formula along the flow.

Lemma 6.7 (Derivative along the flow)**.** *Let $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, ϕ, $F = X_\phi$, $\varepsilon \le \frac12$ as in Lemma 6.5; let $\Omega_1 \subset \Omega$ be open and u holomorphic on Ω_1, and $z \in \Omega^{1/2}$ with $X^t(z) \in \Omega_1$ for $|t| \le 1$. Then $g(t) = u(X^t(z))$ is C^∞ on $[-1,1]$ and $g^{(j)}(t) = (\mathrm{ad}_\phi^j u)(X^t(z))$ for all $j \ge 0$ (each $\mathrm{ad}_\phi^j u$ is holomorphic on Ω_1).*

Source: Pos93 p. 211 ("Taylor's formula" for $h \circ X_\phi^t$); standard.

Proof. **Step S6.14: One derivative.** For v holomorphic on Ω_1, v is real-differentiable with derivative Dv, and $\gamma = X^\cdot(z)$ has derivative $X_\phi(\gamma)$, so $\frac{d}{dt} v(\gamma(t)) = Dv(\gamma)\cdot X_\phi(\gamma) = \{v,\phi\}(\gamma(t))$ (Definition 6.2), which is continuous in t.

Step S6.15: Induction. Apply the previous step to $v = \mathrm{ad}_\phi^j u$, holomorphic on Ω_1 (derivatives of holomorphic functions, [K:`bp:p1:holomorphic-regularity`]). □

Lemma 6.8 (Taylor with integral remainder along the flow)**.** *In the setting of Lemma 6.7, for every $N \ge 1$,*

$$u(X^1 z) = \sum_{j=0}^{N-1} \frac{1}{j!}(\mathrm{ad}_\phi^j u)(z) + R_N(z),$$

$$R_N(z) = \int_0^1 \frac{(1-t)^{N-1}}{(N-1)!}(\mathrm{ad}_\phi^N u)(X^t z)\,dt, \qquad |R_N(z)| \le \frac{1}{N!}\max_{0\le t\le 1}|(\mathrm{ad}_\phi^N u)(X^t z)|.$$

In particular ($N = 1$ for f_0, $N = 2$ for h), if h, f_0 are holomorphic on Ω_1 and the trajectory of z stays in Ω_1,

$$(h + f_0)(X^1 z) = h(z) + \{h, \phi\}(z) + f_0(z) + \int_0^1 \big\{(1-t)\{h,\phi\} + f_0,\ \phi\big\}(X^t z)\, dt. \tag{1}$$

Source: Pos93 p. 211, displayed formula for $H_0 \circ \Phi$.

Proof. **Step S6.16: Taylor.** Apply the Taylor formula with integral remainder for C^N functions of one real variable (external fact: induction on N with one integration by parts per step, [K:`bp:p 1:ext-integral`]) to $g(t) = u(X^t z)$ at 0, evaluated at 1, using $g^{(j)}(0) = (\mathrm{ad}_\phi^j u)(z)$ and $g^{(N)}(t) = (\mathrm{ad}_\phi^N u)(X^t z)$ (Lemma 6.7). The bound uses $\int_0^1 (1-t)^{N-1}/(N-1)!\, dt = 1/N!$.

Step S6.17: The displayed formula. Apply the case $N = 2$ to h ($\mathrm{ad}^2 h = \{\{h,\phi\},\phi\}$) and the case $N = 1$ to f_0 and add: the remainders are $\int_0^1 (1-t)\{\{h,\phi\},\phi\}(X^t z)\, dt + \int_0^1 \{f_0,\phi\}(X^t z)\, dt$, and brackets are bilinear. □

6.4. **Lemma B.5: composition in the weighted norm.**

Lemma 6.9 (Iterated brackets)**.** *Let $\phi \in \mathcal{O}(r', s')$, $u \in \mathcal{O}(r, s)$ have finite norms, $0 < \rho < r$, $0 < \sigma < s$, $r + \rho \le r'$, $s + \sigma \le s'$, and $\eta = 2|\phi|_{r',s'}/(\rho\sigma)$. Then for every $h \ge 1$,*

$$|\mathrm{ad}_\phi^h u|_{r-\rho,s-\sigma} \le \Big(\frac{2h^2}{e\rho\sigma}\Big)^h \frac{h!}{(2h)!}\, |\phi|_{r',s'}^h |u|_{r,s}, \qquad \textit{hence} \qquad \frac{1}{h!}|\mathrm{ad}_\phi^h u|_{r-\rho,s-\sigma} \le \Big(\frac{e\eta}{4}\Big)^h |u|_{r,s}.$$

Source: Pos93 p. 215 (the displayed estimates for $|\mathrm{ad}_\phi^h u|$ with $\tilde\rho = \rho/h$, $\tilde\sigma = \sigma/h$).

Proof. **Step S6.18: Setup.** Fix $h \ge 1$, $\kappa = \rho/h$, $\lambda = \sigma/h$, $r_i = r - i\kappa$, $s_i = s - i\lambda$ ($0 \le i \le h$), so $r_h = r - \rho$, $s_h = s - \sigma$. By induction each $\mathrm{ad}_\phi^i u \in \mathcal{O}(r_i, s_i)$.

Step S6.19: One step. Let $1 \le i \le h$. Since $\mathrm{ad}_\phi^i u = \{\mathrm{ad}_\phi^{i-1} u, \phi\} = -\{\phi, \mathrm{ad}_\phi^{i-1} u\}$, Lemma 5.7 with $(u, v; r_0, s_0; r, s; \rho, \sigma) \to (\phi, \mathrm{ad}_\phi^{i-1} u; r', s'; r_{i-1}, s_{i-1}; \kappa, \lambda)$ applies (note $\kappa < r_{i-1}$, $r_i < r'$, and likewise for s) and gives $|\mathrm{ad}_\phi^i u|_{r_i,s_i} \le \frac{1}{e}\Big(\frac{1}{(r'-r_{i-1}+\kappa)\lambda} + \frac{1}{(s'-s_{i-1}+\lambda)\kappa}\Big)|\phi|_{r',s'}|\mathrm{ad}_\phi^{i-1} u|_{r_{i-1},s_{i-1}}$. Now $r' - r_{i-1} + \kappa = (r'-r) + i\kappa \ge \rho + i\kappa = (h+i)\kappa$ and likewise $s' - s_{i-1} + \lambda \ge (h+i)\lambda$; so the bracket is $\le \frac{2}{e(h+i)\kappa\lambda}$.

Step S6.20: Product. Multiplying over $i = 1, \dots, h$ and using $\prod_{i=1}^h (h+i) = (2h)!/h!$ and $\kappa\lambda = \rho\sigma/h^2$ gives the first estimate. For the second, $(2h)! \ge (2h/e)^{2h}$ (from $e^N \ge N^N/N!$), so $\frac{1}{h!}|\mathrm{ad}_\phi^h u| \le \frac{(\eta h^2/e)^h}{(2h)!}|u| \le \big(\eta\frac{h^2}{e}\cdot\frac{e^2}{4h^2}\big)^h |u| = (\frac{e\eta}{4})^h |u|$. □

Theorem 6.10 (Lemma B.5: $|u \circ X_\phi^1| \le (1-\eta)^{-1}|u|$)**.** *Let $D \subset \mathbb{C}^n$, $0 < \rho < r \le r' - \rho$, $0 < \sigma < s \le s' - \sigma$, $\phi \in \mathcal{O}(r', s')$ with finite norm and $\eta = 2|\phi|_{r',s'}/(\rho\sigma) \le \frac{1}{2}$. Put $\Omega = \mathcal{D}_{r'-\rho,s'-\sigma} \supset \mathcal{D}_{r,s}$ and $\nu = \nu_{\rho,\sigma}$.*

(i) $\nu(X_\phi) \le \eta/2 \le \frac{1}{4}$ on Ω; the flow $X^t = X_\phi^t$, $|t| \le 1$, of Lemma 6.4 is defined on $\Omega^{1/2} \supset \mathcal{D}_{r-\rho,s-\sigma}$, with $X^t(\mathcal{D}_{r-\rho,s-\sigma}) \subset \mathcal{D}_{r,s}$.

(ii) For every $u \in \mathcal{O}(r, s)$ with finite norm, $\sum_{h\ge0} \frac{1}{h!}\mathrm{ad}_\phi^h u$ converges in $|\cdot|_{r-\rho,s-\sigma}$ to a function $S \in \mathcal{O}(r-\rho, s-\sigma)$ and $S = u \circ X_\phi^1$ on $\mathcal{D}_{r-\rho,s-\sigma}$. In particular $u \circ X_\phi^1 \in \mathcal{O}(r-\rho, s-\sigma)$ and

$$|u \circ X_\phi^1|_{r-\rho,s-\sigma} \le \Big(1 - \frac{e\eta}{4}\Big)^{-1} |u|_{r,s} \le \Big(1 - \frac{2}{\rho\sigma}|\phi|_{r',s'}\Big)^{-1} |u|_{r,s}.$$

(iii) The same holds for X_ϕ^t, $0 \le t \le 1$ (and $-1 \le t \le 0$), with η replaced by $|t|\eta$.

(iv) If ϕ and u are real on real points, $u \circ X_\phi^1$ is real on real points of $\mathcal{D}_{r-\rho,s-\sigma}$.

Source: Pos93 pp. 214–215, Lemma B.5 (with $|\phi|_{r_0,s_0} < \rho\sigma/2$ there).
Source gap: Pöschel writes "Consider the Lie series expansion $u\circ X^1_\phi = \sum \frac{1}{h!}\operatorname{ad}^h_\phi u$" and estimates the series; the equality with the composition is not proved on p. 214. It is proved here for $\eta \le \frac12$; see Remark 6.1. The remark that the smallness condition implies $X^t_\phi : V_{r-\rho,s-\sigma} \to V_{r,s}$ is (i).

Proof. **Step S6.21: The field and the domain.** Since $r \le r'-\rho$, $s \le s'-\sigma$: $\mathcal{D}_{r,s} \subset \Omega$. Lemma 6.3 (for $\phi \in \mathcal{O}(r',s')$ and losses ρ,σ) gives $\nu(X_\phi) \le |\phi|/(\rho\sigma) = \eta/2 =: \varepsilon \le \frac14$ on Ω. If $z \in \mathcal{D}_{r-\rho,s-\sigma}$, then $\bar B_\nu(z,\frac12) = \bar P(z;\sigma/2,\rho/2) \subset \mathcal{D}_{r-\rho/2,s-\sigma/2} \subset \mathcal{D}_{r,s} \subset \Omega$ (Definition 5.1(b)), so $z \in \Omega^{1/2}$. By Lemma 6.4(a), $X^t(z) \in \bar P(z;\varepsilon\sigma,\varepsilon\rho) \subset \mathcal{D}_{r-(1-\varepsilon)\rho,s-(1-\varepsilon)\sigma} \subset \mathcal{D}_{r,s}$. This is (i).

Step S6.22: The series. By Lemma 6.9 (applicable: $r+\rho \le r'$, $s+\sigma \le s'$), $\frac{1}{h!}\operatorname{ad}^h_\phi u \in \mathcal{O}(r-\rho,s-\sigma)$ has norm $\le (e\eta/4)^h|u|_{r,s}$, and $e\eta/4 < 1$. The sum of norms is $\le (1-e\eta/4)^{-1}|u|_{r,s}$. By Lemma 5.4(a) the partial sums converge uniformly on $\mathcal{D}_{r-\rho,s-\sigma}$, so S is holomorphic ((E2)) and periodic, and termwise integration ((E3)) shows that its Fourier coefficients are the sums of the coefficients; the triangle inequality in $\sum_k |\hat S_k(I)|e^{2\pi|k|_1(s-\sigma)}$ gives $|S|_{r-\rho,s-\sigma} \le (1-e\eta/4)^{-1}|u|_{r,s}$; and $1-e\eta/4 \ge 1-\eta$.

Step S6.23: The remainder tends to 0. Let $z \in \mathcal{D}_{r-\rho,s-\sigma}$, $t \in [0,1]$, $w = X^t(z)$. By Step S6.21, $w \in \mathcal{D}_{r-\hat\rho,s-\hat\sigma}$ with $\hat\rho = (1-\varepsilon)\rho$, $\hat\sigma = (1-\varepsilon)\sigma$, $0 < \hat\rho < r$, $r+\hat\rho \le r'$, and likewise for s. Lemma 6.9 with losses $(\hat\rho,\hat\sigma)$ and $\hat\eta = 2|\phi|/(\hat\rho\hat\sigma) = \eta/(1-\varepsilon)^2 \le \frac{16}{9}\eta \le \frac89$ gives $|\operatorname{ad}^N_\phi u|_{r-\hat\rho,s-\hat\sigma} \le N!(e\hat\eta/4)^N|u|_{r,s}$, and $e\hat\eta/4 \le \frac{2e}{9} < 1$. By Lemma 5.4(a), $|(\operatorname{ad}^N_\phi u)(X^t z)| \le N!(e\hat\eta/4)^N|u|_{r,s}$. Lemma 6.8 (with $\Omega_1 = \mathcal{D}_{r,s}$, Ω_1-trajectory by Step S6.21) gives $|R_N(z)| \le (e\hat\eta/4)^N|u|_{r,s} \to 0$. Hence $u(X^1 z) = \lim_N \sum_{j<N}\frac{1}{j!}(\operatorname{ad}^j_\phi u)(z) = S(z)$. This proves (ii).

Step S6.24: (iii), (iv). $X^t_\phi = X^1_{t\phi}$ (the field $X_{t\phi} = tX_\phi$ and $s \mapsto X^{ts}_\phi(z)$ solves it), and $|t\phi| = |t||\phi|$, so (iii) is (ii) for $t\phi$. (iv): Lemma 6.4(e) with $F = X_\phi$ real on real points (a real-valued ϕ on real points has real partials there). □

Proposition 6.11 (Lie transform with first-order remainder, in the weighted norm)**.** *In the setting of Theorem 6.10, let $h \in \mathcal{O}(r,s)$ (any norm, possibly ∞) with $w := \{h,\phi\} \in \mathcal{O}(r,s)$ of finite norm, and $f_0 \in \mathcal{O}(r,s)$ with finite norm. Then on $\mathcal{D}_{r-\rho,s-\sigma}$*

$$(h+f_0)\circ X^1_\phi = h+w+f_0+\mathcal{R}, \qquad \mathcal{R} = \sum_{j\ge1}\Big(\frac{1}{(j+1)!}\operatorname{ad}^j_\phi w + \frac{1}{j!}\operatorname{ad}^j_\phi f_0\Big) = \int_0^1 \big\{(1-t)w+f_0,\phi\big\}\circ X^t_\phi\,dt,$$

the series converges in $|\cdot|_{r-\rho,s-\sigma}$, $\mathcal{R} \in \mathcal{O}(r-\rho,s-\sigma)$, and

$$|\mathcal{R}|_{r-\rho,s-\sigma} \le \sum_{j\ge1}\Big(\frac{e\eta}{4}\Big)^j\Big(\frac{|w|_{r,s}}{j+1}+|f_0|_{r,s}\Big) \le \frac{\eta}{1-\eta}\Big(\tfrac12|w|_{r,s}+|f_0|_{r,s}\Big).$$

The integral form is Lemma 6.8, (1); the pointwise identity requires no bound on h itself.
Source: Pos93 p. 211 (the decomposition $H_0\circ\Phi = h + \{h,\phi\} + f_0 + \int\ldots$) and p. 212–213 (estimate of f_+); the series form of the remainder is ours.

Proof. **Step S6.25: Derivatives of h.** Since $\operatorname{ad}^1_\phi h = w$, $\operatorname{ad}^j_\phi h = \operatorname{ad}^{j-1}_\phi w$ $(j \ge 1)$. In Lemma 6.7 (with $\Omega_1 = \mathcal{D}_{r,s}$) $g(t) = h(X^t z)$ has $g^{(j)}(t) = (\operatorname{ad}^{j-1}_\phi w)(X^t z)$; Taylor's formula to order N gives $h(X^1 z) = h(z) + \sum_{i=0}^{N-2}\frac{1}{(i+1)!}(\operatorname{ad}^i_\phi w)(z) + R'_N$, with $|R'_N| \le \frac{1}{N!}\max_t|(\operatorname{ad}^{N-1}_\phi w)(X^t z)|$. By Step S6.23 (Lemma 6.9 with $u = w$ and the losses $\hat\rho,\hat\sigma$) this is $\le \frac{(N-1)!}{N!}(e\hat\eta/4)^{N-1}|w|_{r,s} \to 0$. So $h\circ X^1 = h + \sum_{i\ge0}\frac{1}{(i+1)!}\operatorname{ad}^i_\phi w$ pointwise, the term $i=0$ being w.

Step S6.26: Derivatives of f_0. By Theorem 6.10(ii), $f_0\circ X^1 = \sum_{j\ge0}\frac{1}{j!}\operatorname{ad}^j_\phi f_0$.

Step S6.27: Sum and bound. Adding gives the first identity with $\mathcal{R}$ as stated (terms $j=0$ are w, f_0). Lemma 6.9 bounds $\frac{1}{j!}|\mathrm{ad}_\phi^j w| \le (e\eta/4)^j|w|$, hence $\frac{1}{(j+1)!}|\mathrm{ad}_\phi^j w| \le \frac{1}{j+1}(e\eta/4)^j|w|$, and similarly for f_0. The geometric sum with $\frac{1}{j+1} \le \frac{1}{2}$ and $e\eta/4 \le \eta$ gives the last bound. Convergence in the norm and membership of $\mathcal{O}(r-\rho, s-\sigma)$ are as in Step S6.22.

Step S6.28: Integral form. Equality with the integral is (1), because $\{h,\phi\} = w$; the trajectory stays in $\mathcal{D}_{r,s}$ by Step S6.21. □

Corollary 6.12 (Integral remainders in the weighted norm)**.** *In the setting of Theorem 6.10 let $v_t \in \mathcal{O}(r,s)$ $(0 \le t \le 1)$ have finite norms with $V = \sup_t |v_t|_{r,s} < \infty$, and suppose $(t,z) \mapsto v_t(z)$ and $(t,z) \mapsto Dv_t(z)$ are continuous on $[0,1] \times \mathcal{D}_{r,s}$. Then $G(z) = \int_0^1 v_t(X_\phi^t z)\,dt$ lies in $\mathcal{O}(r-\rho, s-\sigma)$ and $|G|_{r-\rho,s-\sigma} \le (1-\eta)^{-1}V$.*
Source: Pos93 p. 212–213 ("$|u \circ X_\phi^t| \le (1-2|\phi|/\rho\sigma)^{-1}|u|$ uniformly for $0 \le t \le 1$", applied inside the integral).

Proof. **Step S6.29: Integrand.** $g_t = v_t \circ X^t \in \mathcal{O}(r-\rho, s-\sigma)$ with $|g_t| \le (1-t\eta)^{-1}V \le (1-\eta)^{-1}V$ (Theorem 6.10(iii)). The map $(t,z) \mapsto g_t(z)$ is continuous and $D_z g_t = Dv_t(X^t z) D_z X^t(z)$ is continuous (Lemma 6.4(b)).

Step S6.30: Holomorphy and coefficients. By (E1), G is holomorphic; it is periodic. Fubini on $[0,1] \times [0,1]^n$ gives $\hat{G}_k(I) = \int_0^1 (\hat{g_t})_k(I)\,dt$, so by Tonelli $|G|_{I,s-\sigma} \le \int_0^1 |g_t|_{I,s-\sigma}\,dt \le (1-\eta)^{-1}V$. □

6.5. The canonical map $\Phi = X_\phi^1$ on weighted domains.

Proposition 6.13 (Iterative-Lemma package for Φ)**.** *Let $D \subset \mathbb{C}^n$, $\phi \in \mathcal{O}(r,s)$ with finite norm, $0 < \rho < r/2$, $0 < \sigma < s/2$, and $\varepsilon := |\phi|_{r,s}/(\rho\sigma) \le \frac{1}{2}$. Put $\Omega = \mathcal{D}_{r-\rho,s-\sigma}$, $\nu = \nu_{\rho,\sigma}$, $\Phi = X_\phi^1$, $\Psi = X_\phi^{-1}$ (Lemma 6.4). Then:*

- *(a) $\nu(X_\phi) \le \varepsilon$ on Ω; Φ, Ψ are defined and holomorphic on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma} \subset \Omega^{1/2}$ and map it into Ω. For every $t \in [-1,1]$, $\nu(X^t - \mathrm{id}) \le |t|\varepsilon$; explicitly $|\Phi_\theta - \theta|_\infty \le |\phi|_{r,s}/\rho$ and $|\Phi_I - I|_\infty \le |\phi|_{r,s}/(e\sigma)$, and the same for Ψ.*
- *(b) $\Phi(\theta + k, I) = \Phi(\theta, I) + (k, 0)$ for $k \in \mathbb{Z}^n$; if ϕ is real on real points, Φ maps real points to real points.*
- *(c) (Derivative) on $\mathcal{D}_{r-2\rho,s-2\sigma}$: $|D\Phi - \mathbf{1}|_\nu \le 2\varepsilon$, i.e. $|W(D\Phi - \mathbf{1})W^{-1}| \le 2|\phi|_{r,s}/(\rho\sigma)$ for $W = \mathrm{diag}(\sigma^{-1}, \rho^{-1})$.*
- *(d) (Canonical) $(D\Phi)^\top J\, D\Phi = J$, and Corollary 6.6 holds for Φ.*
- *(e) (Inverse) $\Psi \circ \Phi = \mathrm{id} = \Phi \circ \Psi$ on $\mathcal{D}_{r-2\rho,s-2\sigma}$, and $\Phi(\mathcal{D}_{r-2\rho,s-2\sigma}) \subset \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$.*

Source: Pos93 p. 212 ("Equivalently, X_ϕ satisfies $|WX_\phi|_{\mathcal{P}} \le \frac{q\delta}{\alpha\rho\sigma} \le \frac{1}{2}$ … X_ϕ^t maps $V_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ into $V_{r-\rho,s-\sigma}$ … $|W(DX_\phi^t - I_{2n})W^{-1}| \le 2\frac{q\delta}{\alpha\rho\sigma}$ uniformly on $V_{r-2\rho,s-2\sigma}$"), with $|\phi|_{r,s}$ in place of $q\delta/\alpha$.

Proof. **Step S6.31: Field.** Lemma 6.3 gives $\nu(X_\phi) \le \varepsilon \le \frac{1}{2}$ on Ω, and Ω is open. For $z \in \mathcal{D}_{r-c\rho,s-c\sigma}$ $(c \ge \frac{3}{2})$ the ball $\bar{B}_\nu(z, c-1) = \bar{P}(z; (c-1)\sigma, (c-1)\rho)$ lies in Ω (Definition 5.1(b)): so $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma} \subset \Omega^{1/2}$ and $\mathcal{D}_{r-2\rho,s-2\sigma} \subset \Omega^1$.

Step S6.32: (a),(b). Lemma 6.4(a),(b),(d),(e) (with $\tau = (k,0)$: $\Omega + \tau = \Omega$ and X_ϕ is periodic, as ϕ is). $X^t(z) \in \bar{P}(z; \varepsilon\sigma, \varepsilon\rho) \subset \mathcal{D}_{r-\rho,s-\sigma}$ for $z \in \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ because $\varepsilon \le \frac{1}{2}$. The explicit component bounds are $|X_\theta^t - \theta| \le |t| \sup|\partial_I \phi| \le |t||\phi|/\rho$ and $|X_I^t - I| \le |t| \sup|\partial_\theta \phi| \le |t||\phi|/(e\sigma)$ (mean value inequality along the trajectory and Lemma 6.3). For $\Psi = X_\phi^{-1} = X_{-\phi}^1$ the same with $-\phi$.

Step S6.33: (c)–(e). (c) is Lemma 6.4(c) with Ω^1 and Step S6.31. (d) is Lemma 6.5 and Corollary 6.6. (e): Lemma 6.4(c) for $z \in \Omega^1$ gives $\Psi(\Phi z) = z$ and $\Phi(\Psi z) = z$; and $\Phi(z) \in \bar{B}_\nu(z, \varepsilon) \subset \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ for $z \in \mathcal{D}_{r-2\rho,s-2\sigma}$ $(\varepsilon \le \frac{1}{2})$. □

6.6. **Variants for the norm ♯.**

Proposition 6.14 (Composition in the norm ♯)**.** *Lemma 6.3, Lemma 6.9, Theorem 6.10(i)–(iv), Proposition 6.11, Corollary 6.12 and Proposition 6.13 hold with every weighted norm $|\cdot|_,$ replaced by $|\cdot|_,^\sharp$ (Lemma 5.10), with the same constants and the same hypothesis $\eta = 2|\phi|^\sharp_{r',s'}/(\rho\sigma) \le \frac{1}{2}$. For the consumer with coefficient radius $\Sigma = 2\pi s$ and loss $\Delta\Sigma = 2\pi\sigma$ the parameter is $\eta = 4\pi|\phi|^\sharp_,/(\rho\,\Delta\Sigma)$ (Section 5.9).*

Proof. **Step S6.34: Where the norm enters.** The geometry (flow, symplecticity, Taylor, displacement) uses the norm only through sup bounds $|u(z)| \le |u|^\sharp_,$ (Lemma 5.10, B.1(a)), which hold for ♯ as for the pointwise norm; the vector-field bound Lemma 6.3 uses the individual derivative estimates, valid for ♯.

Step S6.35: Series. Lemma 6.9 uses only the bracket estimate; Theorem 6.10 Step S6.22 uses the triangle inequality, valid for ♯ since $\sup_I|\sum_h c_k^h(I)| \le \sum_h \sup_I|c_k^h|$. The integral corollary uses $\sup_I \int_0^1|(\hat{g_t})_k|\,dt \le \int_0^1 \sup_I|(\hat{g_t})_k|\,dt$ and Tonelli. □

Remark 6.15 (Use by Part 6 and track P; what is not provided). Part 6 applies Proposition 6.11 with h the integrable part, $w = \{h,\phi\} = g_+ - f_0$ by the homological equation (so $|w|$ is controlled by $|\chi|$ and $|T_K f|$, not by $|h|$), $f_0 = g + T_K f$, and Theorem 6.10 for $(f - T_K f)\circ X^1_\phi$, with the losses (ρ,σ) from (r,s) to $(r-\rho, s-\sigma)$ (so ϕ lives on $(r+\rho, s+\sigma)$, or one re-indexes as Pöschel: $\phi \in \mathcal{O}(r,s)$, $u \in \mathcal{O}(r-\rho, s-\sigma)$, result on $\mathcal{D}_{r-2\rho,s-2\sigma}$). The smallness used is $\eta = 2|\phi|/(\rho\sigma) \le \frac{1}{2}$; the displacement and derivative bounds of Proposition 6.13 need only $\varepsilon \le \frac{1}{2}$. The sup-norm *fallback* of `PLAN.md` is not used. Not proved here: the stability transfer (Part 7) and the estimate of $|W(\Phi_1\circ\cdots\circ\Phi_N - \mathrm{id})|$ for compositions of the maps of successive steps, which is Part 6's (the displacement bounds of (a) add, since each Φ_i maps into the domain of Φ_{i+1} by the choice of losses).

7. The homological equation, the Iterative Lemma and the Normal Form Lemma (shared by tracks N, P)

This part is Section 2 of Pöschel 1993 (the Normal Form Lemma, pp. 192–196, with the Iterative Lemma, p. 193) and Appendix A (pp. 211–213), in the frozen conventions of Part 5: angles of period 1, weight $e^{2\pi|k|_1 s}$, geometric widths, max norms on $\mathbb{R}^n, \mathbb{C}^n$, and the Lie transform of Part 6. It is shared infrastructure: track P derives CP10's averaging lemma ($\alpha_1 = \alpha_2$, $\Lambda = \{0\}$) from Theorem 7.6, which is therefore stated for an arbitrary set $\Lambda \subset \mathbb{Z}^n$ (including $\{0\}$), arbitrary n, and an arbitrary domain $D \subset \mathbb{C}^n$, with all constants explicit. Nothing but Parts 5 and 6 is used.
Notation. $Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}$ for real $K \ge 0$ (the ℓ^1 ball: it is the cut-off of T_K, Lemma 5.9); $\omega = \partial_I h$ is the frequency; $p > 1$ and $q = p/(p-1)$, so $p^{-1} + q^{-1} = 1$ (Pöschel takes $q = 9$, $p = \frac{9}{8}$; nothing below depends on that choice); for $\alpha > 0$ and $s > 0$

$$\bar\alpha := 2\pi\alpha, \qquad \kappa := 2\pi K s.$$

By Remark 5.11, $\bar\alpha$ and κ are Pöschel's α and Ks (his ω is 2π times ours, his weight $e^{|k|_1 s_P}$ is $e^{2\pi|k|_1 s}$, his $\rho_P\sigma_P$ is our $\rho\sigma$), so every formula of pp. 192–196, 211–213 holds with $\alpha \to \bar\alpha$, $Ks \to \kappa$ and r unchanged in the combination αr; the arithmetic below is nevertheless redone in our variables.
Lattice cut-offs (finding N06). Only the ℓ^1 ball Z_K occurs here, and $|k\cdot v| \le |k|_1|v|_\infty$ is all that is used about it. The Euclidean ball $\{\|k\|_2 \le K_g\}$ of the geometry of resonances (Part 8) is contained in $Z_{\sqrt{n}K_g}$ since $|k|_1 \le \sqrt{n}\|k\|_2$; so wherever the two meet, the analytic cut-off is $K = K_a \ge \sqrt{n}K_g$ (the factor $\sqrt{n}$ is carried by the caller). The results below are insensitive to the choice of K.

7.1. **External facts.**

(E1) *Numerics.* $2/e < 1$; $e^{-3\ln 2} = \frac{1}{8}$; $e^2 < 8$, hence $\ln 2 > \frac{2}{3}$ and $12\ln 2 > 8$; $\sum_{i\ge1} 4^{1-i} = \frac{4}{3}$; $(1 - \frac{1}{2^7})^{-1} = \frac{128}{127}$, $(1-\frac{1}{8})^{-1} = \frac{8}{7}$. Exists.

(E2) *Chain rule for holomorphic maps.* Compositions of holomorphic maps are holomorphic and $D(F\circ G)=DF\circ G\cdot DG$. Exists.

Besides these, [K:`bp:p1:ext-mvt`] (mean value inequality) and the external facts of Parts 5 and 6 are used.

7.2. Setting and the divisor bound.

Definition 7.1 (Resonant setting)**.** Let $D\subset\mathbb{C}^n$, $r>0$, $K\ge 0$, $\alpha>0$, $L\ge 0$, $\Lambda\subset\mathbb{Z}^n$, and let h be holomorphic on V_rD (Definition 5.1), regarded as a θ-independent function on $\mathcal{D}_{r,s}$ for every $s>0$; $\omega=\partial_I h: V_rD\to\mathbb{C}^n$. We say that $(D,r,K,\alpha,L,\Lambda,h)$ satisfies

(N1) ((α,K)-*nonresonant modulo* Λ) $|k\cdot\omega(I)|\ge\alpha$ for all $I\in D$ and $k\in Z_K\setminus\Lambda$ (Pöschel p. 192);
(N2) (*frequency Lipschitz bound*) $|\omega(I)-\omega(I_0)|_\infty\le L|I-I_0|_\infty$ whenever $I_0\in D$, $I\in V_rD$, $|I-I_0|_\infty<r$.

For $\Lambda=\{0\}$ and $\Lambda=\mathbb{Z}^n$ these are the completely nonresonant and the trivial cases. If $0\notin\Lambda$ then (N1) forces $D=\emptyset$ (take $k=0$), which is harmless. For $g=\sum_{j\in J}g_j$ (J finite, $g_j\in\mathcal{O}(r_j,s_j)$ with finite norm, $r_j\ge r$, $s_j\ge s$) and $0<\rho<r$, $0<\sigma<s$ put

$$|g|_*:=\sum_{j\in J}\frac{1}{2}\Big(\frac{\rho}{r_j-r+\rho}+\frac{\sigma}{s_j-s+\sigma}\Big)|g_j|_{r_j,s_j}\qquad\text{(depends on the decomposition).}$$

If $J=\{1\}$, $g_1=g\in\mathcal{O}(r,s)$ then $|g|_*=|g|_{r,s}$.
Source: Pos93 p. 192 (nonresonance), p. 193 ($|g|_*$), p. 211–212 ("on the domain D we have $|k\cdot\omega(I)|\ge\alpha\ldots$ by the mean value theorem and our assumptions on Q and r"): the Lipschitz bound is stated as hypothesis (N2) instead of via $\sup|Q|$, in max norms.

Lemma 7.2 (Lipschitz bound from the Hessian)**.** *If $\omega=\partial_I h$ is holomorphic on V_rD and the operator norm of $D\omega=\partial_I^2 h$ (max norm on both sides) is $\le L$ on V_rD, then (N2) holds. In particular $L=nM_C$ if every entry of the Hessian has modulus $\le M_C$ (Lemma 2.5).*
Source: Pos93 p. 212 (mean value theorem); standard.

Proof. **Step S7.1: Segment.** For $I_0\in D$, $I\in V_rD$ with $|I-I_0|_\infty<r$ the segment $[I_0,I]$ lies in the polydisc $P(I_0,r)\subset V_rD$ (convex, Definition 5.1). [K:`bp:p1:ext-mvt`] gives $|\omega(I)-\omega(I_0)|_\infty\le L|I-I_0|_\infty$. □

Lemma 7.3 (Divisor bound)**.** *Assume (N1), (N2) and $KLr\le\alpha/p$. Then $|k\cdot\omega(I)|\ge\alpha-KLr\ge\alpha/q$ for all $I\in V_rD$ and $k\in Z_K\setminus\Lambda$.*
Source: Pos93 p. 211–212 ("$|k\cdot\omega(I)|\ge\alpha-MKr\ge(1-\frac{1}{p})\alpha=\alpha/q$").

Proof. **Step S7.2: Move to D.** Let $I\in V_rD$ and choose $I_0\in D$ with $|I-I_0|_\infty<r$. Then $|k\cdot\omega(I)|\ge|k\cdot\omega(I_0)|-|k|_1|\omega(I)-\omega(I_0)|_\infty\ge\alpha-KLr$ by (N1), (N2). Finally $\alpha-\alpha/p=\alpha/q$. □

7.3. The homological equation.

Proposition 7.4 (Homological equation)**.** *Assume (N1), (N2) and $KLr\le\alpha/p$. Let $s>0$, $f\in\mathcal{O}(r,s)$ with $\delta=|f|_{r,s}<\infty$, and define*

$$\xi=P_\Lambda T_K f,\qquad \phi=\sum_{k\in Z_K\setminus\Lambda}\frac{\hat f_k(I)}{2\pi i\,k\cdot\omega(I)}\,\chi_k(\theta),\qquad \chi_k(\theta)=e^{2\pi ik\cdot\theta}.$$

Then $\xi,\phi\in\mathcal{O}(r,s)$ are finite sums of characters (trigonometric polynomials of degree $\le K$ in θ), and

(a) $\{\phi,h\}+\xi=T_Kf$ on $\mathcal{D}_{r,s}$, where $\{\phi,h\}=\partial_\theta\phi\cdot\omega$; $P_\Lambda\phi=0$, $P_\Lambda\xi=\xi$, $\xi=P_{\Lambda\cap Z_K}f$;
(b) $|\xi|_{r,s}\le\delta$, $|T_Kf|_{r,s}\le\delta$, and $|\phi|_{r,s}\le q\delta/\bar\alpha=q\delta/(2\pi\alpha)$;

(c) if $\Lambda = -\Lambda$, f is real on real points and $\omega(I) \in \mathbb{R}^n$ for real $I \in V_r D$, then ξ and ϕ are real on real points.

Source: Pos93 p. 211–212 ("$\xi = P_\Lambda T_K f$, $\phi = L^{-1}(T_K f - P_\Lambda T_K f)$, $\phi_k = f_k/(ik\cdot\omega(I))$ ... $|\xi|_{r,s} \le \delta$, $|\phi|_{r,s} \le q\delta/\alpha$"); the 2π in the divisor is the period-1 convention, so that the bound reads $q\delta/\bar\alpha$ as in the source.

Proof. **Step S7.3: Well defined and holomorphic.** For $k \in Z_K \setminus \Lambda$, $k\cdot\omega$ is holomorphic on $V_r D$ and has modulus $\ge \alpha/q > 0$ (Lemma 7.3), so $\hat\phi_k := \hat f_k/(2\pi i k\cdot\omega)$ is holomorphic (Lemma 5.3(a)). The sets $Z_K \cap \Lambda$ and $Z_K \setminus \Lambda$ are finite. Hence ϕ, ξ are finite sums of terms $a_k(I)\chi_k(\theta)$ with a_k holomorphic and $\sup_I \sum |a_k| e^{2\pi|k|_1 s} < \infty$, so they lie in $\mathcal{O}(r,s)$ with the stated coefficients (Lemma 5.3(c)); $\xi = P_\Lambda T_K f = P_{\Lambda\cap Z_K} f$ by Lemma 5.9.

Step S7.4: Norms. $|\hat\phi_k(I)| \le \frac{q}{2\pi\alpha}|\hat f_k(I)|$ by Lemma 7.3; multiplying by $e^{2\pi|k|_1 s}$, summing over $k \in Z_K \setminus \Lambda$ and taking the supremum over I gives $|\phi|_{r,s} \le q\delta/\bar\alpha$. The bounds for ξ and $T_K f$ are Lemma 5.9.

Step S7.5: The equation. By Lemma 5.3(b), $\partial_{\theta_j}\phi$ has coefficients $2\pi i k_j \hat\phi_k$; since h is independent of θ, $\{\phi, h\} = \sum_j \partial_{\theta_j}\phi\,\omega_j$, and multiplication by the θ-independent $\omega_j(I)$ multiplies each coefficient by $\omega_j(I)$ (integrate against χ_{-k}). So $\widehat{\{\phi,h\}}_k = 2\pi i(k\cdot\omega)\hat\phi_k = \hat f_k$ for $k \in Z_K \setminus \Lambda$ and 0 otherwise. Adding $\hat\xi_k = \hat f_k \mathbf{1}_{Z_K\cap\Lambda}(k)$ gives the coefficients of $T_K f$; coefficients determine the function (Lemma 5.3(d)).

Step S7.6: Reality. If f is real on real points, $\overline{\hat f_k(I)} = \hat f_{-k}(I)$ for real I (conjugate the integral defining $\hat f_k$). If $\omega(I)$ is real and $k \notin \Lambda$ with $-k \notin \Lambda$, then $\overline{\hat\phi_k} = \hat f_{-k}/\overline{2\pi i k\cdot\omega} = -\hat f_{-k}/(2\pi i k\cdot\omega) = \hat\phi_{-k}$; the set $Z_K \setminus \Lambda$ is symmetric. A finite sum $\sum a_k\chi_k$ with $\bar a_k = a_{-k}$ is real at real θ. The same for ξ. □

Lemma 7.5 (Bracket with a decomposed g)**.** *Let $\phi \in \mathcal{O}(r,s)$ and $g = \sum_{j\in J} g_j$ as in Definition 7.1, $0 < \rho < r$, $0 < \sigma < s$. Then $\{g,\phi\} \in \mathcal{O}(r-\rho, s-\sigma)$ and $|\{g,\phi\}|_{r-\rho,s-\sigma} \le \dfrac{2}{e\rho\sigma}|g|_*|\phi|_{r,s} \le \dfrac{2}{\rho\sigma}|g|_*|\phi|_{r,s}$.*

Source: Pos93 p. 212 ("$|\{g,\phi\}|_{r-\rho,s-\sigma} \le \frac{q}{\alpha\rho\sigma}|g|_*|f|_{r,s}$"), with the bracket estimate of Lemma 5.7.

Proof. **Step S7.7: One term.** Lemma 5.7 with $(u,v;r_0,s_0) \to (g_j,\phi;r_j,s_j)$ (admissible: $r-\rho < r_j$, $s-\sigma < s_j$) gives $|\{g_j,\phi\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\big(\frac{1}{(r_j-r+\rho)\sigma} + \frac{1}{(s_j-s+\sigma)\rho}\big)|g_j||\phi| = \frac{2}{e\rho\sigma}\cdot\frac{1}{2}\big(\frac{\rho}{r_j-r+\rho} + \frac{\sigma}{s_j-s+\sigma}\big)|g_j||\phi|$.

Step S7.8: Sum. The bracket is bilinear and the norm satisfies the triangle inequality; sum over j and use (E1) ($e > 2$). □

7.4. The Iterative Lemma.

Theorem 7.6 (Iterative Lemma)**.** *Let $(D, r, K, \alpha, L, \Lambda, h)$ satisfy (N1), (N2) with $KLr \le \alpha/p$ ($p > 1$, $q = p/(p-1)$), let $s > 0$, $0 < \rho$, $0 < \sigma$ with $2\rho < r$, $2\sigma < s$, and put $\bar\alpha = 2\pi\alpha$. Let $H = h + g + f$ on $\mathcal{D}_{r,s}$, where $g = \sum_{j\in J} g_j$ is as in Definition 7.1 (any finite decomposition), $f \in \mathcal{O}(r,s)$, and*

$$\delta := |f|_{r,s} \le \frac{\bar\alpha\,\rho\sigma}{4q}, \qquad \textit{equivalently} \qquad u := \frac{q\delta}{\bar\alpha\rho\sigma} \le \frac{1}{4}.$$

Then there is $\phi \in \mathcal{O}(r,s)$ (the one of Proposition 7.4, so $|\phi|_{r,s} \le q\delta/\bar\alpha$) such that the time-one map $\Phi = X_\phi^1$ of Part 6 satisfies:

(a) *Φ is defined and holomorphic on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ with values in $\mathcal{D}_{r-\rho,s-\sigma}\subset\mathcal{D}_{r,s}$; it is canonical: $(D\Phi)^\top J\,D\Phi=J$, $\Phi(\theta+k,I)=\Phi(\theta,I)+(k,0)$ ($k\in\mathbb{Z}^n$), and Φ is injective on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with left inverse X_ϕ^{-1} (Proposition 6.13(d),(e),(b));*

(b) *(displacement) with $\nu=\nu_{\rho,\sigma}$ of Definition 6.2: $\nu(\Phi-\mathrm{id})\le u$ on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$, and $|\Phi_\theta-\theta|_\infty\le\dfrac{q\delta}{\bar\alpha\rho}$, $|\Phi_I-I|_\infty\le\dfrac{q\delta}{e\,\bar\alpha\sigma}$; on $\mathcal{D}_{r-2\rho,s-2\sigma}$: $|D\Phi-\mathbf{1}|_\nu\le 2u$;*

(c) *(normal form) $H\circ\Phi=h+g_+ +f_+$ on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with*

$$g_+=g+P_\Lambda T_K f,\qquad |g_+-g|_{r,s}\le\delta,\qquad f_+\in\mathcal{O}(r-2\rho,s-2\sigma);$$

(d) *(estimate of f_+)*

$$|f_+|_{r-2\rho,s-2\sigma}\le\Big(1-\frac{2q\delta}{\bar\alpha\rho\sigma}\Big)^{-1}\Big(\frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_*+\delta\big)+e^{-2\pi K\sigma}\Big)\delta,$$

with $|g|_*$ computed with these ρ,σ;

(e) *(reality) if $\Lambda=-\Lambda$, and h,g,f are real on real points, then ϕ, g_+, f_+ are real on real points and Φ maps real points to real points.*

Source: Pos93 p. 193 (Iterative Lemma; the hypothesis there is $|f|_{r,s}<\alpha\rho\sigma/2q$, the estimate $|f_+|\le(1-\frac{2q\delta}{\alpha\rho\sigma})^{-1}(\frac{|g|_*+|f|}{\alpha\rho\sigma/q}+e^{-K\sigma})|f|$) and Appendix A, pp. 211–213. Pöschel's $\alpha,\sigma,|W(\Phi-\mathrm{id})|\le\frac{q\delta}{\alpha\rho\sigma}$ correspond to $\bar\alpha,2\pi\sigma$ and u (Remark 5.11). Deviation*: the smallness is $|\phi|\le\rho\sigma/4$ (so $\eta=2|\phi|/(\rho\sigma)\le\frac12$) instead of $<\rho\sigma/2$, because Theorem 6.10 proves the Lie series identity only for $\eta\le\frac12$ (Remark 6.1); the estimate has the extra factor $2/e<1$ from Lemma 5.7. Constants only change.*

Proof. **Step S7.9: The generating function.** By Proposition 7.4, ϕ,ξ exist with $\{\phi,h\}+\xi=T_Kf$, $|\xi|,|T_Kf|\le\delta$, $|\phi|_{r,s}\le q\delta/\bar\alpha$. Put $\varepsilon:=|\phi|_{r,s}/(\rho\sigma)\le u\le\frac14$ and $\eta:=2|\phi|_{r,s}/(\rho\sigma)\le 2u\le\frac12$.

Step S7.10: Items (a), (b). Proposition 6.13 (with $\rho<r/2$, $\sigma<s/2$, $\varepsilon\le\frac12$) gives (a), the displacement bounds of (b) with $|\phi|/(\rho\sigma)\le u$, $|\phi|/\rho\le q\delta/(\bar\alpha\rho)$, $|\phi|/(e\sigma)\le q\delta/(e\bar\alpha\sigma)$, and $|D\Phi-\mathbf{1}|_\nu\le 2\varepsilon\le 2u$.

Step S7.11: Taylor identity. Put $f_0=g+T_Kf\in\mathcal{O}(r-\rho,s-\sigma)$ (finite norm: $g=\sum g_j$ and monotonicity, $|g|_{r,s}\le\sum|g_j|$), $w=\{h,\phi\}=-\{\phi,h\}=\xi-T_Kf$ (Proposition 7.4(a), antisymmetry, Definition 6.2); w has finite norm. Apply Proposition 6.11 with the data $(r,s;r',s')\to(r-\rho,s-\sigma;r,s)$ (admissible: $\rho<r-\rho$ iff $2\rho<r$, $r-\rho\le r'-\rho$; likewise for s; and $\eta\le\frac12$): on $\mathcal{D}_{r-2\rho,s-2\sigma}$, $(h+f_0)\circ\Phi=h+w+f_0+\mathcal{R}$ with $\mathcal{R}=\int_0^1\{(1-t)w+f_0,\phi\}\circ X_\phi^t\,dt$. Here $h+w+f_0=h+g+\xi=h+g_+$, and $(1-t)w+f_0=g+f_t$ with $f_t:=(1-t)\xi+tT_Kf$.

Step S7.12: Decomposition of $H\circ\Phi$. $H=(h+f_0)+(f-T_Kf)$ with $f-T_Kf\in\mathcal{O}(r,s)$, so on $\mathcal{D}_{r-2\rho,s-2\sigma}$

$$H\circ\Phi=h+g_+ +f_+,\qquad f_+:=\mathcal{R}+(f-T_Kf)\circ\Phi.$$

Both terms are holomorphic and periodic (Theorem 6.10(ii), Corollary 6.12), so $f_+\in\mathcal{O}(r-2\rho,s-2\sigma)$ once the norm below is finite. This is (c); $|g_+-g|=|\xi|\le\delta$.

Step S7.13: The integrand. $v_t:=\{g+f_t,\phi\}\in\mathcal{O}(r-\rho,s-\sigma)$ is affine in t, so $(t,z)\mapsto v_t(z)$ and $Dv_t(z)$ are continuous. By Lemma 7.5 and Lemma 5.7 (the single-term decomposition f_t, $|f_t|\le\delta$ since $|\xi|,|T_Kf|\le\delta$),

$$|v_t|_{r-\rho,s-\sigma}\le\frac{2}{e\rho\sigma}\big(|g|_*+\delta\big)|\phi|_{r,s}\le\frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_*+\delta\big)\delta=:V.$$

Step S7.14: The two terms. Corollary 6.12 (data as in Step S7.11) gives $|\mathcal{R}|_{r-2\rho,s-2\sigma}\le(1-\eta)^{-1}V$. By Lemma 5.9 and monotonicity, $|f-T_Kf|_{r-\rho,s-\sigma}\le|f-T_Kf|_{r,s-\sigma}\le e^{-2\pi K\sigma}\delta$, and Theorem 6.10(ii) (same data) gives $|(f-T_Kf)\circ\Phi|_{r-2\rho,s-2\sigma}\le(1-\eta)^{-1}e^{-2\pi K\sigma}\delta$. Add and use $(1-\eta)^{-1}\le(1-2q\delta/(\bar\alpha\rho\sigma))^{-1}$ ($|\phi|\le q\delta/\bar\alpha$). This is (d).

Step S7.15: Reality. If $\Lambda = -\Lambda$ and h, g, f are real, then ω is real at real points (h is real-valued and holomorphic on the open set $V_r D$, so its partial derivatives at real points of $V_r D$ are real), and Proposition 7.4(c) gives ϕ, ξ real; Φ preserves real points (Proposition 6.13(b)); so $g_+ = g + \xi$ is real and $f_+ = H \circ \Phi - h - g_+$ is real on real points of $\mathcal{D}_{r-2\rho, s-2\sigma}$. □

Remark 7.7 (Special cases for track P). (1) *First step, $g = 0$:* $|g|_* = 0$ and (d) reads $|f_+| \le (1 - \frac{2q\delta}{\bar\alpha\rho\sigma})^{-1}\big(\frac{2q\delta}{e\bar\alpha\rho\sigma} + e^{-2\pi K\sigma}\big)\delta$. (2) *Averaging, $\Lambda = \{0\}$:* $P_\Lambda T_K f = \hat f_0 = \langle f \rangle$, the θ-average; (N1) is $|k \cdot \omega(I)| \ge \alpha$ for $0 < |k|_1 \le K$; the transformed Hamiltonian is $h + g + \langle f \rangle + f_+$. (3) *Constants:* no constant depends on f or on H beyond δ, $|g|_*$; the data are $(n, r, s, \rho, \sigma, K, \alpha, L, p)$. The hypotheses involve only δ through $u \le \frac14$.

7.5. Composition of the steps.

Lemma 7.8 (Telescoping). *Let $U_0 \supset U_1 \supset \cdots \supset U_{m+1}$ be subsets of $\mathbb{C}^{2n}$ and $\Phi_i : U_{i+1} \to U_i$ $(0 \le i \le m)$ maps. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_m : U_{m+1} \to U_0$. Then for every $z \in U_{m+1}$ and every seminorm μ on $\mathbb{C}^{2n}$ (e.g. $|\Pi_I \cdot|_\infty$, $|\Pi_\theta \cdot|_\infty$), $\mu(\Psi(z) - z) \le \sum_{i=0}^m \sup_{U_{i+1}} \mu(\Phi_i - \mathrm{id})$. If every Φ_i is holomorphic and $(D\Phi_i)^\top J D\Phi_i = J$, then Ψ is holomorphic, $(D\Psi)^\top J D\Psi = J$; if every Φ_i is injective, so is Ψ.*

Proof. **Step S7.16: Telescoping.** Put $z_{m+1} = z$, $z_i = \Phi_i(z_{i+1}) \in U_i$ (so $z_{i+1} \in U_{i+1}$ for the next step because $U_{i+1} \subset U_i$). Then $\Psi(z) - z = \sum_{i=0}^m (z_i - z_{i+1})$ and $z_i - z_{i+1} = (\Phi_i - \mathrm{id})(z_{i+1})$; apply the triangle inequality. The chain rule (E2) gives $D\Psi = \prod D\Phi_i(z_{i+1})$ and $(D\Psi)^\top J D\Psi = J$ by induction; injectivity is clear. □

7.6. The Normal Form Lemma.

Theorem 7.9 (Normal Form Lemma). *Let $(D, r, K, \alpha, L, \Lambda, h)$ satisfy (N1), (N2), $K > 0$, and $KLr \le \alpha/p$ $(p > 1,\ q = p/(p-1))$. Let $s > 0$ and $H = h + f$ with $f \in \mathcal{O}(r, s)$ of finite norm $\varepsilon := |f|_{r,s}$. Assume*

$$\text{(i)}\quad \varepsilon \le \frac{1}{2^7 q} \cdot \frac{\alpha r}{K}, \qquad\qquad \text{(ii)}\quad \kappa = 2\pi K s \ge 6.$$

Put $r_* = r/2$, $s_* = s/6$, $c = 2qK/(\alpha r)$, $g_0 = P_\Lambda T_K f$. Then there are

- *a holomorphic, injective map $\Psi : \mathcal{D}_{r_*, s_*} \to \mathcal{D}_{r,s}$ with $(D\Psi)^\top J D\Psi = J$ and $\Psi(\theta + k, I) = \Psi(\theta, I) + (k, 0)$ $(k \in \mathbb{Z}^n)$;*
- *$g, f_* \in \mathcal{O}(r_*, s_*)$, with g a trigonometric polynomial in θ with frequencies in $\Lambda \cap Z_K$ (so $P_\Lambda g = g$: $h + g + f_*$ is in Λ-resonant normal form up to f_*),*

such that $H \circ \Psi = h + g + f_*$ on $\mathcal{D}_{r_*, s_*}$ and

(a) $|g - g_0|_{r_*, s_*} \le 16 c\, \varepsilon^2$ $\quad$ ($g = g_0$ if $N = 0$ (Step S7.19) or in the second case of Step S7.18);
(b) $|f_*|_{r_*, s_*} \le e^{-2\pi K s/6}\varepsilon = e^{-\kappa/6}\varepsilon$;
(c) $|\Pi_I \Psi - I|_\infty \le qK\varepsilon/\alpha$ $\big(= \frac12 c\, r\varepsilon\big)$ and $|\Pi_\theta \Psi - \theta|_\infty \le 10\, q\varepsilon/(2\pi\alpha r)$ on $\mathcal{D}_{r_*, s_*}$;
(d) if $\Lambda = -\Lambda$ and h, f are real on real points, then g, f_* are real on real points and Ψ maps real points to real points.

The data on which Ψ, g, f_* are constructed are $(h, f, K, \alpha, L, \Lambda, p)$ and (r, s); r_*, s_*, c depend only on (r, s, K, α, q).

Source: Pos93 p. 192 (Normal Form Lemma) and pp. 193–196 (proof). Changes: (1) frozen conventions ($\bar\alpha = 2\pi\alpha$, $\kappa = 2\pi K s$ are his α, Ks; max norms); (2) $|\Pi_I \Psi - I| \le qK\varepsilon/\alpha$ in the max norm instead of $\|\Pi_I \Psi - I\| \le c\varepsilon r = 2qK\varepsilon/\alpha$ (Euclidean); (3) the smallness $\eta \le \frac12$ of Theorem 6.10 is built into Theorem 7.6; every numerical step below was recomputed (Remark 7.12).

Source gap: Pöschel's text p. 194 ends the second case with "$\le e^{Ks/6}\varepsilon$"; the correct (and used) bound is $e^{-Ks/6}\varepsilon$, as the first line of that display shows.

Proof. Throughout $\bar\alpha = 2\pi\alpha$ and $\kappa = 2\pi Ks$; note $\bar\alpha s = \alpha\kappa/K$ and $K \ge 3/(\pi s)$. *Indexing:* $V_i = \mathcal{D}_{r_i,s_i}$, $f_0 := f$, $g_{-1} := 0$; step i applies Theorem 7.6 to $H_i = h + g_{i-1} + f_i$ on V_i with losses (ρ_i, σ_i), producing $\Phi_i : V_{i+1} \to V_i$, $g_i = g_{i-1} + P_\Lambda T_K f_i$, f_{i+1} on V_{i+1}, $r_{i+1} = r_i - 2\rho_i$, $s_{i+1} = s_i - 2\sigma_i$; $\varepsilon_i := |f_i|_{V_i}$, $\varepsilon_0 = \varepsilon$. ($V_{i+1}$ stands for $\mathcal{D}_{r_i-2\rho_i,s_i-2\sigma_i}$, which is where Φ_i is injective and f_{i+1} lives; Φ_i maps it into $\mathcal{D}_{r_i-\frac32\rho_i,s_i-\frac32\sigma_i} \subset V_i$.)

Step S7.17: Step 0: hypotheses of the Iterative Lemma. Let $\rho_0 = r/8$, $\sigma_0 = s/6$, so $r_1 = \frac34 r$, $s_1 = \frac23 s$, $\rho_0\sigma_0 = rs/48$. Then $2\rho_0 < r$, $2\sigma_0 < s$, $KLr \le \alpha/p$. From (i), (ii): $\varepsilon \le \frac{\alpha r}{2^7 qK} \le \frac{\alpha r}{2^7 q} \cdot \frac{\pi s}{3} = \frac{\pi\alpha rs}{384q} = \frac{\bar\alpha\rho_0\sigma_0}{16q}$. Hence $y_0 := \frac{q\varepsilon}{\bar\alpha\rho_0\sigma_0} \le \frac{1}{16}$ and the hypothesis $u \le \frac14$ of Theorem 7.6 holds. Also $\kappa/6 = 2\pi K\sigma_0 \ge 1$.

Step S7.18: Step 0: two cases. Apply Theorem 7.6(d) with $g = 0$ ($|g|_* = 0$), $\delta = \varepsilon$: $|f_1| \le (1-2y_0)^{-1}\big(\frac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon$, with $(1-2y_0)^{-1} \le \frac87$.

Case A: $e^{-\kappa/6} \le y_0$. Then $|f_1| \le \frac87(\frac2e + 1)y_0\varepsilon \le \frac{16}{7}y_0\varepsilon$, so

$$\varepsilon_1 \le \frac{16}{7}\cdot\frac{48\,q\varepsilon^2}{\bar\alpha rs} \le \frac{2^7 q\varepsilon^2}{\bar\alpha rs} \le \frac{2^7 q\varepsilon}{\bar\alpha rs}\cdot\frac{\alpha r}{2^7 qK}\,\varepsilon = \frac{\varepsilon}{2\pi Ks} = \frac{\varepsilon}{\kappa} \le \frac{\varepsilon}{6} \le \frac{\varepsilon}{4}. \tag{6.1}$$

Case B: $e^{-\kappa/6} > y_0$. Repeat the step with twice the losses $\rho' = r/4$, $\sigma' = s/3$ (so $2\rho' < r$, $2\sigma' < s$, $\bar\alpha\rho'\sigma' = 4\bar\alpha\rho_0\sigma_0$, the hypothesis $u' = y_0/4 \le \frac14$ holds). Then $|f_1|_{r/2,s/3} \le (1-y_0/2)^{-1}\big(\frac{2}{e}\cdot\frac{y_0}{4} + e^{-2\kappa/6}\big)\varepsilon$; as $e^{-2\kappa/6} \le e^{-1}e^{-\kappa/6}$ and $y_0 < e^{-\kappa/6}$, this is $\le \frac{32}{31}\big(\frac{1}{2e} + \frac1e\big)e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$. We stop: $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,s/3} \supset \mathcal{D}_{r_*,s_*}$, $g = g_0$, $f_* = f_1$ restricted to V_{r_*,s_*}. Items (a), (b) hold; items (c), (d) in Step S7.22. From now on *Case A*.

Step S7.19: Number of steps. Let $N = \lfloor \kappa/(12\ln 2)\rfloor \ge 0$, so $N \le \frac{\kappa}{12\ln 2} < N+1$ and $\kappa \ge 12\ln 2\, N > 8N$ ((E1)). If $N = 0$ stop after Step 0: $\Psi = \Phi_0$, $g = g_0$, $f_* = f_1$ on $V_1 \supset \mathcal{D}_{r_*,s_*}$ and $|f_*| \le \varepsilon_1 \le \varepsilon/4 \le e^{-\kappa/6}\varepsilon$, because $\kappa < 12\ln 2$ gives $e^{-\kappa/6} > e^{-2\ln 2} = \frac14$. Let $N \ge 1$ and, for $1 \le i \le N$,

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac34 r - 2(i-1)\rho, \quad s_i = \tfrac23 s - 2(i-1)\sigma;$$

then $r_{N+1} = \frac34 r - \frac{r}{4} = r_*$ and $s_{N+1} = \frac23 s - \frac{s}{2} = s_*$. Note $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, i.e. $e^{-2\pi K\sigma} \le \frac18$, and $\rho\sigma = rs/(32N^2)$. Smallness constants: from (i), $\kappa > 8N$ and $K = \kappa/(2\pi s)$,

$$\varepsilon \le \frac{\alpha r\cdot 2\pi s}{2^7 q\kappa} \le \frac{\bar\alpha rs}{2^{10}qN}, \qquad \text{hence by (6.1)} \quad \varepsilon_1 \le \frac{2^7 q}{\bar\alpha rs}\Big(\frac{\bar\alpha rs}{2^{10}qN}\Big)^2 = \frac{\bar\alpha rs}{2^{13}qN^2} = \frac{\bar\alpha\rho\sigma}{2^8 q}. \tag{6.2}$$

Step S7.20: Induction. *Claim:* for $1 \le i \le N$, step i is admissible and $\varepsilon_{i+1} \le \varepsilon_i/4$. Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ for $1 \le i \le N+1$, and $|g_i - g_{i-1}|_{V_i} = |P_\Lambda T_K f_i| \le \varepsilon_i$ (Lemma 5.9). Induct on i; $\varepsilon_1 \le \varepsilon/4$ is (6.1) (so for $i = 1$ the claim "$\varepsilon_i \le 4^{1-i}\varepsilon_1$" is trivial). Let $i \le N$ and suppose $\varepsilon_j \le 4^{1-j}\varepsilon_1$ for $1 \le j \le i$. *Admissibility.* $r_i \ge r_{N+1} = r/2 > 2\rho$; $s_i \ge s_N = s_* + 2\sigma = \frac{s}{6} + \frac{s}{2N} > 2\sigma$; $KLr_i \le KLr \le \alpha/p$; and $\varepsilon_i \le \varepsilon_1 \le \bar\alpha\rho\sigma/(2^8 q) \le \bar\alpha\rho\sigma/(4q)$ by (6.2). *Size of* $|g_{i-1}|_*$ (computed with (r_i, s_i, ρ, σ) in Definition 7.1). Decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$, the term $j \ge 1$ living on V_j with $r_j - r_i = 2(i-j)\rho \ge 2\rho$, $s_j - s_i \ge 2\sigma$, so its weight is $\le \frac12(\frac13 + \frac13) \le \frac12$; the term $g_0 = P_\Lambda T_K f$ lives on $V_0 = \mathcal{D}_{r,s}$ with norm $\le \varepsilon$ and $r - r_i \ge r - r_1 = \frac{r}{4}$, $s - s_i \ge \frac{s}{3}$, so its weight is $\le \frac12\big(\frac{r/8N}{r/4} + \frac{s/4N}{s/3}\big) = \frac{5}{8N} \le \frac1N$. Thus

$$|g_{i-1}|_* \le \tfrac12(\varepsilon_{i-1} + \cdots + \varepsilon_1) + \tfrac1N\varepsilon \le \tfrac12\cdot\tfrac43\varepsilon_1 + \tfrac{\varepsilon}{N} \le \varepsilon_1 + \frac{\varepsilon}{N} \le \frac{\bar\alpha\rho\sigma}{2^8 q} + \frac{\bar\alpha\rho\sigma}{2^5 q} \le \frac{\bar\alpha\rho\sigma}{2^4 q},$$

using (6.2) and $\varepsilon/N \le \bar\alpha rs/(2^{10}qN^2) = 2^5\bar\alpha\rho\sigma/(2^{10}q)$. *The step.* Theorem 7.6(d) with $\delta = \varepsilon_i$: $\frac{2q}{\bar\alpha\rho\sigma}\varepsilon_i \le 2^{-7}$, so $(1-2^{-7})^{-1} = \frac{128}{127}$, and $\frac{2q}{e\bar\alpha\rho\sigma}(|g_{i-1}|_* + \varepsilon_i) \le \frac{2}{e}(\frac{1}{16}+\frac{1}{2^8}) \le \frac{1}{16}+\frac{1}{2^8}$, $e^{-2\pi K\sigma} \le \frac{1}{8}$:

$$\varepsilon_{i+1} \le \frac{128}{127}\Big(\frac{1}{2^4}+\frac{1}{2^8}+\frac{1}{8}\Big)\varepsilon_i \le \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac{1}{4}\varepsilon_i.$$

Step S7.21: Conclusion of the construction. Put $\Psi = \Phi_0\circ\cdots\circ\Phi_N$, $g = g_N = g_0+\sum_{i=1}^N P_\Lambda T_K f_i$, $f_* = f_{N+1}$, all restricted to $V_{N+1} = \mathcal{D}_{r_*,s_*}$. Since $H_{i+1} = H_i\circ\Phi_i$, $H\circ\Psi = H_{N+1} = h+g+f_*$. g is a finite sum of trigonometric polynomials with frequencies in $\Lambda\cap Z_K$. Lemma 7.8 with $U_i = V_i$ gives the symplectic, periodic (each Φ_i is, Theorem 7.6(a)) and injective Ψ. (b): $|f_*| = \varepsilon_{N+1} \le 4^{-N}\varepsilon_1 \le 4^{-N-1}\varepsilon \le e^{-\kappa/6}\varepsilon$, because $N+1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} = e^{-(N+1)2\ln 2} < e^{-\kappa/6}$. (a): $g-g_0 = \sum_{i=1}^N P_\Lambda T_K f_i$, so $|g-g_0| \le \frac{4}{3}\varepsilon_1 \le \frac{4}{3}\cdot\frac{2^7q\varepsilon^2}{\bar\alpha rs}$ by (6.1); $\bar\alpha s = \alpha\kappa/K$ and $\kappa > 8N \ge 8$ give $\le \frac{4}{3}\cdot\frac{2^7}{8}\cdot\frac{qK\varepsilon^2}{\alpha r} \le 32\frac{qK\varepsilon^2}{\alpha r} = 16c\varepsilon^2$.

Step S7.22: (c) Displacement. By Lemma 7.8 and Theorem 7.6(b) ($|\Phi_I - I| \le q\delta/(e\bar\alpha\sigma)$, $|\Phi_\theta-\theta| \le q\delta/(\bar\alpha\rho)$, $\delta = \varepsilon_i$): in Case A

$$|\Pi_I\Psi - I|_\infty \le \frac{q\varepsilon}{e\bar\alpha\sigma_0} + \sum_{i=1}^N \frac{q\varepsilon_i}{e\bar\alpha\sigma} \le \frac{6q\varepsilon}{e\bar\alpha s} + \frac{q}{e\bar\alpha\sigma}\cdot\frac{4}{3}\varepsilon_1.$$

Now $1/(\bar\alpha s) = K/(\alpha\kappa) \le K/(6\alpha)$; $1/(\bar\alpha\sigma) = 4NK/(\alpha\kappa) \le 4K/(12\ln 2\,\alpha)$ because $N \le \kappa/(12\ln 2)$, and $\varepsilon_1 \le \varepsilon/4$; so the right side is at most $\frac{qK\varepsilon}{e\alpha}\big(1+\frac{4}{12\ln 2}\cdot\frac{4}{3}\cdot\frac{1}{4}\big) \le \frac{qK\varepsilon}{e\alpha}\cdot 1.17 \le \frac{qK\varepsilon}{\alpha}$. Similarly $|\Pi_\theta\Psi-\theta|_\infty \le \frac{q\varepsilon}{\bar\alpha\rho_0}+\frac{q}{\bar\alpha\rho}\cdot\frac{4}{3}\varepsilon_1 = \frac{8q\varepsilon}{\bar\alpha r}+\frac{8Nq}{\bar\alpha r}\cdot\frac{4}{3}\cdot\frac{\varepsilon}{\kappa} \le \frac{q\varepsilon}{\bar\alpha r}\big(8+\frac{32}{3\cdot 12\ln 2}\big) \le \frac{10\,q\varepsilon}{\bar\alpha r}$ (using $\varepsilon_1 \le \varepsilon/\kappa$ from (6.1), $N/\kappa \le 1/(12\ln 2)$). In Case B, $\Psi = \Phi_0'$ with losses $(2\rho_0, 2\sigma_0)$ and $\delta = \varepsilon$, and the bounds are smaller by the factor $\frac{1}{2}$.

Step S7.23: (d) Reality. Each step preserves reality (Theorem 7.6(e)); g_i, f_i, H_i stay real on real points and so do the Φ_i; compositions of real-point-preserving maps preserve real points. □

Corollary 7.10 (Consequence for the action variable)**.** *In Theorem 7.9, if $(\theta', I') = \Psi(\theta, I)$ with $(\theta, I) \in \mathcal{D}_{r_*,s_*}$ then $|I'-I|_\infty \le qK\varepsilon/\alpha$; in particular if $\varepsilon \le \alpha r/(2^7qK)$ (hypothesis (i)) this is $\le r/2^7$.*
Source: Pos93 p. 196, last display.

Proof. Theorem 7.9(c) and hypothesis (i). □

7.7. **The case $2\pi Ks < 6$.**

Proposition 7.11 (No transformation for small Ks)**.** *Let $H = h+f$ on $\mathcal{D}_{r,s}$, $f \in \mathcal{O}(r,s)$ with finite norm ε and $\kappa = 2\pi Ks < 6$. Then, with $\Psi = \mathrm{id}$, $g = 0$, $f_* = f$, one has $H\circ\Psi = h+g+f_*$ on $\mathcal{D}_{r,s}$ and*

$$|\Psi - \mathrm{id}| = 0, \qquad |f_*|_{r,s} = \varepsilon \le e\cdot e^{-\kappa/6}\varepsilon,$$

and $g = 0 = P_\Lambda g$ is in Λ-normal form for every Λ. (No nonresonance, convexity or smallness hypothesis is used.)
Source: Pos93 p. 197, first line ("For $Ks \le 6$ the same reasoning applies in original coordinates, letting $g = 0$ and $f_* = f_\varepsilon$").
Source gap: The source does not say what is claimed about $|f_*|$ in this case. With $f_* = f$, the bound $|f_*| \le e^{-\kappa/6}\varepsilon$ of the Normal Form Lemma is false *for $\kappa < 6$ (take $\Psi = \mathrm{id}$, $f \ne 0$, $K \to 0$); what holds is $|f_*| \le e^{1-\kappa/6}\varepsilon$, since $\kappa/6 < 1$. Part 7 (Proposition 1, the time bound $\frac{sr}{5\varepsilon}e^{Ks/6}$) must therefore carry the factor e^{-1} in this case or use $|f_*| = \varepsilon$ directly; this changes constants only. The trivial statement is recorded here once so that Parts 7 and 9 cite it.*

Proof. **Step S7.24: Trivial.** Identity map, $g = 0$; $|f_*| = \varepsilon$ and $1 \le e^{1-\kappa/6}$ for $\kappa \le 6$. □

7.8. **Deviations, errata, uses.**

Remark 7.12 (Deviations from Pöschel and bookkeeping). (1) *Lie-series range.* Part 6 proves $u \circ X_\phi^1 = \sum_h \frac{1}{h!} \operatorname{ad}_\phi^h u$ for $\eta = 2|\phi|/(\rho\sigma) \le \frac{1}{2}$ only. Hence the Iterative Lemma assumes $|\phi| \le \rho\sigma/4$, which is implied by $\delta \le \bar\alpha\rho\sigma/(4q)$ (Pöschel: $\delta < \alpha\rho\sigma/2q$). All hypotheses on ε in the Normal Form Lemma are already stronger than this: $\varepsilon_0 \le \bar\alpha\rho_0\sigma_0/(16q)$ and $\varepsilon_i \le \bar\alpha\rho\sigma/(2^8 q)$ (Steps S7.17, S7.20). The numerical constants 2^7, 4^{1-i}, $N = \lfloor \kappa/(12\ln 2)\rfloor$, $r_* = r/2$, $s_* = s/6$, $16c\varepsilon^2$ are therefore *the same as Pöschel's* (re-derived above), and the number of iteration steps is unchanged.

(2) *Constants gained.* The factor $2/e < 1$ in the estimate of f_+ (from $\frac{1}{e} < \frac{1}{2}$ in Lemma 5.7); where the source's arithmetic is reused we simply bound $2/e \le 1$.

(3) *Norms.* Max norms on actions and angles replace his Euclidean/sup norms; the divisor bound uses $|k\cdot v| \le |k|_1 |v|_\infty$, which is why Z_K is the ℓ^1 ball and hypothesis (N2) is stated in $|\cdot|_\infty$. The Euclidean bound $\|\Pi_I \Psi - I\| \le c\varepsilon r$ of the source follows from the max-norm statement of Theorem 7.9(c) only with a factor $\sqrt{n}$.

(4) *Not proved in the source, proved here:* the θ-displacement of Ψ and the injectivity of Ψ (from Proposition 6.13(e)); the $|W_0(\Psi - \mathrm{id})|$ bound of p. 195 ("usual telescoping arguments") is replaced by the componentwise bound of Theorem 7.9(c).
Source gap: Pöschel's last display of p. 195 asserts $|W_0(D\Psi - I)W_0^{-1}| \le 2q\varepsilon_0/(\alpha\rho_0\sigma_0)$ by telescoping; the later steps contribute $\sum_{i\ge1} q\varepsilon_i/(\bar\alpha\rho\sigma)$ in the scaled norm $\nu_{\rho,\sigma}$, which is not obviously $\lesssim q\varepsilon_0/(\bar\alpha\rho_0\sigma_0)$ for large N since $\rho\sigma/(\rho_0\sigma_0) = 3/(2N^2)$. It is not used by the stability estimates (only $\Pi_I\Psi$ is), so it is not reproduced.

(5) *The second case of the first step* (his "twice the step size") is Step S7.18, Case B, and ends with the correct sign of the exponent (see the
Source gap:
of Theorem 7.9).

(6) *Domains.* The generating function lives on $\mathcal{D}_{r,s}$, the map Φ on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ and the new remainder on $\mathcal{D}_{r-2\rho,s-2\sigma}$; hence the chain $V_{i+1} = \mathcal{D}_{r_i-2\rho_i,s_i-2\sigma_i}$ and Ψ on $\mathcal{D}_{r/2,s/6}$ as in the source.

Remark 7.13 (Consumers). Part 7 uses Theorem 7.9, Corollary 7.10 and Proposition 7.11; Part 9 fixes $q = 9$, $p = \frac{9}{8}$ and, with Part 8, chooses K (as $K_a \ge \sqrt{n}K_g$) and the domain D. Track P (Part 2 of its plan) takes Theorem 7.6 with $\Lambda = \{0\}$, Remark 7.7(2), the elementary constant bookkeeping being its own.

8. Stability estimates: nonresonant, convex, quasi-convex and resonant (shared by tracks N, I)

This part is Section 3 of Pöschel 1993 (pp. 196–200): the nonresonant Proposition 1, the two energy estimates near a resonance (Lemma 1 for m-convex h, Lemma 2 for ℓ, m-quasi-convex h) and the Resonant Stability Lemma (Proposition 2), which carries these estimates from normal-form coordinates back to the original ones. The idea is that in normal form the action of an orbit moves only through the non-resonant remainder f_* (exponentially small) and through the resonant part g, whose gradient lies in the span of Λ. Energy conservation and convexity of h transverse to that span then bound the motion in the resonant directions by $\sqrt{\zeta/m}$, without any further small-divisor analysis; the span itself is handled because D is close to the exact resonance space.

Conventions. Everything is in the frozen conventions of Parts 5–7: angles of period 1, $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, geometric strip width s, weight $e^{2\pi|k|_1 s}$, max norms $|\cdot| = |\cdot|_\infty$ on actions, $\kappa = 2\pi K s$, $\bar\alpha = 2\pi\alpha$. The convexity argument needs an inner product: there the Euclidean norm $\|\cdot\|_2$ on $\mathbb{R}^n$ is used, with the comparisons of Lemma 2.1 ($|x| \le \|x\|_2 \le \sqrt{n}|x|$). Time and the values of Hamiltonians are those of the frozen system $\dot\theta = \partial_I H$, $\dot I = -\partial_\theta H$ with period-1 angles; no rescaling

to Pöschel's 2π-normalisation is needed, because Part 5 (Remark 5.11) shows that the quantities $s\,r$, $s\,m\,r^2/\omega$, $|\phi|/(\rho\sigma)$ are the same numbers in both normalisations (checked below where they occur, Remark 8.21). We assume $n \geq 1$.

8.1. **External facts.**

(E1) *Banach fixed point theorem.* A contraction of a nonempty closed subset of $\mathbb{R}^N$ into itself has a fixed point (Mathlib `ContractingWith.exists_fixedPoint`). Exists.

(E2) *Taylor, Lagrange form, order one.* If $\varphi : [0,1] \to \mathbb{R}$ is C^2, then $\varphi(1) = \varphi(0) + \varphi'(0) + \frac{1}{2}\varphi''(\xi)$ for some $\xi \in (0,1)$ (Mathlib `taylor_mean_remainder_lagrange`). Exists.

(E3) *Numerics.* $e > 2.7$, so $\frac{1}{12e} < 0.0307$, $\frac{6}{5e} < 0.442$, $\frac{15e}{192} > \frac{1}{5}$; Young's inequality $ab \leq \frac{a^2}{2c} + \frac{cb^2}{2}$ $(c > 0)$; $xe^{-x} \leq e^{-1}$ for $x \geq 0$. Exists.

(E4) *Orthogonal projections.* For a subspace $E \subset \mathbb{R}^n$ with orthogonal projection Π and $\Pi^\perp = 1 - \Pi$: Π is self-adjoint, $\|\Pi v\|_2 \leq \|v\|_2$, $\langle u, \Pi v\rangle = \langle \Pi u, v\rangle$, and $\|\Pi v\|_2 = \operatorname{dist}_2(v, E^\perp)$. Exists.

Track-K and earlier-part results used: [K:`bp:p1:ext-mvt`], [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:real-restriction`]; (E3), Lemma 2.1, Lemma 2.7; Part 4 (Definition 4.2, Lemma 4.19, Theorems 4.11, 4.13, Corollary 4.16, Lemma 4.22, Theorem 4.23); Part 5 (Definition 5.1, 5.2, Lemmas 5.3, 5.4, 5.6, 5.8, 5.9); Lemma 6.5; Part 7 (Definition 7.1, Lemma 7.2, Theorem 7.9, Proposition 7.11).

8.2. **Real systems on a real neighbourhood of D.**

Definition 8.1 (Real neighbourhoods, real systems, orbits)**.** Let $D \subset \mathbb{R}^n$ and $\varrho, s > 0$. Put $U_\varrho D = V_\varrho D \cap \mathbb{R}^n = \{I \in \mathbb{R}^n : \exists J \in D,\ |I - J| < \varrho\}$ (Definition 5.1); it is open, and $W_\varrho := \mathbb{R}^n \times U_\varrho D$ is open in $\mathbb{R}^{2n}$ and equals $\mathcal{D}_{\varrho,s} \cap \mathbb{R}^{2n}$ (real θ lies in T_s). For $0 < \varrho \leq r$, $W_\varrho \subset W_r$. $\mathcal{O}(r,s)^{\mathbb{R}}$ denotes the set of $G \in \mathcal{O}(r,s)$ with $G(x) \in \mathbb{R}$ for $x \in W_r$ ("real on real points"). For such G the *real Hamiltonian* is $G^{\mathbb{R}}(x) = G(x)$ for $x \in W_r$ and $G^{\mathbb{R}}(x) = 0$ otherwise (a total function $\mathbb{R}^{2n} \to \mathbb{R}$; its values off W_r never matter, Remark 4.1). A *G-orbit in W_ϱ on J* ($J \subset \mathbb{R}$ an interval, $0 < \varrho \leq r$) is a $z : \mathbb{R} \to \mathbb{R}^{2n}$ with $\mathtt{IsSolIn}\, X_{G^{\mathbb{R}}}\, W_\varrho\, J\, z$ (Definition 4.2). We write $z = (\theta, I)$ and $I(t) = z(t)_2$. For h holomorphic on $V_r D$ and real on $U_r D$: $h_{\mathbb{R}} = h|_{U_r D}$, $\omega = \nabla h_{\mathbb{R}}$ (real, $= \partial_I h$), $Q(I) = D^2 h_{\mathbb{R}}(I)$ (symmetric).

Source: Ours (bookkeeping for [K:`bp:p1:hamiltonian-real`] and Part 4); Pöschel's $U_r D$, $V_r D$ (p. 189) in max norm.

Lemma 8.2 (Holomorphic real systems)**.** *Let $G \in \mathcal{O}(r,s)^{\mathbb{R}}$, $0 < \varrho \leq r$. Then (a) $G^{\mathbb{R}}$ is C^∞ on W_r and $G^{\mathbb{R}}(\theta + k, I) = G^{\mathbb{R}}(\theta, I)$; (b) at $x \in W_r$, $G^{\mathbb{R}}$ is differentiable and $X_{G^{\mathbb{R}}}(x) = X_G(x) = \big((\partial_{I_i} G(x))_i, (-\partial_{\theta_i} G(x))_i\big) \in \mathbb{R}^{2n}$ (complex partials at the real point); (c) $X_{G^{\mathbb{R}}}$ is C^1 on W_r and 1-periodic in θ; (d) Theorems 4.23, 4.11, 4.13, Corollary 4.16 and Lemma 4.19 apply to $F = X_{G^{\mathbb{R}}}$ on the open set W_ϱ; in particular $G^{\mathbb{R}}$ is constant along every G-orbit on every interval (energy conservation).*

Source: Ours; [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:holomorphic-regularity`]*; the special case $G = h + f$ of N-HEAD-C is Lemma 2.3.*

Proof. **Step S8.1: Smoothness.** G is holomorphic on $\mathcal{D}_{r,s}$, hence C^∞ in the real sense ([K:`bp:p1:holomorphic-regularity`], (E3)); $W_r = \iota^{-1}(\mathcal{D}_{r,s})$ is open and $G^{\mathbb{R}} = G \circ \iota$ on it, so $G^{\mathbb{R}} \in C^\infty(W_r)$, with $D^k G^{\mathbb{R}} = D^k G \circ \iota$ on real vectors. Periodicity is the periodicity in $\mathcal{O}(r,s)$.

Step S8.2: Field. [K:`bp:p1:hamiltonian-real`] with $V = \mathcal{D}_{r,s}$ (open, G holomorphic, real at the real points of V since $V \cap \mathbb{R}^{2n} = W_r$): $G^{\mathbb{R}}$ agrees with $\operatorname{Re} G$ on the neighbourhood W_r of each real point of V, so $X_{G^{\mathbb{R}}} = X_G$ there. The field is C^1 by Lemma 4.22, periodic by (a) and Lemma 4.24.

Step S8.3: (d). Lemma 4.22 ($G^{\mathbb{R}} \in C^2$ on the open set W_ϱ); the cited results need only F C^1 on the open set W_ϱ, periodicity (for the corollary) and, for energy, differentiability of $G^{\mathbb{R}}$ at the points of the orbit. □

Lemma 8.3 (Derivative bound at real points)**.** *Let $u \in \mathcal{O}(\varrho, s)$ have finite norm. Then for all $\theta \in \mathbb{R}^n$, $I \in U_\varrho D$: $\sum_{j=1}^n |\partial_{\theta_j} u(\theta, I)| \le |u|_{\varrho,s}/(e\,s)$; in particular $\|\partial_\theta u(\theta, I)\|_2 \le |u|_{\varrho,s}/(es)$. Also $|u(x)| \le |u|_{\varrho,s}$ for $x \in \mathcal{D}_{\varrho,s}$.*
Source: Pos93 p. 198 ("by the Cauchy estimate of Lemma B.3": $\sup\|\partial_\theta f_*\| \le \frac{1}{es}|f_*|_{r,s}$); the limit $s' \downarrow 0$ in Lemma 5.6(a) is ours.

Proof. **Step S8.4: Strip loss.** For $0 < s' < s$ and fixed I, Step S5.12 gives $\sum_j |\partial_{\theta_j} u|_{I,s'} \le |u|_{I,s}/(e(s-s'))$ (Lemma 5.6(a)). By the pointwise bound in Step S5.7 (Lemma 5.4(a)), $|v(\theta, I)| \le |v|_{I,s'}$ for $\theta \in T_{s'} \supset \mathbb{R}^n$.

Step S8.5: Limit. Let $s' \downarrow 0$; $\|v\|_2 \le \sum_j |v_j|$ (Lemma 2.1). The last claim is Lemma 5.4(a). □

Lemma 8.4 (Resonant gradients lie in the span of Λ)**.** *Let $\Lambda \subset \mathbb{Z}^n$, $\Pi = \Pi_\Lambda$ the Euclidean orthogonal projection onto $\operatorname{span}_{\mathbb{R}} \Lambda$, $\Pi^\perp = 1 - \Pi$. If $g \in \mathcal{O}(r, s)^{\mathbb{R}}$ has finite norm and $P_\Lambda g = g$, then $\partial_\theta g(\theta, I) \in \operatorname{ran} \Pi$, i.e. $\Pi^\perp \partial_\theta g = 0$, for all $(\theta, I) \in W_r$.*
Source: Pos93 p. 198 ("the vector $Q\Delta I$ is given by the integral ... of $\partial_\theta f_*$ only, since g is in Λ-resonant normal form").

Proof. **Step S8.6: Coefficients.** The components of $u = \Pi^\perp \partial_\theta g$ are finite combinations of $\partial_{\theta_j} g \in \mathcal{O}(r, s)$ with coefficients $\hat{u}_k = 2\pi i\, \Pi^\perp k\, \hat{g}_k$ (Lemma 5.3(b)). For $k \in \Lambda$, $\Pi^\perp k = 0$; for $k \notin \Lambda$, $\hat{g}_k = 0$ because $g = P_\Lambda g$ (Lemma 5.9). So $\hat{u}_k = 0$ for all k.

Step S8.7: Vanishing. For $s' < s$, u has finite norm at width s' (Lemma 5.6(a)), so $u = \sum_k \hat{u}_k \chi_k = 0$ on $\mathcal{D}_{r,s'}$ (Lemma 5.3(d)); real points lie in it. □

8.3. **Transfer through a symplectic change of variables.**

Lemma 8.5 (Conjugation of orbits)**.** *Let $H \in \mathcal{O}(r, s)^{\mathbb{R}}$, $\tilde{H} \in \mathcal{O}(\tilde{r}, \tilde{s})^{\mathbb{R}}$, and $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathcal{D}_{r,s}$ holomorphic, real at real points, with $(D\Psi)^\top J D\Psi = J$ and $\tilde{H} = H \circ \Psi$. If $\tilde{z}$ is an $\tilde{H}$-orbit in $W_{\tilde{r}}$ on J, then $z = \Psi \circ \tilde{z}$ is an H-orbit in W_r on J.*
Source: Standard; Corollary 6.6(c) for a single flow map, here for an arbitrary symplectic Ψ.

Proof. **Step S8.8: Real points.** Ψ maps $W_{\tilde{r}} = \mathcal{D}_{\tilde{r},\tilde{s}} \cap \mathbb{R}^{2n}$ into $\mathcal{D}_{r,s} \cap \mathbb{R}^{2n} = W_r$; the real derivative $Y = D\Psi(x)$ at $x \in W_{\tilde{r}}$ is the (real) restriction of the complex one ([K:bp:p1:real-restriction]), and $Y^\top J Y = J$; hence $\det Y = \pm 1$ and $Y J Y^\top = J$ (Step S6.10).

Step S8.9: Chain rule. $\tilde{H}^{\mathbb{R}} = H^{\mathbb{R}} \circ \Psi$ on $W_{\tilde{r}}$, so $\nabla \tilde{H}^{\mathbb{R}}(x) = Y^\top \nabla H^{\mathbb{R}}(\Psi x)$ and, with $X_u = J\nabla u$, $\frac{d}{dt}\Psi(\tilde{z}) = Y\tilde{z}' = Y J Y^\top \nabla H^{\mathbb{R}}(\Psi\tilde{z}) = X_{H^{\mathbb{R}}}(\Psi\tilde{z})$ for $t \in J$ (Lemma 8.2(b)); $z(t) \in W_r$ by Step S8.8. □

Lemma 8.6 (Real preimages of a near-identity map)**.** *Let $D \subset \mathbb{R}^n$, $\tilde{r}, \tilde{s} > 0$, $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathbb{C}^{2n}$ holomorphic, real at real points, $G = \Psi - \mathrm{id}$, with $|G_\theta| \le d_\theta$, $|G_I| \le d_I$ on $\mathcal{D}_{\tilde{r},\tilde{s}}$. Suppose there are $0 < \varrho' \le \tilde{r}$, $0 < \sigma' \le 2\tilde{s}$ with $\kappa_* := \max(d_\theta/\sigma', d_I/\varrho') \le \frac{1}{4}$. Then for every $y = (\theta_y, I_y) \in \mathbb{R}^n \times D$ there is $x = (\theta_x, I_x) \in W_{\tilde{r}}$ with $\Psi(x) = y$, $|\theta_x - \theta_y| \le d_\theta$, $|I_x - I_y| \le d_I$.*
Source: Ours. Pöschel p. 196 and p. 200 assert ("Ψ^{-1} takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$") without proof; a left inverse (Proposition 6.13(e)) does not give this.

Proof. Let $\nu(a, b) = \max(|a|/\sigma', |b|/\varrho')$ on $\mathbb{C}^{2n}$ (the scaled norm of Definition 6.2), so $\nu(G) \le \kappa_*$.

Step S8.10: A closed convex set. $X = \{x \in \mathbb{R}^{2n} : \nu(x - y) \le \kappa_*\}$ is closed, convex, nonempty. For $x \in X$: $|I_x - I_y| \le \varrho'/4 < \tilde{r}$ and $I_y \in D$, so $x \in W_{\tilde{r}}$.

Step S8.11: Self-map. $T(x) = y - G(x)$ is real (Ψ real at real points) and $\nu(T(x) - y) = \nu(G(x)) \le \kappa_*$: $T : X \to X$.

Step S8.12: Contraction. For $x \in X$ the complex ν-ball $\{\nu(z-x) < \frac{1}{2}\}$ lies in $\mathcal{D}_{\tilde r,\tilde s}$: $|\mathrm{Im}\,\theta'| \le |\theta' - \theta_x| < \sigma'/2 \le \tilde s$ and $|I' - I_y| < \varrho'/2 + \varrho'/4 < \tilde r$. On it G is holomorphic with $\nu(G) \le \kappa_*$, so Lemma 5.8 ($R = \frac{1}{2}$) gives $|DG(x)|_\nu \le 2\kappa_* \le \frac{1}{2}$. The real derivative of $G|_{\mathbb{R}^{2n}}$ is the restriction ((E3)); [K:bp:p1:ext-mvt] on the convex set X gives $\nu(T(x) - T(x')) \le \frac{1}{2}\nu(x - x')$.

Step S8.13: Fixed point. (E1) gives $x = y - G(x)$, i.e. $\Psi(x) = x + G(x) = y$, and $y - x = G(x)$ gives the displacement bounds. □

Corollary 8.7 (Preimages for the Normal Form Lemma)**.** *In Theorem 7.9 with $\Lambda = -\Lambda$, h, f real on real points, $\kappa \ge 6$: every $y \in \mathbb{R}^n \times D$ is $\Psi(x)$ for some $x \in W_{r/2}$ with $|I_x - I_y| \le qK\varepsilon/\alpha$ ($\le r/2^7$).*

Proof. **Step S8.14: Apply Lemma 8.6.** Ψ is real at real points (Theorem 7.9(d)); $\tilde r = r/2$, $\tilde s = s/6$, $d_I = qK\varepsilon/\alpha \le r/2^7$ (hypothesis (i)), $d_\theta = 10q\varepsilon/(2\pi\alpha r)$ (item (c)). Take $\varrho' = r/2$, $\sigma' = s/3 = 2\tilde s$: $d_I/\varrho' \le \frac{1}{64}$ and, by (i), $d_\theta/\sigma' = \frac{30q\varepsilon}{2\pi\alpha r s} \le \frac{30}{2\pi 2^7 K s} = \frac{30}{2^7\kappa} \le \frac{30}{768} < \frac{1}{4}$. □

Lemma 8.8 (Transfer of an a priori confinement through a symplectic map)**.** *Let $D \subset \mathbb{R}^n$, $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $\tilde H \in \mathcal{O}(\tilde r,\tilde s)^{\mathbb{R}}$, $\Psi : \mathcal{D}_{\tilde r,\tilde s} \to \mathcal{D}_{r,s}$ as in Lemma 8.5, and $|\cdot|_*$ a norm on $\mathbb{R}^n$ with $|\cdot| \le |\cdot|_*$. Assume*

- *(T1) $|\Pi_I \Psi(x) - I_x|_* \le d_*$ for all $x \in W_{\tilde r}$;*
- *(T2) every $y \in \mathbb{R}^n \times D$ equals $\Psi(x)$ for some $x \in W_{\tilde r}$;*
- *(T3) (*a priori confinement in the new variables*) there are $\varrho > 0$ with $d_* + \varrho < \tilde r$ and $T : D \to (0,\infty]$ such that for every $I_y \in D$, every interval $\tilde J \ni 0$ and every $\tilde H$-orbit $\tilde z$ in $W_{\tilde r}$ on $\tilde J$ with $|\tilde I(0) - I_y|_* \le d_*$: $|\tilde I(t) - \tilde I(0)|_* \le \varrho$ for all $t \in \tilde J$, $|t| \le T(I_y)$.*

Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I_y := I(0) \in D$:

$$|I(t) - I(0)|_* \le \varrho + 2d_* \qquad (t \in J,\ |t| \le T(I_y)).$$

Source: Ours; this is the transfer step of Pöschel pp. 196, 200 ("Going back to original coordinates"), in stopped form so that Part 10 can connect it with existence (Theorem 4.34).

Proof. **Step S8.15: A normal-form orbit.** By (T2) choose $x \in W_{\tilde r}$ with $\Psi(x) = z(0)$; by (T1) $|I_x - I_y|_* \le d_*$. Let $\tilde z$ with maximal interval (T_-, T_+) be the maximal solution of $X_{\tilde H^{\mathbb{R}}}$ in $W_{\tilde r}$ through x (Lemma 8.2(d), Theorem 4.13).

Step S8.16: It lives long enough. (T3) applies on (T_-, T_+). The set $K = \{I : |I - I_x|_* \le \varrho\}$ is compact and, since $|I - I_y| \le |I - I_y|_* \le \varrho + d_* < \tilde r$ and $I_y \in D$, contained in $U_{\tilde r}D$. If some $t \ge 0$ with $t \le T(I_y)$ had $t \ge T_+$, then $T_+ < \infty$ and $\tilde I(t') \in K$ for all $t' \in [0, T_+)$, so $T_+ = \infty$ by Corollary 4.16 (the field is periodic, Lemma 8.2(c)): a contradiction. Likewise for $t \le 0$. Thus $\{|t| \le T(I_y)\} \subset (T_-, T_+)$.

Step S8.17: Identification. $\Psi \circ \tilde z$ is an H-orbit in W_r on (T_-, T_+) (Lemma 8.5) with value y at 0, as is z on J; Theorem 4.11 (with $F = X_{H^{\mathbb{R}}}$, $W = W_r$) gives $z = \Psi \circ \tilde z$ on $J \cap (T_-, T_+)$.

Step S8.18: Estimate. For $t \in J$, $|t| \le T(I_y)$: $I(t) - I(0) = (\Pi_I \Psi(\tilde z(t)) - \tilde I(t)) + (\tilde I(t) - \tilde I(0)) + (\tilde I(0) - I_y)$, of size $\le d_* + \varrho + d_*$ by (T1), (T3) and $\tilde I(0) = I_x$. □

Corollary 8.9 (Stopped form for any extension of the field)**.** *Let $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $|\cdot|_* \ge |\cdot|$ a norm, $0 < \varrho < r$, $T \in (0,\infty]$ and suppose that every H-orbit in W_r on every interval $J \ni 0$ with $I(0) \in D$ satisfies $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$. Let $F : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be any function with $F = X_{H^{\mathbb{R}}}$ on W_r and $z : \mathbb{R} \to \mathbb{R}^{2n}$, $J \ni 0$ an interval with $z'(t) = F(z(t))$ for $t \in J$ and $z(0) \in \mathbb{R}^n \times D$. Then $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$; in particular $z(t) \in W_r$ there.*

Proof. **Step S8.19: The stopped interval.** $\{t \in J : z(t) \in W_r\}$ is relatively open in J and contains 0; let J' be its connected component through 0. z is an H-orbit in W_r on J', so the bound holds on $J' \cap [-T, T]$.

Step S8.20: No exit. If $t_e \in J \cap [-T,T]$ were an endpoint of J' not in J', continuity would give $|I(t_e)-I(0)|_* \le \varrho$, hence $|I(t_e)-I(0)| < r$ and $I(t_e) \in U_r D$ ($I(0) \in D$), so t_e would belong to the component: contradiction. □

Lemma 8.10 (First exit from a norm ball)**.** *Let $|\cdot|_*$ be a norm on $\mathbb{R}^n$, $J \ni 0$ an interval, $z : J \to \mathbb{R}^{2n}$ continuous, $I_0 = I(0)$, $\varrho > 0$. If $|I(t_1)-I_0|_* > \varrho$ for some $t_1 \in J$, then there is $t^* \in J$ between 0 and t_1 with $|I(t^*)-I_0|_* = \varrho$ and $|I(\tau)-I_0|_* \le \varrho$ for all τ between 0 and t^*.*

Proof. **Step S8.21: Exit set.** Apply Lemma 4.19 to $V = \mathbb{R}^n \times \{|I-I_0|_* < \varrho\}$ (open; $z(0) \in V$), forward if $t_1 > 0$, and to $t \mapsto z(-t)$ if $t_1 < 0$ (only continuity is used): t^* exists, $z(\tau) \in V$ before it and $z(t^*) \in \overline{V} \setminus V$, i.e. $|I(t^*)-I_0|_* \le \varrho$ and $\not< \varrho$. □

8.4. **Proposition 1: nonresonant stability.**

Proposition 8.11 (Nonresonant stability estimate)**.** *Let $n \ge 1$, $D \subset \mathbb{R}^n$, $r,s,K,\alpha > 0$, $L \ge 0$, $p > 1$, $q = p/(p-1)$, $\kappa = 2\pi K s$. Let h be holomorphic on $V_r D$, real on $U_r D$, $f \in \mathcal{O}(r,s)^{\mathbb{R}}$ with $\varepsilon := |f|_{r,s} < \infty$, and $H = h + f$. Assume*

*(i) (*completely α, K-nonresonant*) $|k \cdot \omega(I)| \ge \alpha$ for $I \in D$ and $k \in \mathbb{Z}^n$, $0 < |k|_1 \le K$;*
(ii) the Lipschitz bound (N2) for ω with constant L, and $KLr \le \alpha/p$;
(iii) $\varepsilon \le \dfrac{\alpha r}{2^7 q K}$.

Put $T_{\mathrm{nr}} = \dfrac{s\,r}{5\varepsilon} e^{\kappa/6}$ ($= +\infty$ if $\varepsilon = 0$). Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I(0) \in D$:

$$|I(t)-I(0)| \le \tfrac{r}{2} \ \ (\le r) \qquad \text{for all } t \in J,\ |t| \le T_{\mathrm{nr}}.$$

No convexity is used. The constants depend only on (r,s,K,α,p) and ε.
Source: Pos93 p. 196 (Proposition 1: conclusion $\|I(t)-I_0\| \le r$ for $|t| \le \frac{sr}{5\varepsilon} e^{Ks/6}$; hypotheses $\varepsilon \le \frac{1}{2^7 q}\frac{\alpha r}{K}$, $r \le \min(\alpha/(pMK), r_0)$). Deviations: the conclusion is $r/2$ (the proof gives it); the case $\kappa < 6$ (his one line p. 197) is done with $\Psi = \mathrm{id}$ and no loss of width, so the factor e of Proposition 7.11 is absorbed (Remark 8.12); the transfer to the original coordinates includes the surjectivity of Ψ, which the source does not prove.

Proof. Let $\tilde H = h + g + f_*$ be the normal form below.

Step S8.22: Case A, $\kappa \ge 6$: data. Theorem 7.9 with $\Lambda = \{0\}$ ($\Lambda \cap Z_K = \{0\}$) applies: (N1) is (i), (N2) and $KLr \le \alpha/p$ are (ii), $\varepsilon \le \alpha r/(2^7 qK)$ is (iii). It gives $\Psi : \mathcal{D}_{r/2,s/6} \to \mathcal{D}_{r,s}$ holomorphic, symplectic, real at real points, $H \circ \Psi = h + g + f_*$ with g independent of θ, $f_* \in \mathcal{O}(r/2, s/6)^{\mathbb{R}}$, $\eta := |f_*|_{r/2,s/6} \le e^{-\kappa/6}\varepsilon$, and $|\Pi_I \Psi - I| \le d := qK\varepsilon/\alpha \le r/2^7$. Put $\tilde r = r/2$, $\tilde s = s/6$, $\varrho = \frac{15}{32} r$, $d_* = d$, $|\cdot|_* = |\cdot|$, $T = T_{\mathrm{nr}}$. (T1) is the displacement bound, (T2) is Corollary 8.7; $d_* + \varrho \le \frac{61}{128} r < \tilde r$.

Step S8.23: Case A: (T3). Let $\tilde z$ be an $\tilde H$-orbit in $W_{\tilde r}$ on $\tilde J$, $|\tilde I(0) - I_y| \le d$, $I_y \in D$, and suppose $|\tilde I(t_1) - \tilde I(0)| > \varrho$ for some $t_1 \in \tilde J$, $|t_1| \le T_{\mathrm{nr}}$. Lemma 8.10 gives t^*, $|t^*| \le |t_1|$, $|\tilde I(t^*) - \tilde I(0)| = \varrho$, with $\tilde I(\tau)$ in the closed ball B of radius ϱ about $\tilde I(0)$ for τ between 0 and t^*; $B \subset U_{\tilde r} D$ as $\varrho + d < \tilde r$. On this stretch $\dot{\tilde I} = -\partial_\theta \tilde H^{\mathbb{R}} = -\partial_\theta f_*(\tilde z)$ (h and g do not depend on θ; Lemma 8.2(b)), so by Lemma 8.3 $|\dot{\tilde I}| \le \eta/(e\tilde s) = 6\eta/(es)$ and, by the mean value inequality ([K:bp:p1:ext-mvt]),

$$\varrho = |\tilde I(t^*) - \tilde I(0)| \le \frac{6\eta|t^*|}{es} \le \frac{6e^{-\kappa/6}\varepsilon}{es} \cdot \frac{sr\,e^{\kappa/6}}{5\varepsilon} = \frac{6}{5e} r < 0.442\, r < \tfrac{15}{32} r = \varrho,$$

a contradiction ((E3); if $\varepsilon = 0$ then $\eta = 0$ and the left side is 0). So (T3) holds with $T(I_y) = T_{\mathrm{nr}}$.

Step S8.24: Case A: conclusion. Lemma 8.8: $|I(t) - I(0)| \le \frac{15}{32} r + \frac{2}{2^7} r = \frac{31}{64} r \le \frac{r}{2}$.

Step S8.25: Case B, $\kappa < 6$. Take $\Psi = \mathrm{id}$ on $\mathcal{D}_{r,s}$, $\tilde H = H$, $g = 0$, $f_* = f$ (Proposition 7.11), $\tilde r = r$, $\tilde s = s$, $d_* = 0$, $\varrho = r/2$: (T1), (T2) are trivial. For (T3): as in Step S8.23, with $\tilde I(\tau)$ in the

ball of radius $r/2$ about $\tilde I(0)\in D$ ($\subset U_rD$), $|\dot{\tilde I}|\le \varepsilon/(es)$ and $|\tilde I(t^*)-\tilde I(0)|\le \frac{\varepsilon}{es}\cdot\frac{sr\,e^{\kappa/6}}{5\varepsilon}<\frac{r}{5}<\frac{r}{2}$ (as $e^{\kappa/6}<e$): contradiction. Lemma 8.8 gives $|I(t)-I(0)|\le r/2$. □

Remark 8.12 (The factor e in the nonresonant case). Proposition 7.11 gives only $|f_*|_{r,s}\le e\cdot e^{-\kappa/6}\varepsilon$ for $\kappa<6$. Used with the widths $(r/2,s/6)$ of the case $\kappa\ge 6$ it would give the coefficient $31/384<\frac15$ in the time bound. Case B avoids this by not losing width ($\tilde s=s$, $\eta=\varepsilon$): the available time is $\frac{es\,r}{2\varepsilon}>\frac{sr}{5\varepsilon}e^{\kappa/6}$ for all $\kappa<6$, so the stated $T_{\rm nr}$ is attained with room to spare, and no factor e is lost in the final statement.

8.5. **The energy argument.**

Definition 8.13 (Resonance data, convexity, quasi-convexity)**.** Let $D\subset\mathbb{R}^n$, $\Lambda\subset\mathbb{Z}^n$, $\Pi=\Pi_\Lambda$ (Lemma 8.4), $R_\Lambda=\{\omega\in\mathbb{R}^n : k\cdot\omega=0\ \forall k\in\Lambda\}=\ker\Pi$. D is *δ-close to exact Λ-resonances* if $\|\Pi\omega(I)\|_2=\operatorname{dist}_2(\omega(I),R_\Lambda)\le\delta$ for all $I\in D$. For h as in Definition 8.1 and $U\subset U_rD$: h is *m-convex on U* if $Q(I)[\xi,\xi]\ge m\|\xi\|_2^2$ for $I\in U$, $\xi\in\mathbb{R}^n$; *ℓ,m-quasi-convex on U* if for all $I\in U$ and $\xi\in\mathbb{R}^n$: $|\langle\omega(I),\xi\rangle|>\ell\|\xi\|_2$ or $Q(I)[\xi,\xi]\ge m\|\xi\|_2^2$ (equivalently $|\langle\omega,\xi\rangle|\le\ell\|\xi\|_2\Rightarrow Q[\xi,\xi]\ge m\|\xi\|_2^2$, Lemma 2.7(b); never full positive definiteness, review N05). $M\ge m$ denotes a bound $\|Q(I)\|_{2\to2}\le M$ on U_rD (the normalisation $m\le M$ is always achievable by enlarging M).
Source: Pos93 pp. 190, 197.

Lemma 8.14 (Energy–convexity inequality (reusable core))**.** *Let $U\subset\mathbb{R}^n$ be open, $h_{\mathbb{R}}\in C^2(U)$, $g,f\in C^1(\mathbb{R}^n\times U;\mathbb{R})$, $H^{\mathbb{R}}=h_{\mathbb{R}}(I)+g+f$ on $\mathbb{R}^n\times U$ (any values elsewhere), Π an orthogonal projection of $(\mathbb{R}^n,\|\cdot\|_2)$ with $\Pi^\perp\partial_\theta g\equiv0$. Let $J\ni0$ be an interval, z with* `IsSolIn` $X_{H^{\mathbb{R}}}(\mathbb{R}^n\times U)\,J\,z$, *$t_1\in J$, $I_0=I(0)$, $\Delta I=I(t_1)-I_0$, $I_s=I_0+s\Delta I$, $\omega_0=\nabla h_{\mathbb{R}}(I_0)$, and assume $I_s\in U$ for $s\in[0,1]$. Then*

(a) (energy) $h_{\mathbb{R}}(I(t_1))-h_{\mathbb{R}}(I_0)=-\big[(g+f)(z(t_1))-(g+f)(z(0))\big]$;
(b) (resonant directions) $\|\Pi^\perp\Delta I\|_2\le|t_1|\sup_\tau\|\partial_\theta f(z(\tau))\|_2$, τ between 0 and t_1;
(c) (Taylor) $h_{\mathbb{R}}(I(t_1))-h_{\mathbb{R}}(I_0)-\langle\omega_0,\Delta I\rangle=\frac12Q(I_\xi)[\Delta I,\Delta I]$ for some $\xi\in(0,1)$;
(d) if $Q(I_s)[\Delta I,\Delta I]\ge m\|\Delta I\|_2^2$ for all $s\in[0,1]$, then

$$\frac{m}{2}\|\Delta I\|_2^2\le\big|(g+f)(z(t_1))-(g+f)(z(0))\big|+\|\Pi\omega_0\|_2\|\Delta I\|_2+\|\omega_0\|_2\|\Pi^\perp\Delta I\|_2.$$

Source: Pos93 p. 198 ("Following Lochak … expanding around I_0", $\Delta h=\langle\omega_0,\Delta I\rangle+\int_0^1(1-s)\langle Q(I(s))\Delta I,\Delta I\rangle ds$, energy conservation, the vector $Q\Delta I$ …), with the Lagrange remainder instead of the integral one. This is the part wanted by track I (energy confinement).

Proof. **Step S8.26: Energy.** $H^{\mathbb{R}}$ is C^1 on the open set $\mathbb{R}^n\times U$, hence differentiable along the orbit, and Theorem 4.23 on the interval between 0 and t_1 gives $H^{\mathbb{R}}(z(t_1))=H^{\mathbb{R}}(z(0))$; this is (a).

Step S8.27: Resonant projection. $\dot I=-\partial_\theta H^{\mathbb{R}}(z)=-\partial_\theta g-\partial_\theta f$ ($h_{\mathbb{R}}$ is θ-independent). Hence $t\mapsto\Pi^\perp I(t)$ has derivative $-\Pi^\perp\partial_\theta f(z(t))$, of norm $\le\|\partial_\theta f(z(t))\|_2$; the mean value inequality ([K:`b p:p1:ext-mvt`], $\Pi^\perp$ linear) gives (b).

Step S8.28: Taylor and combination. (c) is (E2) for $\varphi(s)=h_{\mathbb{R}}(I_s)$ ($\varphi'(0)=\langle\omega_0,\Delta I\rangle$, $\varphi''=Q(I_s)[\Delta I,\Delta I]$). Under the hypothesis of (d), (c) gives $\frac{m}{2}\|\Delta I\|^2\le h_{\mathbb{R}}(I(t_1))-h_{\mathbb{R}}(I_0)-\langle\omega_0,\Delta I\rangle$; insert (a), and $\langle\omega_0,\Delta I\rangle=\langle\Pi\omega_0,\Delta I\rangle+\langle\omega_0,\Pi^\perp\Delta I\rangle$ ((E4)) with Cauchy–Schwarz. □

The resonant setting (RS).. $n\ge1$, $D\subset\mathbb{R}^n$, $r,s>0$, $\Lambda\subset\mathbb{Z}^n$, h holomorphic on V_rD and real on U_rD, $0<m\le M$ with $\|Q(I)\|_{2\to2}\le M$ on U_rD, $g,f_*\in\mathcal{O}(r,s)^{\mathbb{R}}$ of finite norm with $P_\Lambda g=g$, $\zeta\ge|g|_{r,s}+|f_*|_{r,s}$, $\eta=|f_*|_{r,s}$, $H=h+g+f_*$, $\delta>0$ with D δ-close to exact Λ-resonances, $\delta\le mr/4$, $\zeta\le mr^2/32$, and $\rho:=\delta/(4M)$ $\big(\le r/16\big)$. An initial action I_0 is *admissible* if $\|I_0-J_0\|_2<\rho$ for some $J_0\in D$.

Lemma 8.15 (Admissible initial actions)**.** *In (RS), for admissible I_0 (with J_0): (a) the closed ball $\bar B = \{I : \|I - I_0\|_2 \le r - \rho\} \subset U_r D$; (b) $\|\Pi\omega(I)\|_2 \le \frac{5}{4}\delta + M\|I - I_0\|_2$ for $I \in \bar B$; in particular $\|\Pi\omega(I_0)\|_2 \le \frac{5}{4}\delta$.*

Proof. **Step S8.29: Ball.** $|I - J_0| \le \|I - J_0\|_2 < (r-\rho) + \rho = r$, so $I \in U_r D$; likewise the segment $[J_0, I_0] \subset U_r D$.

Step S8.30: Closeness. By the mean value inequality for ω in $\|\cdot\|_2$ ($D\omega = Q$, $\|Q\| \le M$, convex segments inside $U_r D$): $\|\omega(I_0) - \omega(J_0)\|_2 \le M\rho = \delta/4$, so $\|\Pi\omega(I_0)\| \le \delta + \delta/4$; and $\|\Pi\omega(I)\| \le \|\Pi\omega(I_0)\| + M\|I - I_0\|_2$. □

Lemma 8.16 (No exit under convexity along the segment)**.** *In (RS) let z be an H-orbit in W_r on $J \ni 0$, $I_0 = I(0)$ admissible, and $t_1 \in J \setminus \{0\}$ with*

(i) $\|I(\tau) - I_0\|_2 \le r - \rho$ for τ between 0 and t_1, and $\|I(t_1) - I_0\|_2 = r - \rho$;
(ii) $Q(I_s)[\Delta I, \Delta I] \ge m\|\Delta I\|_2^2$ for all $s \in [0,1]$ ($\Delta I = I(t_1) - I_0$, $I_s = I_0 + s\Delta I$);
(iii) $|t_1|\,\|\omega(I_0)\|_2\,\eta \le s\,m\,r^2/12$.

Then a contradiction arises (so these three cannot hold together).

Source: Pos93 p. 198 (end of the proof of Lemma 1): $\frac{m}{3}\|\Delta I\|^2 \le 2\zeta + \frac{5}{2m}\delta^2 + T\|\omega_0\|\frac{1}{es}|f_*| \le \frac{mr^2}{4} < \frac{m}{3}(r-\rho)^2$; all numbers recomputed.

Proof. Put $\Delta = \|\Delta I\|_2 = r - \rho \ge \frac{15}{16} r$.

Step S8.31: Apply the core. $U = U_r D$, $\bar B \subset U$ (Lemma 8.15) is convex so $I_s \in U$; g, f_* are C^∞ on W_r (Lemma 8.2), $\Pi^\perp \partial_\theta g = 0$ (Lemma 8.4), the orbit is a solution of $X_{H^{\mathbb{R}}}$ in $\mathbb{R}^n \times U$. Lemma 8.14(d) with (ii): $\frac{m}{2}\Delta^2 \le A + B + C$, $A = |\Delta(g + f_*)|$, $B = \|\Pi\omega_0\|\Delta$, $C = \|\omega_0\|\|\Pi^\perp \Delta I\|$.

Step S8.32: Three terms. $A \le 2(|g|_{r,s} + |f_*|_{r,s}) \le 2\zeta \le \frac{mr^2}{16}$ (Lemma 8.3, last claim). $B \le \frac{5}{4}\delta\Delta \le \frac{5}{2m}\delta^2 + \frac{m}{6}\Delta^2$ (Lemma 8.15, Young), and $\frac{5}{2m}\delta^2 \le \frac{5mr^2}{32}$. By Lemma 8.14(b), (i) and Lemma 8.3, $C \le \|\omega_0\||t_1|\frac{\eta}{es} \le \frac{mr^2}{12e} < 0.0307\, mr^2$ by (iii).

Step S8.33: Contradiction. Hence $\frac{m}{3}\Delta^2 \le mr^2(\frac{1}{16} + \frac{5}{32} + \frac{1}{12e}) < \frac{mr^2}{4}$, while $\frac{m}{3}\Delta^2 \ge \frac{m}{3}\cdot\frac{225}{256}r^2 > 0.29\, mr^2$ ((E3)). □

8.6. **Lemma 1 and Lemma 2.**

Lemma 8.17 (Lemma 1: stability for m-convex h in normal form)**.** *In (RS) let h be m-convex on $U_r D$. Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I_0 = I(0)$ admissible:*

$$\|I(t) - I_0\|_2 \le r - \rho \qquad (t \in J,\ |t| \le T_*), \qquad T_* = \frac{s\,m\,r^2}{12\,\|\omega(I_0)\|_2\,\eta},$$

with $T_* = +\infty$ if $\|\omega(I_0)\|_2\eta = 0$. (In particular for $\Lambda = \mathbb{Z}^n$, $f_* = 0$: all $t \in J$.)

Source: Pos93 p. 197 (Lemma 1: $\delta \le mr/4$, $\zeta \le mr^2/32$, $\rho = \delta/4M$, $T_* = \frac{s}{12\|\omega(I_0)\|}\frac{mr^2}{\eta}$). Statement in stopped form ("as long as the action stays in W_r": the orbit is an orbit in W_r on J); with exit analysis by Lemma 8.10.

Proof. **Step S8.34: Exit.** Suppose $\|I(t_1) - I_0\|_2 > r - \rho$ for some $t_1 \in J$, $|t_1| \le T_*$. Lemma 8.10 ($|\cdot|_* = \|\cdot\|_2$) gives $t^* \in J$, $|t^*| \le |t_1|$, with (i) of Lemma 8.16 for t^*.

Step S8.35: Hypotheses (ii), (iii). $I_s \in \bar B \subset U_r D$ (Lemma 8.15), so (ii) is m-convexity; (iii) is $|t^*| \le T_*$. Lemma 8.16 gives the contradiction. □

Lemma 8.18 (Lemma 2: stability for ℓ, m-quasi-convex h in normal form)**.** *In (RS) let h be ℓ, m-quasi-convex on $U_r D$ and assume*

$$\text{(R)} \qquad r \le 2\ell/m \qquad \text{and} \qquad Mr \le \ell/4$$

(the second is an added *hypothesis, Remark 8.20(1)). For admissible I_0 let $\Omega = \max\{\|\omega(I)\|_2 : \|I - I_0\|_2 \le r - \rho\}$. Then for every H-orbit z in W_r on $J \ni 0$ with $I(0) = I_0$ admissible,*

$$\|I(t) - I_0\|_2 \le r - \rho \quad (t \in J,\ |t| \le T_*), \qquad T_* = \frac{s\, m\, r^2}{12\, \Omega\, \eta}\ (= +\infty \text{ if } \Omega\eta = 0).$$

Source: Pos93 p. 199 (Lemma 2: $r \le 2\ell/m$, $\Omega = \sup_{\|I-I_0\|\le r-\rho}\|\omega(I)\|$, proof by contradiction at the first exit time). The hypothesis $Mr \le \ell/4$ repairs a gap of the proof (Remark 8.20(1)).

Proof. **Step S8.36: Exit.** As in Lemma 8.17, a violation yields t^*, $|t^*| \le T_*$ with (i) of Lemma 8.16. Put ΔI, I_s as there, $\Delta = r - \rho \ge \frac{15}{16} r$; $I_s \in \bar{B}$ and $\|\omega(I_s)\| \le \Omega$.

Step S8.37: (iii). $\|\omega(I_0)\| \le \Omega$, so $|t^*|\|\omega(I_0)\|\eta \le |t^*|\Omega\eta \le smr^2/12$.

Step S8.38: The cone condition along the segment. Fix $s' \in [0,1]$. Then $|\langle \omega(I_{s'}), \Delta I\rangle| \le \|\Pi\omega(I_{s'})\|\Delta + \|\omega(I_{s'})\|\|\Pi^{\perp}\Delta I\|$. By Lemma 8.15(b) ($\|I_{s'} - I_0\| \le \Delta \le r$) the first term is $\le (\frac{5}{4}\delta + Mr)\Delta \le (\frac{5}{16} mr + Mr)\Delta$. By Lemma 8.14(b), Lemma 8.3 and Step S8.37 the second is $\le \Omega|t^*|\frac{\eta}{es} \le \frac{mr^2}{12e} \le \frac{16}{15}\cdot\frac{mr}{12e}\Delta < 0.0328\, mr\, \Delta$. With $mr \le 2\ell$ and $Mr \le \ell/4$:

$$|\langle \omega(I_{s'}), \Delta I\rangle| \le \big(0.3453 \cdot 2\ell + \tfrac{\ell}{4}\big)\Delta < 0.941\, \ell\, \|\Delta I\|_2.$$

Step S8.39: Convexity along the segment. Since $|\langle \omega(I_{s'}), \Delta I\rangle| \le \ell\|\Delta I\|_2$, the first alternative of quasi-convexity at $I_{s'} \in U_r D$ fails for $\xi = \Delta I$, so $Q(I_{s'})[\Delta I, \Delta I] \ge m\|\Delta I\|^2$: (ii) of Lemma 8.16 holds. That lemma gives the contradiction. □

8.7. **Proposition 2: the Resonant Stability Lemma.**

Proposition 8.19 (Resonant Stability Lemma)**.** *Let $n \ge 1$, $D \subset \mathbb{R}^n$, $r, s, K, \alpha > 0$, $1 < p \le \frac{3}{2}$ and $q = p/(p-1)$ (≥ 3), $\kappa = 2\pi K s$. Let $\Lambda \subset \mathbb{Z}^n$ be a subgroup, $0 < m \le M$, $\ell, \delta > 0$, h holomorphic on $V_r D$ and real on $U_r D$ with*

$$\|\partial_I^2 h(I)\|_{2\to 2} \le M \quad \text{for all } I \in V_r D \text{ (complex Hessian, operator norm on } (\mathbb{C}^n, \|\cdot\|_2)),$$

h ℓ, m-quasi-convex on $U_r D$, $f \in \mathcal{O}(r,s)^{\mathbb{R}}$ with $\varepsilon = |f|_{r,s} < \infty$, $H = h + f$. Assume:

- *(a) D is α, K-nonresonant modulo Λ: $|k\cdot\omega(I)| \ge \alpha$ for $I \in D$, $k \in Z_K \setminus \Lambda$, and δ-close to exact Λ-resonances;*
- *(b) $\varepsilon \le \dfrac{p}{q}\cdot\dfrac{mr^2}{2^7}$ and $\delta \le \dfrac{mr}{8}$;*
- *(c) $r \le \min\Big(\dfrac{\alpha}{p\sqrt{n}\, MK},\ \dfrac{4\ell}{m}\Big)$ and (added) $Mr \le \ell/2$.*

For $I_0 \in D$ let $\Omega(I_0) = \max\{\|\omega(I)\|_2 : \|I - I_0\|_2 \le r/2\}$ and

$$T(I_0) = \frac{s\, m\, r^2}{288\, \Omega(I_0)\, \varepsilon}\, e^{\kappa/6} \qquad (= +\infty \text{ if } \Omega\varepsilon = 0).$$

Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I_0 = I(0) \in D$,

$$\|I(t) - I_0\|_2 \le \tfrac{r}{2} \ (\text{so } |I(t) - I_0| \le r) \qquad (t \in J,\ |t| \le T(I_0)),$$

and if $\Lambda = \mathbb{Z}^n$ this holds for all $t \in J$.

Source: Pos93 pp. 199–200 (Proposition 2: $\varepsilon \le \frac{p}{q}\frac{mr^2}{2^7}$, $\delta \le \frac{mr}{8}$, $r \le \min(\frac{\alpha}{pMK}, \frac{4\ell}{m}, r_0)$, $q \ge 3$; $\|I(t) - I_0\| \le r$ for $|t| \le \frac{s}{288\Omega}\frac{mr^2}{\varepsilon}e^{Ks/6}$; $\Lambda = \mathbb{Z}^n$: all t). Deviations: $\sqrt{n}$ in the first bound of (c) (Euclidean Hessian bound M against the max-norm Lipschitz bound $L = \sqrt{n}M$ of Part 7); added hypothesis $Mr \le \ell/2$ and the radius $r/2$ for $\kappa < 6$ (Remark 8.20(1),(2)); Ω taken over the radius $r/2$ (the proof confines to $r/2$).

Proof. **Step S8.40: Reductions.** Replace δ by $\delta' = mr/8 \geq \delta$ (D is δ'-close). Then $\rho' := \delta'/(4M) = \frac{mr}{32M}$. Since $\frac{p}{q} = \frac{1}{q-1} \leq \frac{1}{2}$, (b) gives $\varepsilon \leq mr^2/2^8$. By (c) and $m \leq M$: $L := \sqrt{n}M$ satisfies $KLr \leq \alpha/p$, and $\varepsilon \leq \frac{pmr^2}{q2^7} \leq \frac{pMr^2}{q2^7} \leq \frac{\alpha r}{2^7 qK}$. The complex Hessian bound gives (N2) with $L = \sqrt{n}M$ (Lemma 7.2: $|Qv|_\infty \leq \|Qv\|_2 \leq M\|v\|_2 \leq \sqrt{n}M|v|_\infty$). Put $\tilde{r} = r/2$; $M\tilde{r} \leq \ell/4$, $\tilde{r} \leq 2\ell/m$, $\delta' = m\tilde{r}/4$.

Step S8.41: Three cases. (A) $\Lambda \neq \mathbb{Z}^n$, $\kappa \geq 6$: Theorem 7.9 (conditions (N1), (N2), $K > 0$, (i), (ii) hold; $\Lambda = -\Lambda$, reality) gives $\Psi : \mathcal{D}_{r/2,s/6} \to \mathcal{D}_{r,s}$ with $H \circ \Psi = h + g + f_*$, $P_\Lambda g = g$, $\tilde{s} = s/6$, $\eta \leq e^{-\kappa/6}\varepsilon$ and $|g|_{r,s}$-type bound: $|g|_{r/2,s/6} \leq |g_0|_{r,s} + 16c\varepsilon^2$, where $16c\varepsilon^2 = \frac{32qK\varepsilon^2}{\alpha r} \leq \frac{\varepsilon}{4}$ by (i), so $|g|_{r/2,s/6} + \eta \leq \frac{5}{4}\varepsilon + \frac{\varepsilon}{e} < 2\varepsilon =: \zeta$. (B) $\Lambda \neq \mathbb{Z}^n$, $\kappa < 6$: $\Psi = \mathrm{id}$ on $\mathcal{D}_{r/2,s}$, $g = 0$, $f_* = f$, $\tilde{s} = s$, $\eta = \varepsilon$, $\zeta = \varepsilon$. (C) $\Lambda = \mathbb{Z}^n$: $\Psi = \mathrm{id}$, $\tilde{s} = s$, $g = f$ ($P_\Lambda g = g$), $f_* = 0$, $\eta = 0$, $\zeta = \varepsilon$. In all cases $\tilde{H} = h + g + f_*$ on $\mathcal{D}_{\tilde{r},\tilde{s}}$ is in (RS) with $(r, s) \to (\tilde{r}, \tilde{s})$ and δ': $\delta' = m\tilde{r}/4$, and $\zeta \leq 2\varepsilon \leq mr^2/128 = m\tilde{r}^2/32$.

Step S8.42: (T1), (T2). Case A: $\|\Pi_I \Psi - I\|_2 \leq \sqrt{n}\frac{qK\varepsilon}{\alpha} \leq \frac{q\varepsilon}{pMr} \leq \frac{mr}{2^7 M} = \frac{\rho'}{4} =: d_*$ (Theorem 7.9(c), $\alpha \geq p\sqrt{n}MKr$, (b)); (T2) is Corollary 8.7. Cases B, C: $d_* = 0$, trivial. Norm $|\cdot|_* = \|\cdot\|_2$; $\varrho := \tilde{r} - \rho'$; then $d_* + \varrho < \tilde{r}$.

Step S8.43: (T3). Let $\tilde{z}$ be an $\tilde{H}$-orbit in $W_{\tilde{r}}$, $\|\tilde{I}(0) - I_y\|_2 \leq d_* < \rho'$, $I_y \in D$: $\tilde{I}(0)$ is admissible. Lemma 8.18 (hypotheses (R) at radius $\tilde{r}$) gives $\|\tilde{I}(t) - \tilde{I}(0)\|_2 \leq \varrho$ for $|t| \leq \tilde{T}_* = \frac{\tilde{s}\, m\tilde{r}^2}{12\tilde{\Omega}\eta}$, $\tilde{\Omega} = \max_{\|I - \tilde{I}(0)\|_2 \leq \varrho} \|\omega\|_2$. Since $\varrho + d_* \leq \tilde{r} - \rho' + \rho'/4$, the ball lies in $\{\|I - I_y\|_2 \leq r/2\}$ so $\tilde{\Omega} \leq \Omega(I_y)$. Case A: $\tilde{T}_* \geq \frac{(s/6)(mr^2/4)}{12\Omega\, e^{-\kappa/6}\varepsilon} = T(I_y)$. Case B: $\tilde{T}_* = \frac{smr^2}{48\Omega\varepsilon} \geq T(I_y)$ because $e^{\kappa/6} < e < 6$. Case C: $\tilde{T}_* = \infty$.

Step S8.44: Conclusion. Lemma 8.8: $\|I(t) - I_0\|_2 \leq \varrho + 2d_* \leq \tilde{r} - \rho' + \rho'/2 < \frac{r}{2}$ for $|t| \leq T(I_0)$ ($t \in J$), and all t in case C. Finally $|\cdot| \leq \|\cdot\|_2$. □

8.8. **Errata, deviations, dictionary, consumers.**

Remark 8.20 (Suspected defects in Pos93 (to the ledger)).

(1) *p. 199, proof of Lemma 2*: "$|\langle \omega(I(s)), P\Delta I\rangle| \leq \frac{5}{4}\delta\|\Delta I\|$". The closeness $\frac{5}{4}\delta$ was derived (p. 198) only for $U_\rho D$, but $I(s)$ ranges over the ball of radius $r - \rho$ about I_0, where the mean value theorem gives only $\|P\omega(I(s))\| \leq \frac{5}{4}\delta + M\|I(s) - I_0\|$. Without a bound on Mr the cone condition $|\langle \omega(I(s)), \Delta I\rangle| \leq \ell\|\Delta I\|$ need not hold, so Lemma 2 (and Proposition 2, which uses it) is not proved as printed. Repair: the hypothesis $Mr \leq \ell/4$ (Lemma 8.18), resp. $Mr \leq \ell/2$ in Proposition 8.19. In Theorem 1 (p. 190) one has $r \leq r_0/(AK)$, $r_0 \leq 4\ell/m$, $A = 11M/m$, so $Mr \leq \frac{4\ell}{11K}$, and Part 9 must check $Mr \leq \ell/2$ there ($K \geq 1$ suffices). The lemma may be true without it; we do not claim so.

(2) *p. 200, Proposition 2, case $Ks < 6$*: "we simply apply those two lemmata to the given hamiltonian H with $g = 0$ and $f_* = f_\varepsilon$" at radius r, but Lemma 2 needs $r \leq 2\ell/m$ while only $r \leq 4\ell/m$ is assumed. Repaired by applying them at radius $r/2$ (case B).

(3) *p. 196, Proposition 1 and p. 200*: "Ψ^{-1} takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$" needs surjectivity of Ψ onto a neighbourhood of $D \times \mathbb{T}^n$; not argued (Lemma 8.6 supplies it by a contraction argument using the displacement bounds of Theorem 7.9(c)).

(4) *p. 197, first line*: "For $Ks \leq 6 \ldots f_* = f_\varepsilon$": the bound $|f_*| \leq e^{-Ks/6}\varepsilon$ is false there (Part 7, Proposition 7.11); not used (Remark 8.12).

(5) Misprint-free checks: the numbers $\frac{m}{3}\|\Delta I\|^2 \leq \frac{mr^2}{4} < \frac{m}{3}(r - \rho)^2$ (p. 198) and T_*, $\frac{s}{288\Omega}$ (p. 200) were recomputed and agree; the transfer constants in Proposition 8.19 are sharper than printed (Ψ-displacement $qK\varepsilon/\alpha$ instead of $2qK\varepsilon/\alpha$ in the max norm).

Remark 8.21 (Dictionary with Pöschel's normalisation). With $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (Remark 5.11): $s_P = 2\pi s$, $r_P = r/2\pi$, $\omega_P = 2\pi\omega$, $m_P = 4\pi^2 m$, $\eta, \varepsilon, \zeta, \delta_P = \delta$ unchanged up to the Euclidean/max conventions. Then $s_P r_P = sr$, $s_P m_P r_P^2/\omega_P = s\,m\,r^2/\omega$, so T_*, T_{nr} and T have the printed form; and $\partial_\varphi f_* = \partial_\theta f_*/2\pi$ makes $|\partial_\theta f_*| \le \eta/(es)$ the same estimate as his $|\partial_\theta f_*| \le \eta/(es_P)$.

Remark 8.22 (Consumers). Part 9 (Theorem 1 of Pöschel) takes $q = 9$, $p = \frac{9}{8}$ ($\frac{p}{q} = \frac{1}{8}$, $q \ge 3$), Λ a maximal K-lattice, D a block of Part 8 (with $K = K_a \ge \sqrt{n}K_g$, review N06), and uses Proposition 8.11 for $\Lambda = \{0\}$ and Proposition 8.19 otherwise. With Part 1: $M = nM_{\mathbb{C}}$ (Lemma 2.5), $\ell, m = \mu$ from Lemma 2.7(b); the Hessian bound on $V_r D$ requires $V_r D \subset B(0,R)_s$ (Lemma 2.2(c)). Part 10 connects the stopped form with existence through Corollary 8.9 (take $F = X_{H_{h,f}}$, which equals $X_{H^{\mathbb{R}}}$ on W_r when $U_r D \subset \mathcal{B}_s(R)$) and Theorem 4.34. Track I takes Lemma 8.14 and Lemmas 8.16, 8.17, 8.18, which use no property of h, g, f_* beyond the stated ones.